A Quantitative Convergence Law for the Connes–Consani–Moscovici Zeta Spectral Triple

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Abstract

The Connes–Consani–Moscovici (CCM) construction produces finite real symmetric operators whose spectra approximate the ordinates of the nontrivial Riemann zeta zeros. We develop a caller-facing convergence law for the first ordinate as a function of the prime cutoff λ^2, basis size N, and requested precision P. Intrinsically, the relative error separates into finite-basis and finite-cutoff channels,

$$E(N,\lambda^2) \le \varepsilon_{\mathrm{modes}}(N,\lambda^2) + \varepsilon_{\mathrm{primes}}(\lambda^2),$$

while the observed number of matching decimal digits is, up to a bounded noncancellation term,

$$D(\lambda^2,N,P) = \min\bigl(D_{\mathrm{Wprec}}(P), D_{\mathrm{nModes}}(N,\lambda^2), D_{\mathrm{Primes}}(\lambda^2)\bigr) + O(1).$$

Precision is an instrument-resolution cap, not a third intrinsic error mechanism. On the calibrated domain $7 \le \lambda^2 \le 100$, the operational total intrinsic edge submodel is

$$D^{\mathrm{fit}}_{\mathrm{edge}} = D_{\mathrm{Primes}} \tanh\left[\left(\frac{cN}{\lambda^2 \ln \lambda^2}\right)^{q_\gamma(\lambda^2)}\right], \qquad q_\gamma(\lambda^2) = \frac{2}{2\gamma + \ln\ln \lambda^2}, \qquad c = \frac{4}{\pi}.$$

With these published constants fixed, an independent numerical reimplementation reproduces complete ramps at $\lambda^2 = 7, 13, 20$ with RMS errors 0.47, 0.66, and 1.18 digits, respectively. A new four-point sweep at $\lambda^2 = 1000$ gives the complementary result: the same closed q_γ submodel overpredicts the first-root accuracy by 118–167 digits. Depending on the observable and fitting convention, the measured effective exponent lies in the range 0.74–0.80, rather than $q_\gamma(1000) = 0.648$. Thus the three-budget law, the stretched-tanh ramp family, and the coordinate $\lambda^2 \ln \lambda^2$ remain the working structure, while the large-cutoff dependence of the exponent, onset coefficient, or transition coordinate is explicitly open.

The finite-cutoff channel has the measured form $D_{\mathrm{Primes}} = K\lambda^2 - P_m \log_{10} \lambda^2 - c_m$. Its leading coefficient $K = 4\pi/\ln 10$ coincides with the classical fourth-index prolate singular-value asymptotic; transfer to the constrained CCM h_0–h_4 trial state and the reverse inequality remain open. The retained lattice has the exact edge $T_{\max} = 2\pi N/\ln \lambda^2$, and the Riemann–von Mangoldt law gives a geometric zero-capacity estimate. The empirical law is stated only for validated matrix assemblies. Explicit rigor tiers separate proved identities and bounds, conditional root-transfer formulas, calibrated laws, and open large-cutoff models. This work has been shared with the CCM authors, including their lead author, Professor Alain Connes. The indexed extension retains a mode budget that depends on the requested zero, together with a sufficient branch-resolution condition. An exact dual-residual identity identifies the additional estimates needed to justify root-rate doubling, and a derived nonlinear local predictor improves selected high-index displacements. A new complete first-1000 stored-state test has an exact finite ordinal count and all 1000 fixed operator-predictor errors below $1.70 \cdot 10^{-7}$ in relative displacement. An exact lower-mode counterexample and

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controlled precision perturbations demonstrate distinct resolution limits. A separate zero-shift pencil now has a certified first-1000 analytic root prefix and an exact indexed error decomposition. Uniform channel bounds and a closed law in the input parameters remain open.

1 Introduction

The Connes–Consani–Moscovici (CCM) construction [1], building on the Connes–van Suijlekom truncated Weil form [2], produces a family of finite self-adjoint operators $D_{\log}^{(\lambda,N)}$ whose spectra approximate the imaginary parts γ_k of the nontrivial Riemann zeta zeros to remarkable numerical accuracy: 55 digits at the headline configuration of [1], which we extended to 1019.0 measured matching digits in an independent reproduction [50]. The construction has two integer-like control parameters, a prime cutoff λ^2 (the primes $p \le \lambda^2$ enter the Weil form) and a basis size N (the matrix is $(2N{+}1) \times (2N{+}1)$), together with the caller-requested precision $P = P_{\text{req}}$ in decimal digits. The rate at which the eigenvalue converges in the basis size N is named the central open problem in [1]; Groskin [7] measured a Galerkin convergence exponent empirically but disclaims deriving it, and Śliwiński [15] proves an inverse-logarithmic lower bound on the mean spectral error. We previously tested two proposed rate laws against high-precision data and found them unsupported [51].

The author's starting point is a belief the Hilbert–Pólya conjecture already expresses: that apparent randomness in the zeros is structure not yet written down. The CCM construction gives that belief a computable form. What remained open is the rate at which the computation converges, and characterizing that rate explicitly is the subject of this paper.

Definition 1. $D(\lambda^2, N, P) := -\log_{10}\big(|\nu_1 - \gamma_1|/|\gamma_1|\big)$ *is the* matching-digit count*: the number of correct leading decimal digits of the first Riemann zero γ_1 reproduced by ν_1, the first ordinate produced by the CCM construction, computed from the caller-requested decimal target $P = P_{\text{req}}$ via MPFR/GMP. The reference γ_1 is computed to* 2500 *significant digits by rigorous interval arithmetic (Arb [52]), far exceeding the largest match reported here (the historical calibration reaches* ≈ 1070 *digits at* $\lambda^2 = 200$*; the retained-state continuation reaches 1175 digits), so the reported counts are genuine and not reference-limited; its leading digits agree with the standard tabulations of [53].*

Notation (each object is defined where cited; collected here for reference)	
$E = \|\nu_1 - \gamma_1\|/\|\gamma_1\|$	relative error of the first ordinate; $D = -\log_{10} E$ (Def. 1)
$P = P_{\text{req}}$	caller-requested decimal target used by the operational law
P_{work}	internal decimal working precision after the guard reserve
$\varepsilon_{\text{modes}},\ \varepsilon_{\text{primes}}$	intrinsic error channels, eq. (7)
D_{Wprec}, D_{nModes}, D_{Primes}	per-source digit budgets (Sections 4–6)
$\varepsilon^{\infty}(\lambda)$	smallest eigenvalue of the continuum form at cutoff λ
$D_{\text{eig}} = -\log_{10}\varepsilon^{\infty}$	eigenvalue floor (Section 6)
$D_{\text{edge}}^{\text{fit}} = D_{\text{Primes}} \cdot g$	empirical total intrinsic edge (Section 3.1)

The same number is read as a *ceiling* or a *floor* depending on the direction of view. D_{Primes} is a ceiling on the achievable digit count and, equally, the depth of the accuracy floor at the cutoff; both usages refer to the one object.

Our central result, the *Andrews CCM Convergence Law*, has two equivalent forms. Its intrinsic (spectral) form is the two-channel decomposition

$$\boxed{E(N, \lambda^2) \le \varepsilon_{\text{modes}}(N, \lambda^2) + \varepsilon_{\text{primes}}(\lambda^2)} \tag{1}$$

with the channels defined in Section 3, (7). Their signed displacements add exactly before absolute values are taken; Prop. 1 records the unconditional one-sided bound and the quantitative condition needed to exclude cancellation. The two channels are modeled as the finite-N

transition and extreme tail of a proposed common prolate-type spectral edge (Section 3.1); convergence $\nu_1 \to \gamma_1$ is exactly $E \to 0$. Its finite-precision measurement is the matching-digit count

$$\boxed{D(\lambda^2, N, P) = \min\big(D_{\mathrm{Wprec}}(P),\, D_{\mathrm{nModes}}(N, \lambda^2),\, D_{\mathrm{Primes}}(\lambda^2)\big) + O(1)} \tag{2}$$

under the noncancellation condition of Prop. 1, and with an unconditional lower bound in its absence. Here D_{Wprec} is the instrument resolution alone. Since D is the relative error on a logarithmic scale, the law is a quantitative convergence statement: $D \to \infty$ is precisely $\nu_1 \to \gamma_1$, and the three budgets are the independent mechanisms that drive (or stall) that convergence (Prop. 8). Six points distinguish it from prior empirical fits:

1. **The minimum is forced, not fitted** (Section 3). In digit space D is a negative logarithm of an error bounded by a sum of operationally independent channels, so the largest binds and D is at least the minimum of the three per-source budgets, unconditionally, and equals it up to an $O(1)$ term absent cancellation (Prop. 1). This is the structural reason earlier single-formula digit fits failed. The precision budget is measurement, not mathematics: removing it leaves the single intrinsic edge (1), whose two channels (modes, primes) are the only mathematical content.

2. **Each budget is analyzed from the construction**, not guessed. The precision budget follows from elementary round-off analysis; the measured prime ceiling is compared with the fourth-index prolate asymptotic, with the CCM-to-prolate transfer stated explicitly as open; and the mode budget is related to the analyticity of the Weil-form eigenvector.

3. **Every claim carries an explicit rigor level** (Table 8), separating the rigorously proved parts from those derived and validated against high-precision computation. We do not label the whole law a theorem; we state precisely which pieces are.

4. **It separates fixed-cutoff convergence from the growing-cutoff rate** (Prop. 8). Under explicit simple/isolated/even hypotheses, the finite zero branch converges to a continuum-transform zero at fixed λ^2; if that limit differs from γ_1, increasing N and P cannot make that branch converge to γ_1. The proposed growing-cutoff rate is a separate, explicitly conditional statement.

5. **It identifies the finite spectral reach of the truncation** (Section 2). The retained scaling lattice has the exact outer frequency $T_{\max} = 2\pi N / \ln \lambda^2$. Combining this edge with the Riemann–von Mangoldt law gives a smooth estimate $K^{(0)}_{\max}$ for the number of zeta ordinates geometrically contained in the finite band. This is a capacity estimate, not a guarantee of accurate or correctly ordered recovery.

6. **No prior unified law, to our knowledge.** Earlier attempts are single closed-form fits of the digit count, and they fail: we tested two proposed rate laws against high-precision data and found them unsupported [51]. Groskin [7] likewise reports that no single smooth-convergence law fits his data and declines to assert a rate, and Śliwiński [15] establishes only a lower bound on the mean error. The unified characterization given here, the intrinsic two-channel edge (1) and its finite-precision measurement (2), is, to the best of the author's knowledge, the first working closed-form convergence law for the construction. Its unity is the point: the data and classical asymptotics support a common spectral-edge hypothesis (Section 7) for the mode ramp and prime ceiling, which offers a mechanism for why a single expression can hold where earlier per-fits could not.

The empirical claim has three distinct levels. First, the minimum-budget structure is the exact bookkeeping skeleton. Second, the stretched-tanh family describes the observed mode ramps. Third, the particular closed map $q_\gamma(\lambda^2) = 2/(2\gamma + \ln\ln\lambda^2)$ together with $c = 4/\pi$ is

a fixed operational submodel. The first level is rigorous in its one-sided form; the transition family remains an empirical description rather than a safe accuracy bound. The third is useful on the calibrated domain but is now known not to extrapolate unchanged to $\lambda^2 = 1000$. Stating those levels separately is essential: a measured failure of the large-cutoff parameter map is a constraint on the formula, not a retraction of the three-budget law or of the stretched-transition family.

The empirical total-edge ramp (Section 5) motivates a mode-error model whose exponent is connected conditionally to a coefficient-decay law for the eigenvector (Gevrey where $q \le 1$). Proving that regularity and its transfer to the root is an explicit research program, not an end-to-end derivation. A data-coordinate comparison explains how an eigenvalue-space local exponent of the kind measured by Groskin [7] can drift across a stretched-exponential window; equality of the two asymptotic observables remains conditional. Section 7 separates the rigorous Riemann–von Mangoldt counting backbone from the stronger conjecture that the mode ramp and prime ceiling are two scales of one CCM spectral edge.

2 The construction and the object D

We recall the construction of [1, 2]. Let $\lambda > 1$ and $L = 2\ln\lambda = \ln\lambda^2$. On $\mathcal{H}_\lambda = L^2([\lambda^{-1}, \lambda], du/u)$ the functions $V_n(u) = L^{-1/2} e^{2\pi i n \ln(\lambda u)/L}$ form an orthonormal basis. The Weil quadratic form QW_λ, restricted to $\mathrm{span}\{V_n : |n| \le N\}$, is a real symmetric $(2N{+}1) \times (2N{+}1)$ matrix whose archimedean part carries the local factor at infinity and whose arithmetic part is a finite sum over prime powers $p^j \le \lambda^2$. Reflection symmetry splits the matrix exactly into even and odd sectors. When the lowest even eigenvalue is simple and lies below the lowest odd eigenvalue, the global ground state ξ is even; this is the critical finite-section hypothesis in the CCM construction, not an identity valid without qualification. The positive zeros of the associated Fourier–Mellin function $\widehat{\xi}$ then approximate the ordinates γ_k [1].

The prime-power cutoff is likewise structural rather than a numerical quadrature convention. Compact logarithmic support exhausts the arithmetic translations beyond $p^j = \lambda^2$. As $u = \log\lambda^2$ varies, Groskin's finite-path analysis proves that the matrix is continuous and piecewise real analytic, with rank-one first-derivative jumps precisely at $u = \log(p^j)$ [9]. Thus the arithmetic cutoff, basis cutoff N, working precision, and any archimedean quadrature cutoff are four distinct controls. Groskin's finite Guinand–Weil dictionary gives an exact coefficient-vector-to-test-function translation and an explicit archimedean tail budget [8]; it therefore supplies an independent audit of finite-cutoff matrix assembly, without by itself proving convergence of the selected CCM zero.

Finite spectral reach and zero capacity. The retained scaling frequencies are $2\pi n/L$, $|n| \le N$, so their exact outer edge is

$$\boxed{T_{\max}(N, \lambda^2) = \frac{2\pi N}{L} = \frac{2\pi N}{\ln\lambda^2} = \frac{N\pi}{\ln\lambda}.} \tag{3}$$

This is a structural identity, not an empirical fit: it is the largest unperturbed scaling frequency retained by E_N. It is already implicit in the CCM definition $|t| \le N\pi/\ln\lambda$ and in the external carrier spectrum of [1, Sec. 5.6]; Groskin also writes the same band edge as ρN, with $\rho = 2\pi/\ln\lambda^2$, in a different archimedean-tail problem [8]. We call it $T_{\max}$ only to make its role as the finite truncation's retained Fourier boundary explicit. It is not an upper bound on the heights of accurate movable roots; Section 8.23 gives a measured example beyond that band and the separate polynomial degree obstruction.

Let $N_\zeta(T)$ count positive nontrivial zeta zeros through height T, with multiplicity. The corresponding geometric zero capacity is

$$K_{\max}^{\mathrm{geom}}(N, \lambda^2) := N_\zeta(T_{\max}). \tag{4}$$

The Riemann–von Mangoldt formula [14] gives

$$N_\zeta(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8} + S(T) + O(T^{-1}),$$

where $S(T) = \pi^{-1} \arg \zeta(\frac{1}{2} + iT)$ in the standard convention, and substitution of (3) therefore yields the smooth capacity estimate

$$\boxed{K_{\max}^{(0)}(N, \lambda^2) = \frac{N}{\ln \lambda^2} \left[\log\left(\frac{N}{\ln \lambda^2} \right) - 1 \right] + \frac{7}{8}.} \tag{5}$$

Equivalently, geometric inclusion of the kth ordinate requires $\gamma_k \le T_{\max}$, or

$$N \ge \frac{\gamma_k \ln \lambda^2}{2\pi}. \tag{6}$$

After a scan of the existing CCM/CvS and adjacent literature, no prior work deriving $K_{\max}^{(0)}$ was found; to the author's knowledge, this particular formula and its zero-capacity interpretation may be novel. The qualification is important: $T_{\max}$ and $K_{\max}$ describe geometric availability in the finite band. They do not prove that every included zero is recovered, positive, correctly ordered, or accurate; those properties additionally depend on the quality of the finite eigenstate and are not claimed here.

Two facts about $\widehat{\xi}$ are used below: $\widehat{\xi}$ is entire, and (in the critical case) all its zeros are real and coincide with the spectrum. Both are proven *self-contained* in [1]: the closed form of the transform is [1, Prop. 5.9] (rederived from scratch in Appendix A), and entirety/reality is [1, Thm. 5.10], whose proof chain (Lemmas 5.2 and 5.4 there: the matrix structure $\tau_{ij} = (b_i - b_j)/(i - j)$, an elementary commutator identity, and self-adjointness in the induced inner product) is complete within [1]. The Carathéodory–Fejér–type reality mechanism *originates* in the Connes–van Suijlekom paper [2], which we cite as the historical source; no statement used in this paper depends on it (Appendix A). An independent classical foundation for the truncated Weil form predates both: Weil's positivity criterion, Yoshida's unconditional positivity theorem for test functions of sufficiently small support [16], Bombieri's study of the truncated quadratic form and the eigenvalues of its finite sections [17], and Suzuki's screw-function framework [20, 21, 5], which unifies all of these by means of continuous functions.

For the configurations used in the reported above-floor analysis, the selected eigenvector ξ is even and the smallest eigenvalue is simple, with the even-symmetry verified numerically across 38 configurations spanning $\lambda^2 = 13$–1000 in [50].

The object of study, $D(\lambda^2, N, P)$, is a digit count: $D = -\log_{10}(\text{relative error})$. The relative error itself, $E = |\nu_1 - \gamma_1|/|\gamma_1|$, is the intrinsic object of the law (1); D is its measurement in digits, and the arithmetic of that measurement is what makes the minimum structure of (2) inevitable.

3 The three-budget skeleton

The computed approximation ν_1 carries error from three sources: finite working precision (round-off), finite basis size (mode truncation), and the displacement of the fixed-cutoff continuum limit. We now *define* the channels, rather than posit their additivity. At a cutoff satisfying the

spectral hypotheses of Section 8.2, write $\nu_1(N, \lambda^2)$ for the selected positive zero branch of the exact (infinite-precision) finite-N operator and $\nu_1(\infty, \lambda^2)$ for its $N \to \infty$ limit, and set

$$\varepsilon_{\text{modes}}(N, \lambda^2) := \frac{|\nu_1(N, \lambda^2) - \nu_1(\infty, \lambda^2)|}{|\gamma_1|}, \qquad \varepsilon_{\text{primes}}(\lambda^2) := \frac{|\nu_1(\infty, \lambda^2) - \gamma_1|}{|\gamma_1|}, \tag{7}$$

the finite-N displacement at fixed cutoff and the cutoff displacement at infinite basis size, with $\varepsilon_{\text{round}}(P)$ the round-off perturbation of the computed ν_1 relative to the exact finite-N value. The triangle inequality then gives, for the computed relative error,

$$E \;\le\; \varepsilon_{\text{round}}(P) + \varepsilon_{\text{modes}}(N, \lambda^2) + \varepsilon_{\text{primes}}(\lambda^2),$$

an inequality, not an identity: the signed displacements can in principle cancel. The channels are *independent* in a precise operational (not probabilistic or structural) sense: each answers to its own parameter (round-off to P, truncation to N, prime content to λ^2), so any one can be driven below the others at will. (Structurally they are coupled: $\varepsilon_{\text{modes}}$ depends on λ^2 through both the exponent q and the Shannon coordinate of Section 5, and below the precision floor round-off corrupts the eigenvector nonlinearly, Section 4.1; the operational sense is all Prop. 1 uses.)

Proposition 1 (The minimum is forced)**.** *Let $\varepsilon_1, \varepsilon_2, \varepsilon_3 \ge 0$ and let $E \ge 0$ satisfy $E \le \Sigma := \sum_i \varepsilon_i$. Then, unconditionally,*

$$D := -\log_{10} E \;\ge\; -\log_{10} \Sigma \;\ge\; \min_i\big(-\log_{10} \varepsilon_i\big) - \log_{10} 3,$$

since $\Sigma \le 3 \max_i \varepsilon_i$. If in addition no cancellation occurs — quantitatively, $E \ge \theta \max_i \varepsilon_i$ for some fixed $\theta \in (0, 1]$ — then also $D \le \min_i(-\log_{10} \varepsilon_i) + \log_{10}(1/\theta)$, whence $D = \min(D_{\text{Wprec}}, D_{\text{nModes}}, D_{\text{Primes}}) + O(1)$.

The lower bound, that the computed digit count is at least the binding budget minus $\log_{10} 3 < 0.48$ of a digit, is a rigorous consequence of the triangle inequality alone. The matching upper bound requires the explicit noncancellation condition $E \ge \theta \max_i \varepsilon_i$. Its additive constant is $\log_{10}(1/\theta)$; it is not bounded by $\log_{10} 3$ without an independent lower bound on θ. Cancellation is most plausible near crossover seams, where two channels have comparable magnitude, but the triangle inequality alone neither confines it to those seams nor excludes an isolated cancellation elsewhere. Empirically, no seam in the recorded starvation sweeps shows D exceeding the binding minimum by more than the fitted bounded offset. Accordingly, the equality form used below is a quantitatively testable noncancellation model, while the one-sided bound is unconditional. Its content is that the *binding mechanism* changes across the parameter space (round-off, mode truncation, then prime content), not that D is literally nonsmooth (the negative logarithm of a signed near-sum can be smooth and can also exhibit cancellation spikes).

This is why, in digit space, D is a minimum rather than a sum or product, and why global single-expression fits of the digit count fail in practice: a single elementary term cannot simultaneously hold a hard precision wall, a prime ceiling, and a λ^2-coupled mode ramp. In the intrinsic form the error is, by contrast, the two-channel decomposition (1); the digit-space minimum is what the negative logarithm of that near-sum becomes, the largest term binding. The junctions differ in character. The precision junction is a hard wall, since round-off is a clamp, so its corner is sharp. The mode→prime junction, by contrast, is observed as a wide, smooth saturation (the tanh of Section 5). The measured digit count approaches the fixed-cutoff limit as $N \to \infty$ along the selected branch. Neither a pointwise inequality $D_{\text{nModes}} \le D_{\text{Primes}}$ nor monotone root accuracy is asserted: finite-N displacement can partially cancel the cutoff displacement.

The starvation protocol. Each budget is isolated experimentally by *starvation*: to read D_i directly, set the other two parameters large enough that their budgets exceed D_i, so the channel of interest dominates E outright and the cancellation caveat of Prop. 1 is vacuous. This protocol is the experimental mirror of the minimum structure. It is what makes the digit form falsifiable rather than descriptive, since each budget's functional form is measured in a regime where it alone binds. We use it throughout Sections 4–6.

Validity premise for empirical calibration. The three budgets begin after the finite CCM matrix has been assembled correctly. Matrix-assembly error is not a fourth intrinsic convergence mechanism and is not added to the caller-facing law. It is instead an admissibility condition on calibration and validation data. Deep spectra can be stable under increased arithmetic precision while still being wrong because of an inadequate archimedean cutoff or a precision-stable special function defect. Accordingly, a state enters the empirical archive only when its assembly is supported by an independent cutoff or closed-form check, structural symmetry tests, residual verification, and recorded reference-zero provenance. This follows the verified-computation principle that numerical work should check the hypotheses of an analytic bound rather than infer correctness from a small residual alone [38]; the finite Guinand–Weil tail analysis [8] and the independent quadrature and closed-form audits [39, 40] make the requirement concrete for the present matrix family.

3.1 The two forms of the law

Of the three budgets, only two (mode truncation and prime cutoff) are properties of the finite operator; round-off is a property of the computation. Separating them gives the two forms stated in Section 1.

Intrinsic (spectral) form. The mathematical content is the two-channel decomposition (1), with $\varepsilon_{\text{modes}} \sim \exp(-aN^q) \to 0$ as $N \to \infty$ (Section 5) and $\varepsilon_{\text{primes}} \to 0$ as $\lambda^2 \to \infty$ (Section 6). It makes no reference to precision. Under (H^*_λ), the mode displacement from the selected fixed-cutoff continuum zero vanishes as $N \to \infty$; the decay of the continuum-zero displacement as $\lambda^2 \to \infty$ is the separate growing-cutoff premise tested in Section 6. Vanishing of both channels is sufficient for $E \to 0$. The converse is not automatic for signed errors because finite-section and cutoff displacements may cancel. At a fixed cutoff satisfying (H^*_λ), however, the selected branch converges to $\nu_\infty(\lambda) \neq \gamma_1$, so increasing N alone cannot make that branch converge to γ_1 (Prop. 8).

Observed (digit) form. A computation with caller-requested target $P = P_{\text{req}}$, realized internally at P_{work}, resolves E only down to its round-off floor, so the measured matching-digit count is

$$D(\lambda^2, N, P) = \min\big(D_{\text{Wprec}}(P),\, -\log_{10} E(N, \lambda^2)\big) = \min\big(D_{\text{Wprec}}, D_{\text{nModes}}, D_{\text{Primes}}\big) + O(1), \quad (8)$$

where the second equality is conditional on a parameter-uniform noncancellation bound as in Prop. 1. Here $D_{\text{Wprec}}(P)$ is the instrument resolution: it is no property of ν_1, γ_1, or the operator, and as $P \to \infty$ it disappears, returning the intrinsic form in digit units. Thus the boxed law (2) is the finite-precision *measurement* of the intrinsic law (1).

Empirical total-edge parametrization. Dividing the measured intrinsic depth by the cutoff ceiling motivates the fitted quantity $D^{\text{fit}}_{\text{edge}} = D_{\text{Primes}} g$, with $g = \tanh((cN/(\lambda^2 \ln \lambda^2))^q)$. It saturates at the cutoff depth and is distinct from the genuine mode depth D_{nModes}, which tends to infinity as the fixed-cutoff mode error tends to zero. Only under an additional same-sign total-error model can one define

$$D^{\text{fit}}_{\text{mode}} = -\log_{10}\Big(10^{-D^{\text{fit}}_{\text{edge}}} - 10^{-D_{\text{Primes}}}\Big).$$

For $D^{\text{fit}}_{\text{edge}} = D_{\text{Primes}} \tanh x$ and $D_{\text{Primes}} > 0$, this extracted depth is $D_{\text{Primes}} + 2x/\ln 10 - \log_{10}(2D_{\text{Primes}} \ln 10) + o(1)$. The correction separates the observables; it does not repair a

failed numerical fit. A common underlying spectral edge remains the comparison hypothesis of Section 7.

4 The precision budget D_{Wprec}

The caller specifies a decimal target $P = P_{\mathrm{req}}$. The implementation converts that target to an internal precision P_{work} by allocating $\lceil P_{\mathrm{req}} \ln 10 / \ln 2 \rceil + 64$ bits; the guard reserve contributes approximately 19.27 decimal digits before accumulation and conditioning losses. For fixed (λ^2, N), a backward-stable computation has unit round-off proportional to $10^{-P_{\mathrm{work}}}$. Its forward root error is bounded by an effective condition factor κ_{eff} that includes matrix assembly, separation of the selected eigenpair, transform evaluation, and the inverse slope at the selected simple zero. If these factors and the algorithmic backward-error constant remain independent of the requested target, then

$$D_{\mathrm{Wprec}}(P_{\mathrm{req}}) = P_{\mathrm{req}} - \log_{10} \kappa_{\mathrm{eff}} + O(1) = P_{\mathrm{req}} + O(1), \tag{9}$$

where the guard reserve is absorbed in the implementation-dependent constant. This is a fixed-problem stability statement, not a claim that the condition factor is uniform as N or λ^2 grows. At $\lambda^2 = 13$ and $N = 80$, the validation sweep $P \in \{30, 45, 90, 180\}$ gives $D \in \{49.238, 55.654, 55.654, 55.654\}$, exhibiting the switch from the working-precision budget to the fixed-cutoff prime ceiling. Isolating this budget at $\lambda^2 = 50$ and $N = 400$, a sweep over $P \in \{50, 100, 150, 200\}$ gives an endpoint slope of 0.99970 and an ordinary least-squares slope of 1.00205 with $R^2 = 0.99994$, consistent with the predicted unit precision slope. The exact offset is not a property of the construction and we do not assign it physical meaning. *Status: the unit slope is T1 conditional on a fixed, backward-stable problem with a simple selected root; backward stability of the complete implementation is supported by the precision sweep and independent replay checks, not proved here.*

4.1 Symmetry, positivity, and the precision floor

A question the precision floor settles, and one easy to answer wrongly without it, is whether the even symmetry of the ground eigenvector, and the positivity of ε_N, are genuine features of the operator or merely of the regime above the floor. The CCM construction [1] posits a simple smallest eigenvalue with an *even* eigenvector (its Step 1) and enforces this with a forced-even projection. The reproduction of [50] tested that projection directly: across 38 configurations ($\lambda^2 = 13$–1000, $N = 10$–800, spanning both sides of the precision floor) with the projection *disabled*, the natural smallest eigenvector converged to the same even state produced by the projection, with numerically equivalent eigenvalues, at every tested configuration above the floor. The forced-even projection is thus empirically a no-op over that tested range: even symmetry is exhibited by the unprojected finite operator there, not manufactured by the projection.

The qualifier "above the floor" is essential, and it is precisely where the budget of this section bites. When the target eigenvalue lies below D_{Wprec}, smaller than the round-off that the working precision can resolve, the computed eigenvector is dominated by numerical noise: it can present spurious mixed symmetry, and the smallest computed eigenvalue can carry a spurious sign. These are artifacts of precision starvation, not properties of the operator; raising P past the floor dissolves them, and the natural eigenvector is again even, into the deep near-zero spectrum. (This is also the regime in which the round-off channel of Section 3 stops acting as an additive perturbation of ν_1: below the floor it corrupts the eigenvector nonlinearly, so the decomposition (7), like the symmetry and positivity statements of this subsection, is an above-floor statement.)

This even, positive ground state was documented in our reproduction [50]. In subsequent correspondence, and in a revision of his own paper [7], Groskin independently corroborated

it by a different route: the sign changes his separate quadrature implementation had shown among the deepest eigenvalues at a finite archimedean cutoff T were identified as a finite-T artifact that vanishes as T is increased, consistent with the naturally even, positive ground state of [50]. Working precision P and archimedean cutoff T are therefore independent controls: a deep-spectrum value is trustworthy only when it is stable across two settings of *each*, not when it merely holds steady in one. Across the tested and independently corroborated configurations, the even, positive ground state persists into the deep near-zero spectrum, while observed breakdowns disappear when the relevant precision or cutoff is raised. A uniform statement for the full growing-cutoff path remains the sector-gap problem of Section 10.

5 The mode budget D_{nModes}

The genuine mode channel is the displacement from the selected fixed-cutoff limiting root. Its quantitative transfer is given in Section 8.22; a uniform parameter law remains open. The historical saturation fit instead describes the total intrinsic edge. Write $g = D^{\mathrm{fit}}_{\mathrm{edge}}/D_{\mathrm{Primes}} \in [0,1]$. On the calibrated closed-submodel domain $7 \le \lambda^2 \le 100$ we use

$$\begin{aligned} D^{\mathrm{fit}}_{\mathrm{edge}}(N,\lambda^2) &= D_{\mathrm{Primes}}(\lambda^2)\,\tanh\Big(\big(cN/(\lambda^2\ln\lambda^2)\big)^{q_\gamma(\lambda^2)}\Big), \\ q_\gamma(\lambda^2) &= \frac{2}{2\gamma+\ln\ln\lambda^2}, \qquad c=\frac{4}{\pi}, \qquad 7\le\lambda^2\le 100. \end{aligned} \tag{10}$$

The stretched-tanh family and the coordinate $\lambda^2\ln\lambda^2$ are tested more broadly; the statement above is the fixed caller-facing parameter map, not a claim that q_γ and c persist unchanged at arbitrary cutoff. The literal mode depth diverges when the fixed-cutoff mode error tends to zero; the fitted edge above instead saturates. These observables are not identified. We build the model in four steps: the stretched-exponential meaning of the ramp (5.1), the candidate archimedean rate (5.2), the prime correction giving q (5.5), and the Shannon coordinate (5.6).

5.1 The ramp is a stretched exponential

Proposition 2 (Stretched-exponential reparametrization)**.** *Suppose the genuine mode-channel depth has the asymptotic form*

$$D_{\mathrm{nModes}}(N,\lambda^2) = \frac{a}{\ln 10}N^q + r(N), \qquad r(N)=o(N^q),$$

for some $a=a(\lambda^2)>0$ and exponent $q=q(\lambda^2)$. Then:

(i) (Stretched-exponential.) *The finite-section root error obeys*

$$\varepsilon_{\mathrm{modes}}(N,\lambda^2) = \exp\big(-aN^q + o(N^q)\big).$$

(ii) (Local power-law coordinate.) *If r is differentiable as a function of $\ln N$ and $dr/d\ln N = o(N^q)$, then the logarithmic local power-law exponent satisfies*

$$2s_{\mathrm{loc}}(N) := -\frac{d\ln\varepsilon_{\mathrm{modes}}}{d\ln N} = a\,q\,N^q + o(N^q).$$

Proof. Part (i) follows from $\ln\varepsilon_{\mathrm{modes}} = -(\ln 10)D_{\mathrm{nModes}} = -aN^q + o(N^q)$. Differentiating this identity with respect to $\ln N$ and using the stated derivative control on r gives part (ii). □

The proposition is an exact change of coordinates for the stipulated mode error; it does not identify the continuum eigenvector's regularity. The Gevrey class G^μ, $\mu \ge 1$, is characterized by a Fourier upper bound $Ce^{-a|j|^{1/\mu}}$ for suitable positive C, a, with strip analyticity at $\mu = 1$

[30, 41]. However, Proposition 6 below excludes a global bound with $q \geq 1$ for a nonzero CCM eigenstate. Thus the $q > 1$ fit at the smallest cutoff is a finite-window statement, not an ultra-analytic continuum regularity claim. Even a global $0 < q < 1$ bound would force endpoint flatness; whether the actual eigenstate has that regularity remains open.

The expanded eigenstate archive supplies a direct stress test of this interpretation. Applying one fixed fractional-window fit to the stored coefficient envelopes does not recover a cutoff-only exponent: at $\lambda^2 = 250$, for example, the effective exponent falls from about 1.72 at $N = 60$ to about 0.99 at $N = 1500$, and it also moves when the fitted part of the finite vector is changed. This does not refute a sharp law for the continuum eigenvector, because the available vectors are finite sections and their edges move with N. It does rule out treating the exponent fitted from the root-error ramp as an already measured coefficient exponent. A complete explanation must establish a continuum tail law compatible with the endpoint obstruction and quantify its finite-window crossover. The fixed-cutoff transfer of a given tail is supplied below; finite coefficient plots alone do not identify that tail.

The force of part (i) is that it converts the open problem of *measuring* a convergence rate into the analytic question of *explaining* the observed exponent: the single exponent q fixes the entire pre-saturation curve. It also explains why a fixed power-law fit drifts across an actually stretched-exponential window and resists extrapolation (Section 9).

Part (ii) gives a direct data-level reconciliation with the local exponent $s(c)$ measured by Groskin [7], provided the two experiments are sampling the same asymptotic error channel; this comparison is taken up in Section 9. Separately, for a simple isolated eigenvalue the Rayleigh-quotient error is second order in the phase-aligned eigenvector error, $|\varepsilon - \varepsilon_N| = O(\|\xi - \xi_N\|^2)$ [37, 42, 43, 44]. That result concerns the Rayleigh eigenvalue. It does not by itself square the displacement of a zero of $\widehat{\xi}_N$; the latter requires the quantitative tail-to-transform estimates and the simple-zero perturbation mechanism stated below. *Status: the reparametrization and local-slope identities are T1; their eigenvector-regularity and cross-observable interpretations remain conditional.*

5.2 The candidate archimedean rate $\delta = \pi/4$

The exact archimedean input is the completed real Gamma factor $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$. Its logarithmic derivative $\psi(\frac{1}{4} + \frac{iz}{2}) \sim \log(z/2)$ supplies the large-frequency $\log|z|$ symbol identified by Suzuki [5], while Stirling gives the real-axis envelope $|\Gamma(\frac{1}{4} + \frac{iz}{2})| \asymp |z|^{-1/4} \exp(-\pi|z|/4)$. Thus the exact archimedean envelope rate is

$$\delta = \pi/4. \tag{11}$$

The same value follows differentially. On the critical line the archimedean log-derivative has the large-z expansion

$$\psi\big(\tfrac{1}{4} + \tfrac{iz}{2}\big) = \log\frac{z}{2} + i\,\frac{\pi}{2} + o(1), \qquad z \to +\infty, \tag{12}$$

since the argument $\frac{1}{4} + \frac{iz}{2}$ runs in the imaginary (critical-line) direction, so $\arg\big(\frac{1}{4} + \frac{iz}{2}\big) \to \pi/2$. Its real and imaginary parts carry complementary information:

- the *real* part $\operatorname{Re}\psi(\frac{1}{4} + \frac{iz}{2}) \to \log(z/2)$ is the large-frequency symbol, Suzuki's $\log|z|$ multiplier [5];
- the *imaginary* part $\operatorname{Im}\psi(\frac{1}{4} + \frac{iz}{2}) \to \frac{\pi}{2}$ sets the envelope, through

$$\frac{d}{dz}\log\big|\Gamma_{\mathbb{R}}(\tfrac{1}{2} + iz)\big| = -\tfrac{1}{2}\operatorname{Im}\psi\big(\tfrac{1}{4} + \tfrac{iz}{2}\big) \longrightarrow -\tfrac{1}{2}\cdot\tfrac{\pi}{2} = -\tfrac{\pi}{4}.$$

Both statements arise from the one Gamma factor. The elementary identity

$$\delta = \frac{\pi}{4} = \frac{1}{2} \cdot \frac{\pi}{2} = \tfrac{1}{2} \arg(i), \tag{13}$$

records the half-angle generated by the critical-line direction. It is an envelope identity, not a proof of an analyticity strip for the finite CCM ground state.

Two transfer steps remain open. First, the finite ground-state transform, which is of exponential type $L/2$, is not literally the completed xi function; one must prove that its phase-aligned coefficient tail inherits the $\Gamma_{\mathbb{R}}$ envelope in the relevant growing window. Second, one must prove how that tail controls the selected transform-zero displacement, including whether the observed exponent is doubled. The measured finite coefficient rate (≈ 0.71 at $\lambda^2 = 100$) is therefore a separate, prime-modulated quantity rather than an instance of a proved $\pi/4$ law. The exact Gamma calculation identifies a compelling candidate value for the open realization step; it does not close that step. Conditional on global RH, Section 8.3 identifies the finite eigenvalue-to-root transfer as a first-order divided-difference identity. That result settles the form of the root-transfer mechanism but not the fixed-cutoff convergence rate, which still requires the Galerkin estimates and limiting norming control stated below.

Lemma 1 (Lattice sampling of the coefficients)**.** *The Fourier–Mellin transform of the N-mode eigenvector has the closed form $\widehat{\xi}_N(z) = 2L^{-1/2}\sin(zL/2)\sum_{|j|\le N}\xi_j/(z-2\pi j/L)$ ([1, Prop. 5.9]; rederived from scratch in Appendix A). On the frequency lattice $z_k = 2\pi k/L$,*

$$\widehat{\xi}_N(2\pi k/L) = (-1)^k\sqrt{L}\,\xi_k \quad (|k|\le N), \qquad \widehat{\xi}_N(2\pi k/L) = 0 \quad (|k|>N),$$

so $|\xi_k| = L^{-1/2}\left|\widehat{\xi}_N(2\pi k/L)\right|$, with $L = \ln\lambda^2$ the period of the basis $\{V_n\}$.

Proof. At $z_k = 2\pi k/L$ the factor $\sin(zL/2) = (-1)^k\sin\big((z-z_k)L/2\big)$ has a simple zero, so every $j \neq k$ term vanishes and the $j = k$ pole is removable: $\lim_{z\to z_k} 2L^{-1/2}(-1)^k\sin\big((z-z_k)L/2\big)/(z-z_k) = (-1)^k\sqrt{L}$. For $|k| > N$ there is no $j = k$ term and the common factor $\sin(z_kL/2) = 0$. □

Lemma 1 rigorously fixes the lattice spacing $2\pi/L$. If the coefficient tail inherits the candidate Gamma envelope, $\exp(-\delta(2\pi/L)N)$, and if the selected-zero displacement has the observed doubled exponent, then the leading digit slope is

$$\frac{\partial D_{\mathrm{nModes}}}{\partial N} = \frac{2\delta\,(2\pi/L)}{\ln 10} = \frac{\pi^2}{\ln 10\ \ln\lambda^2} \approx \frac{4.286}{\ln\lambda^2}, \tag{14}$$

a parameter-free conditional prediction. At $\lambda^2 = 7$, where few primes enter and $q \approx 1$, (14) gives 2.203 against a measured small-N slope of 2.264, agreement to 3% with no free parameters. At larger λ^2 the measured early slope runs above (14) exactly as the $q < 1$ concavity (below) requires.

What *is* rigorous is the final transfer from a locally uniform transform estimate to displacement of a simple zero. We record it next. The open content before that step includes the coefficient law, quantitative Galerkin transfer, and the observed doubling.

Lemma 2 (Simple-zero displacement under uniform convergence)**.** *Let $a \in \mathbb{C}$, and let F be holomorphic on a neighborhood of the closed disc $\overline{D}(a,r)$, with a simple zero at a and no other zero in the disc. Shrink r if necessary so that $|F(z)| \ge \frac{1}{2}|F'(a)|\,|z-a|$ throughout the disc; this follows by applying continuity to the nonvanishing holomorphic quotient $F(z)/(z-a)$, extended at a by $F'(a)$. Let G be holomorphic on the disc and set $\eta := \sup_{\overline{D}(a,r)}|G-F|$. If $\rho := 2\eta/|F'(a)| < r$, then G has exactly one zero b in $D(a,r)$, and*

$$|b-a| \ \le\ \frac{2\,\eta}{|F'(a)|}.$$

Proof. If $\eta = 0$, the conclusion is immediate. Otherwise, on $|z-a| = r$ the hypothesis $\rho < r$ gives $|F(z)| > \eta \geq |G(z) - F(z)|$, so Rouché's theorem gives G exactly one zero b in $D(a,r)$. For every s with $\rho < s < r$, the same strict inequality holds on $|z-a| = s$. Thus $b \in D(a,s)$ for every such s, and letting $s \downarrow \rho$ gives $|b-a| \leq \rho$. □

At fixed cutoff, assume $\widehat{\xi}_\infty$ has a simple isolated zero $\nu_\infty(\lambda)$ as in (H^*_λ). Applied with $a = \nu_\infty(\lambda)$, $F = \widehat{\xi}_\infty$, and $G = \widehat{\xi}_N$, the lemma converts uniform convergence near the continuum zero into finite-section zero displacement with the explicit constant $2/|\widehat{\xi}'_\infty(\nu_\infty)|$. It does not, by itself, compare $\nu_\infty(\lambda)$ with the zeta ordinate γ_1.

5.3 An exact logarithmic diagonal and the late-resolution error

The following decomposition identifies the principal operator in the actual CCM basis, including the correction from zero extension at the endpoints. It requires neither RH nor positivity of the Weil form. All operators in this subsection use exact assembly; numerical assembly errors are separate.

Proposition 3 (Bounded logarithmic remainder)**.** *Set $c = \lambda^2 > 1$, $L = \log c$, and let A_c be the self-adjoint operator of the full Weil form, expressed in the basis V_j. Then $A_c = D_L + K_c$, where*

$$D_L V_j = d_j V_j, \qquad d_j = \tfrac{1}{2}\log\big(\tfrac{1}{4} + (2\pi j/L)^2\big) - \log(2\pi), \qquad \mathrm{Dom}(A_c) = \mathrm{Dom}(D_L),$$

and K_c is bounded self-adjoint with $\|K_c\| \leq \mathcal{C}(c)$, where

$$\begin{aligned}\mathcal{C}(c) = \pi - \log(1 - c^{-1/2}) + \gamma + \tfrac{\pi}{2} + \log 2 \\ + \log c + \sqrt{c} - c^{-1/2} + 2\sum_{p^k \leq c} \frac{\log p}{p^{k/2}}.\end{aligned} \tag{15}$$

The same decomposition restricts to either parity sector.

Proof. Transport V_j to $U_j(x) = L^{-1/2} e^{2\pi i j x/L}$ on $(0, L)$ by [1, Prop. 3.2]. For $a > 0$, the multiplier $m_a(t) = \frac{1}{2}\log(a^2 + t^2)$ satisfies

$$m_a(t) = \log a + \int_0^\infty e^{-as} \frac{1 - \cos(ts)}{s}\, ds.$$

Subtract its periodic realization from its whole-line realization applied to zero-extended functions. The local terms cancel and the difference is the integral operator

$$H_{a,L}(x,y) = \frac{1}{2} \sum_{k \in \mathbb{Z}\setminus\{0\}} \frac{e^{-a|x-y+kL|}}{|x-y+kL|}.$$

This identity first holds as a quadratic-form identity on trigonometric polynomials, by truncating the singular integral and taking its limit. Each of the two adjacent images is bounded in norm by $\pi/2$: reflect one endpoint and apply weighted Schur to $1/(x+y)$ with weight $x^{-1/2}$, using $\int_0^\infty u^{-1/2}(1+u)^{-1}\, du = \pi$. The images $|k| = m+1$, $m \geq 1$, together have norm at most e^{-amL}/m. Hence $\|H_{a,L}\| \leq \pi - \log(1 - e^{-aL})$. Positive values of this kernel are not being used to assert positive semidefiniteness.

The archimedean multiplier is $h(t) = \mathrm{Re}\,\psi(1/4 + it/2) - \log\pi$ [1, eq. (3.19)]. Choose $a = 1/2$. With $z = 1/4 + it/2$ its remainder after subtracting $m_{1/2}(t) - \log(2\pi)$ is $\mathrm{Re}(\psi(z) - \log z)$. The digamma integral [54, eq. 5.9.13] gives

$$\psi(z) - \log z = -\int_0^\infty \left(\frac{1}{1 - e^{-s}} - \frac{1}{s}\right) e^{-zs}\, ds.$$

The parenthesis is positive, so the absolute value of the real remainder is at most $\log(1/4) - \psi(1/4) = \gamma + \pi/2 + \log 2$. Plancherel gives the same norm bound after compression to the interval. The pole kernel $2\cosh((x-y)/2)$ has eigenvalues $L \pm 2\sinh(L/2)$ and norm $L + 2\sinh(L/2)$. Each active prime power is $-(\log p)p^{-k/2}$ times a compressed translation and its adjoint, with norm at most $2(\log p)p^{-k/2}$. These estimates give (15).

Trigonometric polynomials are a form core for D_L and for the Weil form [1, Prop. 3.4]. A bounded perturbation preserves the form domain; equality on the core therefore identifies the closed forms and their associated operators. Reflection about $L/2$ commutes with every component. □

Proposition 4 (Quantitative tail stability at fixed cutoff)**.** *Suppose A_c has a simple normalized ground state u, eigenvalue ε_∞, and spectral gap $g > 0$. Let u_N be a normalized ground state of $P_N A_c P_N$, phase aligned with u, and put $q_N = \|(I - P_N)u\|_2^2$. For all sufficiently large N,*

$$\|u_N - u\|_2^2 \le 2\left(1 + \frac{\mathcal{C}(c)^2}{g^2}\right) q_N. \tag{16}$$

For centered logarithmic coordinates and every complex z,

$$|\widehat{u}_N(z) - \widehat{u}(z)| \le \sqrt{2L\,(1 + \mathcal{C}(c)^2/g^2)\,q_N}\, e^{L|\operatorname{Im} z|/2}. \tag{17}$$

Proof. Write $p = P_N u$, $w = (I - P_N)u$. The projected eigenvalue equation is $(P_N A_c P_N - \varepsilon_\infty)p = -P_N K_c w$. By min–max, every retained eigenvalue other than the first is at least $\varepsilon_\infty + g$. Projecting this equation orthogonally to u_N therefore gives $\|(I - |u_N\rangle\langle u_N|)p\|_2 \le \mathcal{C}(c)\sqrt{q_N}/g$. Consequently $1 - |\langle u, u_N\rangle|^2 \le (1 + \mathcal{C}(c)^2/g^2)q_N$. Phase alignment and $2(1-t) \le 2(1-t^2)$ for $0 \le t \le 1$ prove (16). Cauchy–Schwarz in the Fourier integral proves (17). The ground Ritz value eventually lies below $\varepsilon_\infty + g$ by form-core convergence, making it simple. □

Proposition 5 (Late eigenvalue error as logarithmic tail energy)**.** *Under the preceding spectral hypotheses, let $E_N = \sum_{|j|>N} d_j |u_j|^2$, $v = P_N u/\sqrt{1 - q_N}$, $\delta_N = \langle v, A_c v\rangle - \varepsilon_\infty$, and $B = \mathcal{C}(c) + |\varepsilon_\infty|$. If $q_N < 1$ and $\delta_N < g$, then*

$$\frac{E_N - Bq_N}{1 - q_N} - \frac{\mathcal{C}(c)^2 q_N}{(1 - q_N)(g - \delta_N)} \le \varepsilon_N - \varepsilon_\infty \le \frac{E_N + Bq_N}{1 - q_N}. \tag{18}$$

If $E_N > 0$, $d_{N+1} > 0$, and $\delta_N \le g/2$, this implies

$$\left|\frac{\varepsilon_N - \varepsilon_\infty}{E_N} - 1\right| \le \frac{q_N}{1 - q_N} + \frac{B + 2\mathcal{C}(c)^2/g}{(1 - q_N)d_{N+1}}. \tag{19}$$

In particular $\varepsilon_N - \varepsilon_\infty \sim E_N$ at fixed c unless the eigenvector has finite support, in which case convergence terminates.

Proof. The exact projection identity is $\delta_N = \langle w, (A_c - \varepsilon_\infty)w\rangle/(1 - q_N)$; its difference from $E_N/(1-q_N)$ is bounded by $Bq_N/(1-q_N)$. Rayleigh–Ritz gives the upper bound. The retained residual of v is $-(I - |v\rangle\langle v|)P_N K_c w/\sqrt{1 - q_N}$, of squared norm at most $\mathcal{C}(c)^2 q_N/(1 - q_N)$. The retained second eigenvalue is at least $\varepsilon_\infty + g$, so the Temple residual inequality bounds $\langle v, A_c v\rangle - \varepsilon_N$ by that squared norm divided by $g - \delta_N$. This proves the lower bound. Finally $E_N \ge d_{N+1} q_N$ gives (19). Since $u \in \mathrm{Dom}(D_L)$, $E_N, q_N \to 0$, while $d_{N+1} \to \infty$. □

These results supply a quantitative fixed-cutoff Galerkin transfer, but their constants can grow rapidly as the spectral gap closes. They neither establish a coefficient asymptotic nor control the coupled c, N limit. For example, if one could prove $|u_j|^2 \sim A(c)^2/[j^2(\log|j|)^3]$ for an even state, then (19) would imply $\varepsilon_N - \varepsilon_\infty \sim 2A(c)^2/[N(\log N)^2]$. The coefficient premise and

its nonzero amplitude remain unproved; this is not a replacement fitted law for the large-cutoff ramp.

A second, high-value transfer runs from the coefficient tail to the transform. The finite-section eigenvector is not the truncation of the limit eigenvector, so the transform difference carries both a discarded-tail term and a retained Galerkin error. We preserve the complete route while making its quantitative Galerkin input explicit.

Lemma 3 (Tail-to-transform transfer at fixed cutoff)**.** *Fix λ^2, assume the continuum ground state is simple and isolated, and choose all eigenvectors with unit L^2 norm and aligned phases. Suppose its coefficients satisfy $|\xi_j^{(\infty)}| \le C\exp(-a|j|^q)$ for some $a, q > 0$. Then, on every fixed compact subset K of the complex plane,*

$$\sup_{z\in K} |\widehat{\xi}_N(z) - \widehat{\xi}_\infty(z)| \le \exp\bigl(-aN^q + o(N^q)\bigr).$$

Proof. An integral comparison gives $\sum_{|j|>N} |\xi_j^{(\infty)}|^2 \le \exp(-2aN^q + o(N^q))$. Apply Proposition 4; the interval, gap, compact set, and $\mathcal{C}(\lambda^2)$ are fixed. In particular, no avoidance of removable frequency-lattice poles is needed. □

For comparison with the earlier formulation, the quantitative Galerkin conditions

$$\begin{aligned} \|P_N A_c (I - P_N)\xi^{(\infty)}\|_2 &\le \mathcal{C}(c)\|(I - P_N)\xi^{(\infty)}\|_2, \\ \operatorname{dist}(\varepsilon^\infty, \sigma(P_N A_c P_N) \setminus \{\varepsilon_N\}) &\ge g(c), \\ \sup_{z\in K} |\widehat{T}[e](z)| &\le \sqrt{L}\exp\left(\tfrac{L}{2}\sup_{z\in K}|\operatorname{Im} z|\right)\|e\|_2 \end{aligned} \tag{QG$_\lambda$}$$

now follow from Proposition 3, min–max, and Cauchy–Schwarz. They are no longer additional fixed-cutoff assumptions. The second line applies for sufficiently large N, when the ground Ritz value is simple. Constants remain cutoff dependent, and boundary-value normalization requires its own nonvanishing condition if values rather than zeros of the transforms are compared.

Corollary 1. *Under the hypotheses of Lemma 3, if the limit transform has a simple isolated zero $\nu_1(\infty, \lambda^2)$, then the corresponding finite zero satisfies $|\nu_1(N, \lambda^2) - \nu_1(\infty, \lambda^2)| \le \exp(-aN^q + o(N^q))$.*

Proof. Apply Lemma 2 on a fixed isolating disc and then Lemma 3. Scaling an eigenvector does not change its transform zeros, so unit-L^2 normalization is sufficient for this conclusion. □

5.4 An endpoint obstruction to global analytic decay

The logarithmic endpoint term imposes a restriction on possible coefficient laws. It is stronger than the observation that finite-window fitted exponents drift, and does not require an eigenfunction to have one sign.

Proposition 6 (Smooth eigenfunctions must be flat at the boundary)**.** *Let $c > 1$. If an eigenfunction u of the exact full CCM operator has a C^∞ representative on $[0, L]$, $L = \log c$, then $u^{(m)}(0) = u^{(m)}(L) = 0$ for every $m \ge 0$. Consequently no nonzero eigenfunction is analytic on a neighborhood of $[0, L]$, and no nonzero eigenfunction has periodic Fourier coefficients bounded by $Ce^{-a|j|^q}$ for any $a > 0$, $q \ge 1$.*

Proof. For the zero-extension realization Q_a of $m_a(t) = \frac{1}{2}\log(a^2 + t^2)$, the symmetric translation formula gives

$$\begin{aligned} Q_a u(x) = (\log a)u(x) &+ \frac{1}{2}\int_0^L e^{-a|x-y|}\frac{u(x) - u(y)}{|x - y|}\,dy \\ &+ \frac{1}{2}u(x)\bigl(E_1(ax) + E_1(a(L - x))\bigr), \qquad 0 < x < L. \end{aligned}$$

The two exponential integrals are exactly the contributions from outside the interval. By the expansion of E_1 at zero [54, eq. 6.6.2], $E_1(ax) + \log x$ is analytic near zero. Split the integral at $y = x$; the divided difference extends smoothly through the diagonal, since

$$\frac{u(x) - u(y)}{x - y} = \int_0^1 u'(y + s(x - y))\, ds.$$

It follows that $Q_a u(x) = -\frac{1}{2}u(x)\log x + G_a(x)$ with G_a smooth up to the left endpoint whenever u is smooth on $[0, L]$.

For $a = 1/2$, the remaining archimedean kernel in Proposition 3 is explicitly

$$-b(2|x - y|)e^{-|x-y|/2}, \qquad b(s) = \frac{1}{1 - e^{-s}} - \frac{1}{s}.$$

This follows by Fourier inversion of the digamma integral used there. The singularity of b at zero is removable, with $b(s) = \frac{1}{2} + s/12 + O(s^3)$. Splitting at $x = y$ again shows that this remainder maps smooth functions to functions smooth at the endpoint. The pole kernel is analytic. In a sufficiently small one-sided neighborhood of zero, each of the finitely many prime translations is either zero or an interior translate of u. A translation of length L is zero on the open interval. Thus the full eigenvalue equation is

$$0 = (A_c - \varepsilon)u(x) = -\tfrac{1}{2}u(x)\log x + G(x), \qquad G \in C^\infty([0, \eta])$$

for some $\eta > 0$. If m is the first nonvanishing derivative of u at zero, the mth derivative of $u(x)\log x$ contains $u^{(m)}(0)\log x$, which is unbounded. This contradicts the smoothness of G. Hence u is flat at zero, and reflection gives flatness at L. An analytic function flat at an endpoint is identically zero on the connected interval. A Fourier bound with $q = 1$ gives a holomorphic strip extension, while $q > 1$ gives a holomorphic extension to every finite strip; both are therefore excluded. □

This does not rule out $0 < q < 1$: smooth nonanalytic functions can be flat at both endpoints. It also does not prove the nonzero logarithmic-boundary amplitude needed for the conditional $1/[N(\log N)^2]$ rate above.

Corollary 2 (No exact termination of the CCM Ritz sequence)**.** *For every finite N, the smallest exact Ritz value satisfies $\varepsilon_N > \varepsilon_\infty$.*

Proof. Equality in the variational principle would put a nonzero trigonometric polynomial in the continuum ground eigenspace. Such a polynomial is analytic, contradicting Proposition 6. □

The calibrated parameter $q_\gamma(c) = 2/(2\gamma + \log\log c)$ satisfies $q_\gamma(c) \geq 1$ up to $c_* = \exp(\exp(2 - 2\gamma)) = 10.27077047\ldots$, wherever its denominator is positive. In particular $q_\gamma(7) = 1.0988038\ldots$ and $q_\gamma(9) = 1.0300643\ldots$. Its proposed identification with a global continuum coefficient exponent is therefore false at those cutoffs. The finite-window root-error fit and its measured accuracy are separate: a finite band can decay much faster than the eventual endpoint tail.

Problem 1 (Boundary regime, coefficient decay, and zero displacement)**.** *For the exact continuum ground state at fixed $c = \lambda^2$:*

(i) Determine its endpoint regularity and sharp Fourier tail. A stretched-exponential upper bound could only have $0 < q_\infty(c) < 1$ and would force a smooth function flat at both endpoints. Neither that regime nor a nonzero logarithmic boundary amplitude is established.

(ii) Determine how the finite coefficient envelope and its observed approach toward the Gamma-factor scale $\pi/4$ connect to the actual tail, including the crossover and its dependence on c. The Gamma envelope alone does not identify an asymptotic coefficient class.

(iii) Given the actual coefficient law, establish the optimal displacement rate of a selected simple continuum transform zero and identify when, if ever, its exponent is doubled.

The earlier global conjecture identified the coefficient exponent with the root-ramp parameter $q_\gamma(c)$ at every calibrated cutoff. Its clause (i) is incompatible with Proposition 6 at $c = 7$ and 9; it is withdrawn as a global statement, rather than repaired by clamping the exponent. The measured finite-band rate near 0.71 at $c = 100$, compared with $\pi/4 \approx 0.785$, remains an empirical observation. The same is true of the rapid root-error improvement at $c = 7$. Neither identifies the infinite coefficient tail.

Proposition 4 supplies the fixed-cutoff Galerkin transfer of a given tail. Conditional on a stretched-exponential coefficient bound with admissible q, Lemma 3 and Corollary 1 give the single exponent at a simple limiting zero. The measured doubling still requires more. The conditional Weil divided-difference identity in Proposition 10 is first order in ε_N, but obtaining a rate for $\nu_k(N) - \nu_k(\infty)$ requires subtracting the limiting identity and controlling the corresponding norming and remote-zero terms.

5.5 The prime correction: $q(\lambda^2)$

The archimedean candidate gives the baseline $q = 1$. The arithmetic part of the localized Weil form is exactly a finite prime-power translation/cosine operator, with frequencies $k \log p$ and weights involving $(\log p)p^{-k/2}$ for $p^k \le \lambda^2$. It is *not* known that the CCM ground-state transform factorizes as a Gamma factor times a truncated Euler product. To formulate a testable fluctuation model without assuming such a factorization, isolate the first prime harmonics $S_1(z) = \sum_{p \le \lambda^2} p^{-1/2} \cos(z \ln p)$. Their long-interval mean square is

$$\langle S_1^2 \rangle = \tfrac{1}{2} \sum_{p \le \lambda^2} p^{-1} = \tfrac{1}{2}(\ln\ln \lambda^2 + M) + o(1)$$

by Mertens' theorem (M the prime Mertens constant). A separate classical identity, Mertens' product theorem, gives $\sum_{p \le \lambda^2} -\ln(1 - 1/p) = \gamma + \ln\ln \lambda^2 + o(1)$. The data show the fitted exponent decreasing as prime content grows. Motivated by these two scales, but not derived from them, model the reciprocal exponent as a baseline plus the prime fluctuation,

$$\frac{1}{q} = c_0 + c_1 \ln\ln \lambda^2, \qquad c_0 = \gamma \text{ (candidate baseline)}, \quad c_1 = \tfrac{1}{2} \text{ (candidate variance coefficient)}, \tag{20}$$

which gives $q(\lambda^2) = 2/(2\gamma + \ln\ln \lambda^2)$. In this model the numerator 2 is $1/c_1$, while Mertens supplies the $\ln\ln$ scale. Neither Mertens theorem supplies the missing operator map from prime fluctuations to a sharp coefficient-decay index. The fixed operational expression is used on the calibrated domain $7 \le \lambda^2 \le 100$. Its denominator need not even remain positive as $\lambda^2 \downarrow 1$, and the large-cutoff test of Section 5.7 shows that it also cannot be extrapolated unchanged as $\lambda^2 \to \infty$.

We test (20) against per-λ^2 fits of the full form (10) (Table 1). Restricting the quantitative regression to the six cutoffs whose complete ramp configurations are preserved by the reproduction suite, a linear regression $1/q = c_0 + c_1 \ln\ln \lambda^2$ over $\lambda^2 \in \{7, 13, 20, 25, 50, 100\}$ yields

$$c_0 = 0.545 \ (\gamma = 0.577), \qquad c_1 = 0.489 \ (\tfrac{1}{2} = 0.500), \qquad R^2 = 0.977.$$

Euler–Mascheroni γ appears as the intercept of a regression that does not assume it. The offset of the parameter-free curve is systematic, not centered scatter. Taking $c_0 = \gamma$ exactly, the idealized $q = 2/(2\gamma + \ln\ln \lambda^2)$ runs 5–7% *below* the per-λ^2 fits across the whole range (e.g. 1.10 vs 1.185 at $\lambda^2 = 7$; 0.75 vs 0.785 at $\lambda^2 = 100$), and the regression intercept 0.545 sits about 5.6% under $\gamma = 0.577$, always on the same side. We therefore report both curves side by side — the provenance-complete fitted $(c_0, c_1) = (0.545, 0.489)$ and the idealized $(\gamma, \frac{1}{2})$ —

Table 1: Per-λ^2 fits of (10) (full tanh form, mode-limited points). The onset constant c clusters near $4/\pi = 1.273$; the exponent q decreases through 1 near $\lambda^2 \approx 13$, marking a finite-window crossover. The 150 and 200 columns are retained as historical reference values, but their exact ramp configurations are not preserved; both are excluded from the quantitative regression.

λ^2	7	13	20	25	50	100	150†	200†
q	1.185	0.985	0.900	0.880	0.835	0.785	0.760	0.805
c	1.20	1.25	1.28	1.29	1.31	1.28	1.26	1.31

†Historical reference value; exact ramp provenance is pending and the value is not used in regression or formal validation. Its numerical consistency with the later large-cutoff effective range does not restore that missing provenance.

and the identification of the intercept with γ is a structural hypothesis the data supports to that one-signed accuracy, not a measurement centered on it. *Status: T3; the remaining lemma is a rigorous variance-to-exponent map is now a finite-window problem: explain how the prime phases and archimedean operator generate the observed root-ramp coordinate before the endpoint tail takes over. The fixed-cutoff global analytic interpretation is excluded by Proposition 6. The empirical identity* $1/q_\gamma = \gamma + \frac{1}{2}\log\log\lambda^2$ *is not thereby promoted to a continuum coefficient theorem.*

5.6 The Shannon coordinate

Lemma 4 (Conditional linear crossover scale)**.** *Assume the candidate leading prime ceiling and the candidate small-N linear slope of Sections 6 and 5.2. Their linear intersection occurs at*

$$N_{\text{lin}} = \frac{D_{\text{Primes}}}{\partial D/\partial N} = \frac{(4\pi/\ln 10)\lambda^2}{\pi^2/(\ln 10\,\ln\lambda^2)} = \frac{4}{\pi}\,\lambda^2\ln\lambda^2.$$

The λ^2 comes from the prime ceiling (Section 6); the $\ln\lambda^2$ from the inverse of the archimedean slope (14). The combination $\lambda^2\ln\lambda^2$ is therefore the natural coordinate within this two-input model. It is Shannon-like: the factor $L = \ln\lambda^2$ converts a physical-frequency scale to a number of lattice modes through the spacing $2\pi/L$. A literal identification with the Shannon number of a specific prolate operator requires the still-open CCM/prolate comparison; the classical time–band limiting results [34, 33, 32] motivate that interpretation but do not supply it automatically.

The crossover multiplier and the fitted tanh coefficient must not be conflated. Equation (10) places c *inside* the argument, so its unit-argument scale is $N_{\tanh} = \lambda^2\ln\lambda^2/c$. Thus the observation $c \approx 4/\pi$ does not follow from $N_{\text{lin}} = (4/\pi)\lambda^2\ln\lambda^2$; one occurrence is a fitted argument coefficient and the other is a crossover multiplier. Their relation also depends on the stretched exponent and the chosen saturation convention. Measured saturation occurs at $N/(\lambda^2\ln\lambda^2) \approx 2.0$–$2.3$, while the fitted onset coefficient clusters near $c \approx 1.27$ (Table 1). A second, independent motivation comes from the geometric capacity of Section 2. Ignoring the 7/8 and oscillatory terms and setting $K^{(0)}_{\max}(N,\lambda^2) \approx N$ gives

$$\log\left(\frac{N}{\ln\lambda^2}\right) - 1 \approx \ln\lambda^2, \qquad \text{hence} \qquad N \approx e\,\lambda^2\ln\lambda^2.$$

This is a constraint-counting heuristic: it says that the transition should occur when the number of geometrically available ordinate constraints is of the same order as the basis dimension. It derives the coordinate and an $O(1)$ multiplier from the paper's own capacity formula, but it does not prove independence of the constraints or identify the empirical tanh convention. Deriving c, or even proving its exact value is $4/\pi$, remains open. *Status: the algebra of the crossover lemma*

is T1 conditional on its T3 rate inputs; the capacity crossover is heuristic; the coordinate and the fitted onset constant remain T3.

5.7 Operational validation and large-cutoff constraint

The accuracy of the caller-facing formula must be assessed with its constants fixed. Per-cutoff refits test the flexibility of the stretched-tanh family; they do not test the formula a caller actually uses. We therefore report the fixed submodel

$$D_1^{\mathrm{pred}}(\lambda^2, N) = D_{\mathrm{Primes}}(\lambda^2) \tanh\left[\left(\frac{(4/\pi)N}{\lambda^2 \ln \lambda^2}\right)^{q_\gamma(\lambda^2)}\right], \quad D_{\mathrm{Primes}} = \frac{4\pi}{\ln 10}\lambda^2 - 5\log_{10}\lambda^2 - 9.5.$$

On an independent exact-assembly reimplementation, the fixed formula gives the following errors; residual means formula minus measurement.

Table 2: Cross-implementation validation of the fixed published formula on the calibrated domain. The independent implementation does not refit c, q_γ, or the cutoff ceiling at any displayed cutoff; all values use the published fixed constants.

λ^2	points	measured span	mean residual	RMS	maximum
7	23	15.14	+0.33	0.47	0.87
13	15	43.29	+0.12	0.66	1.07
20	16	74.36	−0.61	1.18	2.12

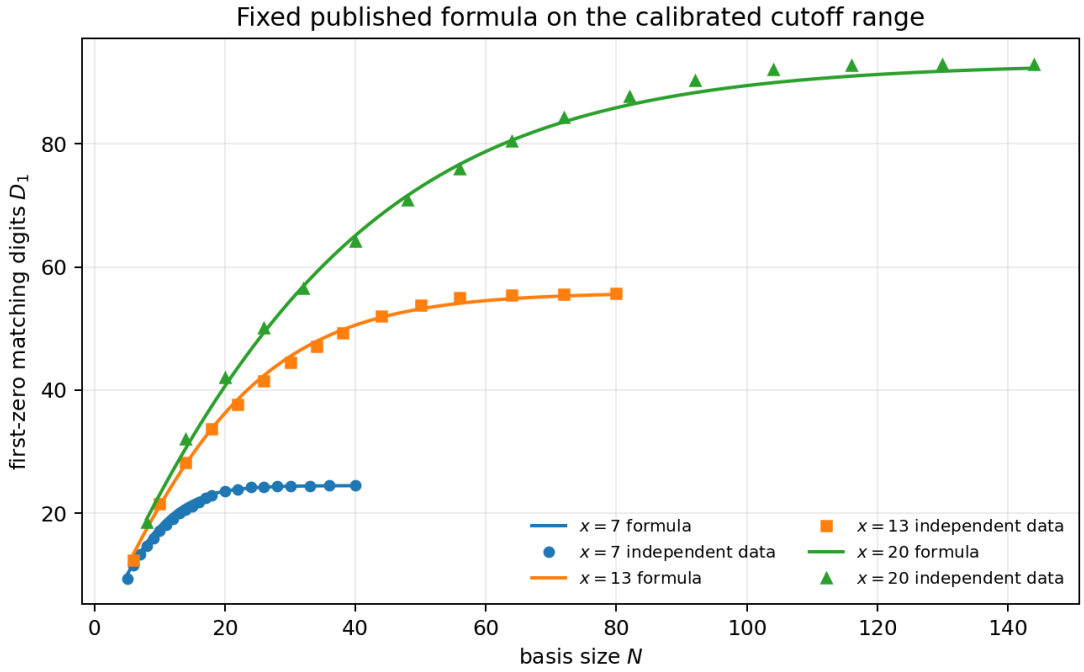


Figure 1: Independent measurements and the caller-facing formula with all published constants fixed. The close agreement tests the formula, not merely the flexibility of the tanh family.

Paper 1 supplies a separate fixed-$N = 120$ cross-check. At $\lambda^2 = 20, 30, 50, 100$, the fixed formula has signed residuals $-1.54, -1.66, +2.13, +6.03$ digits against measured values ranging from 92.817 to 211.29 digits [50]. The growing positive tail is small relative to the total depth, but it is the first visible warning that the closed parameter map is beginning to drift.

That warning becomes decisive at $\lambda^2 = 1000$. An independent four-point sweep gives:

Table 3: Large-cutoff test of the unchanged closed submodel. The conversion gap is $D_{\mathrm{eig},N} - D_1$; residual is formula minus measured D_1.

N	$D_{\mathrm{eig},N}$	D_1	conversion gap	formula	residual
60	180.88	175.36	5.52	293.24	+117.88
90	247.60	243.06	4.54	381.08	+138.02
120	308.77	304.69	4.08	458.81	+154.12
150	366.41	362.60	3.81	529.75	+167.15

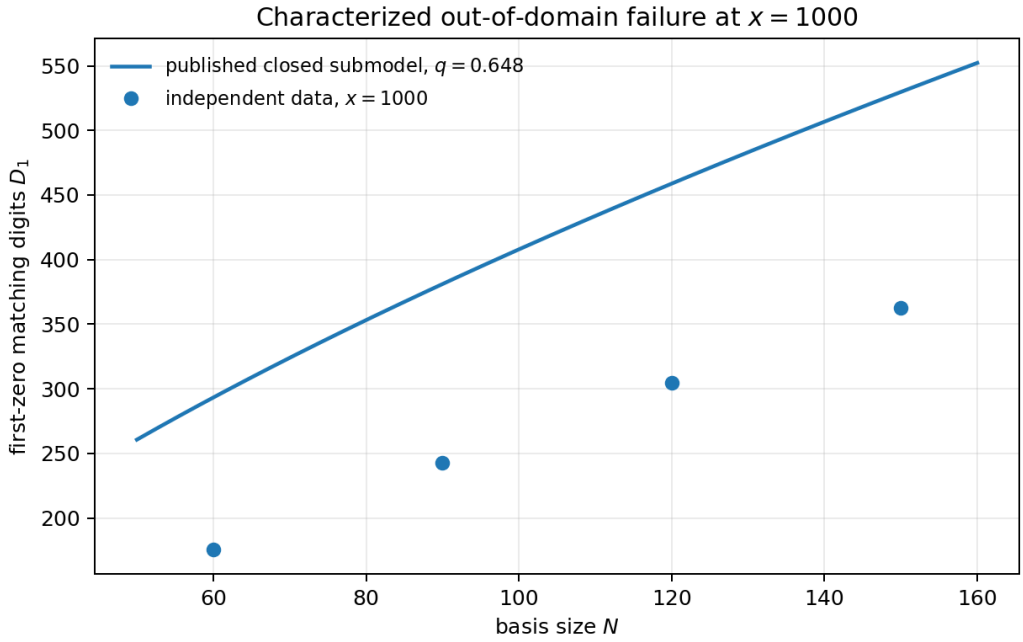


Figure 2: The closed calibrated-range parameter map overpredicts the new $\lambda^2 = 1000$ ramp. This is a characterized out-of-domain failure of q_γ and $c = 4/\pi$, not a failure of the minimum-budget structure or of the stretched-transition family.

The inferred effective exponent is convention dependent: a log–log fit of D_1 gives 0.793, a fit of $D_{\mathrm{eig},N}$ gives 0.770, fixing $c = 4/\pi$ and inverting the level at $N = 120$ gives 0.756, and jointly fitting (c, q) gives $(1.512, 0.792)$. A wide-decade eigenvalue-depth comparison to the $N = 800$ artifact gives approximately 0.743. The robust conclusion is therefore only

$$q_\gamma(1000) = 0.648 \text{ is too small}, \qquad q_{\mathrm{eff}}(1000) \approx 0.74\text{–}0.80,$$

while the data do not yet decide whether the missing correction belongs to q, to c, or to the transition coordinate. The next formula-selection pass will compare, by whole-cutoff holdout: the present q_γ model; the fitted linear law for $1/q$; a saturated exponent; a logarithmically corrected Kulikov-type plunge coordinate; and a two-stage eigenvalue-depth minus conversion-gap model. No replacement enters the boxed law until it improves whole-cutoff holdouts while preserving the calibrated-domain accuracy.

Status: the fixed-form validation and the $\lambda^2 = 1000$ failure are T3 measurements from an independent numerical implementation. The four large-cutoff points and the conversion identities below should be replayed through the primary Rust artifact stack before publication.

6 The prime ceiling D_{Primes}

The prime channel of the intrinsic law (1) is the accuracy at infinite N and P, set solely by which primes $p \le \lambda^2$ enter the form. It is the relative error of the first zero at the cutoff,

$$\varepsilon_{\mathrm{primes}}(\lambda^2) = \frac{|\nu_1(\infty, \lambda^2) - \gamma_1|}{|\gamma_1|}, \tag{21}$$

with $\nu_1(\infty, \lambda^2)$ the first zero at infinite basis size and fixed λ^2: it is what remains once modes and precision are exhausted, and its digit measurement $D_{\mathrm{Primes}} = -\log_{10} \varepsilon_{\mathrm{primes}}$ is the ceiling the minimum (2) consumes. We fix its leading rate through the plunge *eigenvalue*, the tractable route, kept here in digit space to reinforce the intrinsic statement with the measured quantity.

The plunge eigenvalue gives an *eigenvalue floor* $D_{\mathrm{eig}} = -\log_{10} \varepsilon^{\infty}(\lambda)$, the digit value of the smallest eigenvalue. Its prolate comparison requires two indexing conventions to be kept separate. Let $\mu_n(c)$ denote the finite-Fourier singular value in the full PSWF indexing and let $\Lambda_n(c) = \mu_n(c)^2$ be the corresponding concentration eigenvalue. The function $h_{4,\lambda}$ has full index $n = 4$; in the even-sector notation of Connes [4, §6.3], its positive finite-Fourier eigenvalue is $\chi_2 = \mu_4$. Thus it is not the principal $n = 0$ quantity χ_0.

Fuchs' fixed-index theorem [31], at $c = 2\pi\lambda^2$, gives

$$1 - \Lambda_4(c) \sim \frac{2^{15}\sqrt{2}\,\pi^5}{3}\,(\lambda^2)^{9/2} e^{-4\pi\lambda^2}.$$

Since $\mu_4 = \sqrt{\Lambda_4} \to 1$, $1 - \mu_4 = (1 - \Lambda_4)/(1 + \mu_4)$, and therefore

$$1 - \chi_2(2\pi\lambda^2) \sim \frac{2^{14}\sqrt{2}\,\pi^5}{3}\,\exp\!\big(-4\pi\lambda^2 + \tfrac{9}{2}\ln\lambda^2\big). \tag{22}$$

Equivalently, the classical prolate digit scale is

$$\begin{aligned} D_{\mathrm{prol}}(\lambda^2) &:= -\log_{10}(1 - \chi_2(2\pi\lambda^2)) \\ &= K\lambda^2 - \frac{9}{2\ln 10}\ln\lambda^2 - C_0 + o(1), \qquad K = \frac{4\pi}{\ln 10} = 5.45750\ldots. \end{aligned} \tag{23}$$

where $C_0 = \log_{10}\!\big(2^{14}\sqrt{2}\,\pi^5/3\big) = 6.3736\ldots$. This fourth-index singular-value asymptotic is classical and T1. Groskin [7] independently evaluates the same §6.4 prefactor and obtains $\log_{10} \approx 6.37$. Hedenmalm's independent theta-kernel asymptotic also contains the same 9/2 logarithmic order [6]; it does not supply the CCM-to-prolate transfer. The separate assertion that the CCM eigenvalue floor D_{eig} follows this scale is the prolate-transfer step discussed next; it is strongly supported by the empirical data but is not part of Fuchs' theorem.

The variational part of the proposed transfer can be anchored rigorously, and we record exactly that part as a proposition. Recall that the bottom of the spectrum of the continuum form is a variational infimum, $\varepsilon^{\infty}(\lambda) = \inf_{\varphi} QW_{\lambda}(\varphi, \varphi)/\|\varphi\|^2$, so *any* trial vector bounds it from above, irrespective of the sign of ε^{∞}.

Proposition 7 (Variational trial bound)**.** *Let h_{λ} be the suitably normalized, zero-integral linear combination of $h_{0,\lambda}$ and $h_{4,\lambda}$ used in the CCM prolate construction, and let $k_{\lambda} = E(h_{\lambda})$ be its image under the CCM Poisson-summation map ([1, §7]; [4, §6.4]). Then, unconditionally,*

$$\varepsilon^{\infty}(\lambda) \;\le\; R[k_{\lambda}] := \frac{QW_{\lambda}(k_{\lambda}, k_{\lambda})}{\|k_{\lambda}\|^2}.$$

Proof. The bottom of the spectrum is the infimum of the Rayleigh quotient over the form domain; any admissible trial vector bounds the infimum from above. □

The proposition is the variational principle verbatim and is T1. It does not by itself identify the size of $R[k_\lambda]$. On the localized form's exact split $QW_\lambda = A_{\text{arch}} - A_{\text{prime}}$ (archimedean part $w_{02} - w_r$, prime part the prime-power sum w_p; [1, §6]), define

$$R[k_\lambda] = R_{\text{arch}}[k_\lambda] - \rho_p[k_\lambda], \qquad \rho_p[k_\lambda] := \frac{\langle A_{\text{prime}} k_\lambda, k_\lambda \rangle}{\|k_\lambda\|^2}. \tag{24}$$

Lemma 5 (Conditional CCM-to-prolate floor bound)**.** *Suppose, as $\lambda \to \infty$, that the constrained CCM trial vector satisfies*

$$R_{\text{arch}}[k_\lambda] \le (1 - \chi_2(2\pi\lambda^2))(1 + o(1)) \quad \textit{and} \quad \rho_p[k_\lambda] \ge 0.$$

Then

$$\varepsilon^\infty(\lambda) \le (1 - \chi_2(2\pi\lambda^2))(1 + o(1)).$$

Proof. By Prop. 7 and (24), $\varepsilon^\infty(\lambda) \le R[k_\lambda] \le R_{\text{arch}}[k_\lambda]$ by the prime-sign hypothesis, and the stated comparison with $1 - \chi_2$ finishes the proof. □

The first hypothesis is the currently open CCM-to-prolate quotient comparison. Connes [4, §6.4] reports the numerical similarity between $\varepsilon^\infty(\lambda)$ and $1 - \chi_2(\lambda)$ and describes k_λ as an educated approximation to the ground state; neither statement is an exact Rayleigh-quotient identity. The second hypothesis is a sign on the prime quotient evaluated on this same constrained trial vector. The paper's measured prime contributions on the computed ground state support the sign but do not prove it for k_λ. Andrade's exact Loewner formula proves a termwise non-positive prime contribution for the specific even vector $\cosh(x/2)$ and exposes mixed signs in the odd sector [10]. It does not prove the sign for the constrained h_0–h_4 vector k_λ, but it sharpens the open task: control the sign or weighted correlation of the exact Fourier symbol of this particular trial state, rather than estimate the prime matrix entrywise.

Under both hypotheses, Eq. (22) gives the conditional one-sided floor bound

$$D_{\text{eig}}(\lambda^2) \;\ge\; K\lambda^2 - \frac{9}{2\ln 10}\ln\lambda^2 - C_0 + o(1), \qquad K = \frac{4\pi}{\ln 10},$$

whenever $\varepsilon^\infty(\lambda) > 0$. The empirical CCM calculations used throughout this paper are unchanged and align with this fourth-index law; what remains open is its analytic transfer from the constrained h_0–h_4 trial construction to the full CCM floor. The reverse inequality, that the ceiling cannot be beaten, i.e. a *lower* bound on ε^∞ and on $|\nu_1(\infty, \lambda^2) - \gamma_1|$ of the same exponential order, remains open, and is the archimedean–prime cancellation of Section 9.

Where finite-λ^2 control (rather than asymptotics) is needed, for instance the curvature that produces the empirically steeper matching-floor slope $P_m \approx 5$ below, the modern non-asymptotic prolate literature supplies two-sided eigenvalue bounds with explicit constants: the three-region estimates of Bonami–Jaming–Karoui [22], the improved bounds of Karnik–Romberg–Davenport [23], the upper bounds of Osipov [24], the super-exponential decay estimates of Bonami–Karoui [25], and Kulikov's exponential lower bound before the plunge region [26]. These replace the asymptotic-only inputs wherever a quantified inequality at a specific cutoff is wanted.

The eigenvalue depth is a scaffold, not the object. What the law consumes is the zero error $\varepsilon_{\text{primes}}$, which sits a few digits below the eigenvalue floor: converting the plunge depth ε^∞ into the zero displacement $|\nu_1 - \gamma_1|$, and normalizing by $|\gamma_1|$ (the $\log_{10}|\gamma_1| \approx 1.15$ digits of the relative error), gives the matching floor

$$D_{\text{Primes}}(\lambda^2) = K\lambda^2 - P_m \log_{10}\lambda^2 - c_m, \qquad P_m \approx 5, \quad c_m \approx 9.5. \tag{25}$$

The leading $K\lambda^2$ is shared with the eigenvalue floor, the same plunge rate, so the intrinsic content is a *single* channel, not two competing ceilings; the subleading P_m, c_m are the eigenvalue-to-zero

conversion together with the $|\gamma_1|$ normalization. The expanded artifact archive permits a more direct test of that conversion. For each numerical configuration define

$$\mathrm{conv}_N(\lambda^2) := -\log_{10}|\varepsilon_N| - D_N,$$

pairing the stored smallest Weil eigenvalue with the stored relative error of the first refined root. On the deepest precision-clear configuration at each of thirteen cutoffs, this gap grows from 2.499 digits at $\lambda^2 = 7$ to 2.925 at 100, then is 3.024–3.029 throughout $\lambda^2 = 500$–1200. Thus the earlier algebraic extrapolation $(P_m - \frac{9}{2})\log_{10}\lambda^2 + (c_m - C_0)$ is not supported as a growing conversion law. The paired data instead favor a conversion approaching a constant $C_{\mathrm{conv}} \approx 3.03$ digits. If this persists in the saturated continuum limit, then the asymptotic logarithmic coefficient returns to $P_m = 9/2$, while $c_m = C_0 + C_{\mathrm{conv}} \approx 6.3736 + 3.03 = 9.40$, close to the finite-range fit $c_m \approx 9.5$. This is a sharper hypothesis, not yet a theorem: several large-cutoff configurations are mode-limited, and a uniform target-transform comparison is still required. The leading rate K is derived for the classical fourth-index deficiency and is observed for the CCM floor; the transfer between those two quantities is the conditional step in Lemma 5. The finite-range fit, hence P_m and c_m, remains empirical; of it, the ≈ 1.15-digit part of c_m is the explicit $|\gamma_1|$ normalization and the remainder is the depth-to-zero factor. Conditional on global RH, the finite conversion factor is no longer merely a fitted constant: equation (33) below expresses $D_{\mathrm{eig},N} - D_k$ exactly through the finite Stieltjes factor, divided difference, and remote-zero term. That conditional finite identity does not prove the saturated continuum limit, the large-cutoff constancy of C_{conv}, or the unconditional comparison needed in an RH argument. A structural unconditional route to the limiting factor would require an additional target-transform comparison, which we record as a proposal rather than claim as a derivation. Specifically, let Z_λ be a compatibly normalized holomorphic target transform on a disc about γ_1, with a simple zero at γ_1, and suppose one can prove

$$\eta_\lambda := \sup_{\overline{D}(\gamma_1, r)} |\widehat{\xi}_\infty(z) - Z_\lambda(z)| \quad \text{is sufficiently small.}$$

Lemma 2, now applied with $a = \gamma_1$, $F = Z_\lambda$, and $G = \widehat{\xi}_\infty$, would then give

$$|\nu_\infty(\lambda) - \gamma_1| \le \frac{2\eta_\lambda}{|Z'_\lambda(\gamma_1)|}.$$

The determinant identity $\det_{\mathrm{reg}}(D - z) = -i\lambda^{-iz}\widehat{\xi}(z)$ of [1, Thm. 5.10] describes the finite transform, but does not itself construct Z_λ or prove this uniform estimate. Establishing that comparison, connecting η_λ to the plunge depth, and deriving the λ-scaling of the ratio would promote (P_m, c_m) from empirical to structural. Until then the row remains T4.

The data indicate that the arithmetic does not move the eigenvalue-floor slope. A formal large-$|t|$ model motivated by Mertens' theorems and prime-number-theorem partial summation has the scalar form $S(t) = \frac{1}{2}t^2 - \gamma|t| + c_0$. Turning this scalar asymptotic into a statement about the compressed CCM prime operator requires its own operator limit: parity, the mean-zero constraint, and the compression Q do not by themselves justify assigning the three displayed terms separate actions on the finite-section ground state. The screw-function decomposition of Suzuki [5] suggests a useful comparison with the metric operator, while the exact finite prime-power translations provide a second, independent route through quadratic-form or Loewner estimates. Either route could explain why the arithmetic changes lower-order conversion terms without changing the prolate prefactor power, but that conclusion is not yet a theorem. Conditional on the transfer in Lemma 5, the prefactor power is the Fuchs index $\frac{9}{2}$ of the fourth-index prolate model (Section 6), while the arithmetic contribution enters the lower-order conversion. The archimedean part of the form does not by itself carry the plunge (it is $O(1)$-indefinite; see the rate-origin discussion in Section 9). The empirically steeper matching-floor slope $P_m \approx 5$ is a finite-λ^2 curvature effect, not a change in the asymptotic slope.

Status: the classical fourth-index prolate coefficient K and slope $\frac{9}{2}$ are T1. Their transfer to the CCM eigenvalue floor is T3: it is validated by the data, while the constrained-trial comparison and the prime sign in Lem. 5 remain open. The reverse inequality requires new archimedean–prime cancellation estimates. The matching-floor offset (P_m, c_m), which also contains the eigenvalue-to-zero conversion, remains empirical (T4). The paired artifacts now isolate a candidate constant $C_{\mathrm{conv}} \approx 3.03$ and favor the asymptotic value $P_m = 9/2$; the structural route remains the target-transform comparison named above.

A smooth envelope with arithmetic microstructure, not a staircase. A natural hypothesis, that D_{Primes} changes by macroscopic steps when a prime power enters, is falsified directly. With N and P held above floor ($N = 300$, $P = 200$), sweeping λ^2 in unit steps from 4 to 30 gives an overall rise of ≈ 5.3 digits per unit λ^2; at that resolution the per-unit rate at prime-power entries is statistically indistinguishable from non-prime entries. This does not mean the exact matrix path is smooth across an entry event. Groskin proves that its first derivative has a rank-one jump at every prime power [9]; small local bumps or kinks are therefore expected. The empirical conclusion is the more precise one: Eq. (25) describes a smooth macroscopic envelope, not a distinct digit staircase, while the arithmetic events survive as finer path structure.

Floor anchors. Floor-isolated measurements (both N and P above floor) give $D_{\mathrm{Primes}}(100) = 526.08$ at $(N, P) = (1000, 700)$ and a saturated $D_{\mathrm{Primes}}(200) \approx 1070.4$ at $N \geq 2400$. The latter sits at the matching floor $K \cdot 200 - 5 \log_{10} 200 - 9.5 = 1070.5$, confirming the two-floor structure at the largest cutoff measured. Regarding provenance, the $\lambda^2 = 200$ anchor was measured against the 2500-digit Arb reference of Definition 1, not a shorter tabulation. A matching-digit count cannot exceed the length of its reference: against a ~ 1000-digit reference the measurement would *saturate* near 999 and could never read 1070.4, so the observed value is itself incompatible with a reference-limited artifact. Such an artifact would present as a suspicious plateau at the reference length, not a reading 70 digits past it.

7 The Riemann–von Mangoldt backbone and the spectral-edge hypothesis

Sections 5 and 6 approach the mode ramp and the prime ceiling by two separate classical routes: the Landau–Widom transition and the Fuchs prolate tail. Their shared scales and the data suggest that they are two regimes of one CCM spectral edge. This section separates the rigorous smooth counting backbone from that stronger, offensive hypothesis.

7.1 The rigorous archimedean counting backbone

Let

$$\vartheta(T) = \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{iT}{2}\right) - \frac{T}{2} \log \pi$$

be the Riemann–Siegel phase in the standard continuous branch. The Riemann–von Mangoldt formula can be written

$$N_\zeta(T) = \frac{\vartheta(T)}{\pi} + 1 + S(T),$$

away from ordinates, with the usual limiting convention at a zero. Stirling's formula gives the unconditional smooth density

$$\rho^{(0)}_{\mathrm{RvM}}(T) := \frac{\vartheta'(T)}{\pi} = \frac{1}{2\pi} \log \frac{T}{2\pi} + O(T^{-2}). \tag{26}$$

Thus the archimedean Gamma phase supplies the smooth Riemann–von Mangoldt backbone, while $S(T)$ contains the oscillatory correction. This statement is independent of RH and does not require an equilibrium-measure interpretation. Evaluating the integrated smooth density at the exact finite-band edge $T_{\max} = 2\pi N/\ln\lambda^2$ recovers the leading terms of $K^{(0)}_{\max}$ in (5). The reach formula and its smooth capacity therefore remain rigorous independently of the conjectural edge picture.

The earlier formal identification of (26) with the minimizer of an unscaled unit-mass logarithmic energy is not valid: the density has infinite mass on the half-line, and differentiating a weighted-energy Euler–Lagrange condition produces a singular Hilbert-transform equation rather than $\rho = Q'/\pi$. A genuine equilibrium formulation must specify a scaling window, a normalized empirical measure, its external field, and which spectral objects are being counted. Weighted logarithmic-energy theory [35] and its orthogonal-polynomial/random-matrix realization [36] provide candidate tools once those CCM-specific objects are derived; their general results do not supply the missing identification by themselves.

7.2 A scaled equilibrium conjecture

For a fixed $0 < a < 1$, a candidate CCM zero measure in a high-frequency window is

$$\mu^{\mathrm{CCM}}_{T,a} := \frac{1}{K_{T,a}} \sum_{aT<\nu_k\le T} \delta_{\nu_k/T},$$

where $K_{T,a}$ is the number of selected transform zeros in the window. Its smooth Riemann–von Mangoldt target is the probability measure

$$d\mu^{(0)}_{T,a}(x) := \frac{T\rho^{(0)}_{\mathrm{RvM}}(Tx)\,\mathbf{1}_{[a,1]}(x)\,dx}{\int_{aT}^{T}\rho^{(0)}_{\mathrm{RvM}}(t)\,dt}.$$

These definitions do not assert that the CCM zeros form a logarithmic gas; they state a normalized object on which such a theorem could be formulated.

Conjecture 1 (Scaled CCM equilibrium and common edge)**.** *Along a joint regime in which the cutoff and finite-section reach tend to infinity:*

(i) the selected CCM zero measures have the same macroscopic limit as $\mu^{(0)}_{T,a}$;

(ii) there are scaled external fields for which these measures are asymptotically characterized by a weighted logarithmic-energy problem; and

(iii) the observed mode transition and prime tail arise as two scales of one associated CCM spectral edge.

The most direct offensive route uses the determinant identity $\det_{\mathrm{reg}}(D'' - z) = -i\lambda^{-iz}\widehat{\xi}(z)$. Numerical and regularized Fredholm determinant theory supplies templates for controlling finite determinants and their product anomalies [47, 48, 49]. A limit for the normalized subharmonic functions

$$\frac{1}{K}\log\left|\det_{\mathrm{reg}}(D'' - z)\right|$$

would determine a limiting zero measure through its distributional Laplacian and could identify the external field and logarithmic potential. This route addresses transform zeros directly and supplies the missing bridge between the finite determinant and a potential-theoretic measure.

A complementary route begins with the archimedean matrix. Its explicit phase rate $\vartheta'(t) = \frac{1}{2}\log(t/2\pi) + O(t^{-2})$ suggests a Toeplitz-, Kac–Murdock–Szegő-, or pseudodifferential-type spectral law [28, 29]. To make that route rigorous one must first identify the exact matrix class

and normalization, handle the unbounded logarithmic symbol, verify form-core and spectral-distribution hypotheses, and obtain a uniform estimate for the arithmetic perturbation. Even then, a separate theorem must connect the resulting matrix eigenvalue measure to the transform-zero measure. The discreteness result of [1, Thm. 3.6] motivates this program but does not by itself establish all of these quantitative steps.

7.3 One-edge conjecture: two observed scales

In the parameter range where the relevant Weil form is positive, the computed spectrum descends from an $O(1)$ bulk into a deep plunge. Two classical facts make a common-edge interpretation compelling:

- the Landau–Widom transition width [32] supplies the natural crossover scale near the Shannon number $\lambda^2 \ln \lambda^2$, matching the observed mode ramp;
- the fourth-index Fuchs asymptotic [31] supplies the extreme prolate tail, matching the observed leading scale of the prime ceiling.

The classical plunge is commonly written as a Fermi–Dirac/logistic profile; after centering and rescaling, a logistic is algebraically a shifted tanh. This makes the empirical choice of a hyperbolic tangent less ad hoc, but Landau–Widom's index variable is not yet the CCM root-error coordinate. Modern pre-plunge estimates sharpen that warning: before the transition, the natural variable contains a logarithmically corrected distance to the Shannon edge rather than a pure normalized power [45, 46]. The measured failure of the unchanged q_γ model at $\lambda^2 = 1000$ therefore elevates a log-corrected plunge coordinate from an optional analogy to a leading empirical competitor. Identifying the CCM variable, including whether the correction should be assigned to q, to c, or to the coordinate itself, remains part of the open edge comparison rather than a consequence of the classical profile. The offensive hypothesis is that these are the transition and extreme tail of the same CCM edge, explaining the algebra $D^{\mathrm{fit}}_{\mathrm{edge}} = D_{\mathrm{Primes}}\, g$. This is not yet a consequence of the Riemann–von Mangoldt bulk density: the Weil-matrix spectrum, the perturbed operator spectrum, the transform zeros, and the prolate concentration spectrum must first be connected by explicit comparison theorems. The microscopic universality class (Airy, Bessel, or Landau–Widom), the determinant-to-measure bridge, and the exact deep-edge constant remain open. *Status: T3 — a structural conjecture supported by two classical asymptotic regimes and the measured convergence surface; proving it requires new potential-theoretic and spectral-comparison analysis.*

8 The complete formula

8.1 The assembled law

Collecting Sections 4–7, the law stands in the two forms of the introduction. *Intrinsically*, the relative error obeys the two-channel decomposition

$$E(N, \lambda^2) \le \varepsilon_{\mathrm{modes}}(N, \lambda^2) + \varepsilon_{\mathrm{primes}}(\lambda^2) \tag{27}$$

(an equality up to sign cancellation; Prop. 1). Separately, the empirical total-edge model is

$$D^{\mathrm{fit}}_{\mathrm{edge}}(N, \lambda^2) = D_{\mathrm{Primes}} \tanh\left[\left(\frac{cN}{\lambda^2 \ln \lambda^2}\right)^{q_\gamma}\right], \qquad D_{\mathrm{Primes}} = K\lambda^2 - P_m \log_{10} \lambda^2 - c_m.$$

With the operational precision cap, its prediction is

$$D^{\mathrm{fit}}(\lambda^2, N, P) = \min\Big(D^{\mathrm{fit}}_{\mathrm{Wprec}}(P), D^{\mathrm{fit}}_{\mathrm{edge}}(N, \lambda^2)\Big), \qquad D^{\mathrm{fit}}_{\mathrm{Wprec}} = P_{\mathrm{req}}. \tag{28}$$

The candidate $K = 4\pi/\ln 10$ is exact for the fourth-index prolate deficiency; its transfer to the CCM root ceiling is open. The calibrated T3 map uses $q_\gamma = 2/(2\gamma + \ln\ln\lambda^2)$ and $c = 4/\pi$ on $7 \le \lambda^2 \le 100$. Its historical large-cutoff failures are retained. Equation (28) is a point predictor, not an equivalent restatement or a guaranteed upper-error consequence of (27). The rigorous mode depth is not the saturating $D^{\mathrm{fit}}_{\mathrm{edge}}$. This is the Andrews CCM Convergence Law. The law is the budget skeleton and its empirically measured channel functions; the particular closed $(q_\gamma, 4/\pi)$ map is its calibrated caller-facing realization, not a construction-wide theorem. Of the law's parts, the minimum bound, the precision slope, the stretched-exponential rate unification (Prop. 2), and the general simple-zero displacement (Lemma 2) are rigorous; every remaining component carries the explicit tier, open item, and open-item type recorded in Table 8, from the conditional transfer estimates through the T3 exponent and onset constant to the empirical matching-floor terms (T4).

Operational model card. The following compact statement is the version a caller can apply and a reviewer can falsify.

Item	Operational statement
Inputs	cutoff λ^2, basis size N, requested decimal target P_{req}
Output	predicted relative first-zero matching digits D_1
Precision map	$D_{\mathrm{Wprec}} = P_{\mathrm{req}} + O(1)$; the toolkit adds a 64-bit internal reserve
Closed mode submodel	equation (10), validated for $7 \le \lambda^2 \le 100$
Cutoff ceiling	equation (25), calibrated and floor-tested through $\lambda^2 = 200$
Independent validation	fixed-form RMS 0.47, 0.66, 1.18 digits at $\lambda^2 = 7, 13, 20$ (Table 2)
Known failure	unchanged closed mode map overpredicts the $\lambda^2 = 1000$ ramp by 118–167 digits (Table 3)
Data admissibility	validated matrix assembly, reference provenance, structural checks, and precision/cutoff stability

To the best of the author's knowledge, the proposed complete convergence model of (28) is new: the assembly of the three budgets into a single law, the calibrated closed-form mode exponent $q_\gamma(\lambda^2) = 2/(2\gamma + \ln\ln\lambda^2)$, the Shannon-like coordinate $\lambda^2 \ln\lambda^2$, the stretched-exponential parametrization of the mode budget, and the proposed common-edge interpretation of the mode ramp and prime ceiling appear here for the first time. The rate it models is precisely the problem named open in [1]: Groskin [7] measures a Galerkin exponent but disclaims deriving it, and Śliwiński [15] establishes only a lower bound, so no prior work supplies the functional form given here. The novelty claimed is the law as a whole and an explicit, falsifiable program for identifying its mechanism; the underlying construction [1, 2], the Paley–Wiener strip mechanism [7], and the classical prolate and Mertens inputs are credited to their sources throughout.

8.2 Convergence criterion

Proposition 8 (Fixed-cutoff branch convergence)**.** *Fix $\lambda > 1$. Assume the spectral and zero-branch hypothesis*

$$(\mathrm{H}^*_\lambda): \quad \begin{cases} QW_\lambda \text{ has a simple, isolated, positive lowest eigenvalue } \varepsilon^\infty(\lambda), \\ \text{its normalized continuum eigenvector } \xi_\infty \text{ is even}, \\ \widehat{\xi}_\infty \text{ has a simple isolated positive zero } \nu_\infty(\lambda) \neq \gamma_1. \end{cases} \tag{29}$$

Let ξ_N be the phase-aligned finite-section ground states and let ν_N denote the zero of $\widehat{\xi}_N$ selected in a sufficiently small disc about ν_∞. Then:

(i) The lowest Rayleigh values converge monotonically to $\varepsilon^\infty(\lambda)$, while $\xi_N \to \xi_\infty$ in norm and $\widehat{\xi}_N \to \widehat{\xi}_\infty$ compact-uniformly.

(ii) For all sufficiently large N, the selected zero ν_N is unique in that disc and $\nu_N \to \nu_\infty(\lambda)$.

(iii) If the numerical root perturbation vanishes as $P \to \infty$, then

$$\lim_{\substack{N\to\infty \\ P\to\infty}} D(\lambda^2, N, P) = D_{\mathrm{Primes}}(\lambda^2) := -\log_{10} \frac{|\nu_\infty(\lambda) - \gamma_1|}{|\gamma_1|} < \infty.$$

Consequently, at this fixed cutoff the selected branch cannot converge to γ_1 merely by increasing N and P.

Proof. The Rayleigh-value statement is [1, Prop. 3.4]. Under the simple/isolated/even hypotheses in (29), [2, Thm. 6.1] gives convergence of the finite-section eigenvectors and compact-uniform convergence of their transforms. Choose a closed disc about ν_∞ containing no other zero of $\widehat{\xi}_\infty$. Since that zero is simple, Hurwitz's theorem (equivalently the local Rouché argument of Lemma 2) gives one zero ν_N in the disc for all sufficiently large N and $\nu_N \to \nu_\infty$. Vanishing numerical perturbation then gives the same limit as $P \to \infty$, and continuity of $-\log_{10}|\cdot|$ away from zero proves part (iii). □

Growing cutoffs. For a sequence $\lambda_j \to \infty$, convergence to γ_1 follows if the three separately defined errors vanish:

$$|\nu_{N_j}(\lambda_j) - \nu_\infty(\lambda_j)| \to 0, \qquad \varepsilon_{\mathrm{round}}(P_j) \to 0, \qquad |\nu_\infty(\lambda_j) - \gamma_1| \to 0.$$

This is a sufficient pathwise criterion from the triangle inequality. The budget choices $N_j \gtrsim (4/\pi)\lambda_j^2 \ln \lambda_j^2$ and $P_j \gtrsim K\lambda_j^2$, and the rate $D \sim K\lambda_j^2$, are the quantitative model tested in this paper; they do not follow from the qualitative fixed-cutoff theorem above. Conversely, ruling out convergence for an arbitrary family of cutoffs in a bounded interval would require a uniform separation $\inf_\lambda |\nu_\infty(\lambda) - \gamma_1| > 0$ on that interval, which is not claimed here.

Status of the hypothesis (H^*_λ)**.** We state (H^*_λ) explicitly because finite-cutoff Weil positivity is *not* known unconditionally for all λ: [1] is explicit that positivity of the continuum measure cannot be asserted in general, and positivity for all λ is precisely the open conjecture linking the construction to RH. What is unconditional is the small-support range: Yoshida proved the Weil hermitian form positive definite for test functions of sufficiently small support [16], Bombieri reproved and quantified this in his study of the truncated functional [17], Connes–Consani gave the operator-theoretic strengthening at the archimedean place via the Sonin space [18], and the ζ-cycles paper [19] tracks finite-λ positivity prime by prime through the first cutoffs. Suzuki [5, Thm. 1.4] also proves a positive, simple, even continuum ground state for sufficiently small aperture. Andrade extends the Perron argument to the *pole-free* continuum operator at every aperture and reduces the remaining pole term, which is rank one in each parity sector, to a scalar resolvent/nodal question [11]. The same obstruction can be sharpened to a scalar inequality between two Herglotz/Weyl functions [12]. These are precise attacks on full even-simplicity, not proofs of it. These results address only part of (H^*_λ): simplicity of the selected continuum-transform zero remains an explicit hypothesis. Beyond the rigorous small-aperture regime, positivity, simplicity, and evenness are supported numerically at every cutoff measured in this program ($\lambda^2 \le 200$, and $\lambda^2 \le 1000$ in [50]). The finite matrices $(\lambda^2, N) = (13, 120), (29, 160), (53, 200)$ additionally have independent Arb certificates that the lowest even-sector eigenvalue is simple and strictly below the lowest odd-sector eigenvalue [13]. Those certificates strengthen three finite data points but do not establish uniformity or the continuum limit. The noncoincidence $\nu_\infty(\lambda) \neq \gamma_1$ is likewise numerical and is what makes the fixed-cutoff limiting digit count finite.

The qualitative division of labor is now explicit. CCM Prop. 3.4 supplies monotone convergence only for the lowest Rayleigh values. Connes–van Suijlekom Thm. 6.1 supplies conditional

eigenvector and compact-uniform transform convergence, and simplicity of the selected continuum zero supplies the zero branch. No monotonicity of root accuracy and no pointwise bound $D \le D_{\mathrm{Primes}}$ for every finite N is asserted: finite-N and cutoff displacements can partially cancel. The quantitative growing-cutoff rate inherits the separate rigor of D_{Primes}: the classical fourth-index prolate asymptotic is T1, while its transfer to the constrained CCM trial quotient is T3 and the continuum plunge remains an assumption.

Conceptually, along the selected branch the fixed-cutoff limit isolates the cutoff displacement once precision and Galerkin errors are driven to zero. Reaching γ_1 therefore requires a growing-cutoff path on which the continuum-transform zero itself approaches γ_1. The measured model $D \sim K\lambda^2$ quantifies the proposed cost along such a path — each additional digit demands a fixed increment $\ln 10/(4\pi)$ in λ^2 — but this quantitative rate is not supplied by the qualitative fixed-cutoff theorem.

8.3 Conditional Gram representation and divided-difference root transfer

The operational law above is empirical. This subsection gives a separate, conditional explanation of the finite eigenvalue-to-root conversion that the law currently records through a conversion gap. It is placed after the formula and its validation because the formula does not depend on the hypothesis used here.

> ($\mathrm{H_{RH}}$) *The Riemann Hypothesis. Ordinates γ are counted with multiplicity. Simplicity is required only where a simple-zero derivative or a one-to-one zero label is used.*

The positive Gram representation in this historical subsection uses a global hypothesis; finite-height verification does not replace it as a positive representation over all zeros. The later dated construction in Section 8.5 instead retains the signed-complex tail and bounds it using a finite verification height. Its numerical substitutions are point audits, not certified assembly or root-matching results.

Proposition 9 (Conditional Gram representation)**.** *Let $V_N = \mathrm{span}\{V_j : |j| \le N\}$ and let $f \in V_N$ have $\|f\| = 1$. The finite Guinand–Weil dictionary and support exhaustion at cutoff λ^2 give the explicit-formula zero sum for $F = f \star \widetilde{f}$ [8]. Under* ($\mathrm{H_{RH}}$) *it becomes*

$$QW_\lambda(f,f) = \sum_{\gamma\in\pm\Gamma} |\widehat{f}(\gamma)|^2, \qquad \varepsilon_N(\lambda) = \min_{\substack{f\in V_N\\ \|f\|=1}} \sum_{\gamma\in\pm\Gamma} |\widehat{f}(\gamma)|^2. \tag{30}$$

The series is absolutely convergent for the finite-mode test functions used here.

Equation (30) is a Christoffel-type extremal formulation of the finite eigenvalue depth. It unifies the eigenvalue-depth analogues of the two intrinsic channels: changing N changes the trial space, while changing λ^2 changes the exponential type and the exhausted prime content. Root digits still require a separate eigenvalue-to-root transfer.

Proposition 10 (Divided-difference root transfer)**.** *Assume* ($\mathrm{H_{RH}}$). *Let ξ be the unit ground vector of the finite form, with*

$$\widehat{\xi}_N(z) = 2L^{-1/2}\sin(zL/2)G_N(z), \qquad G_N(z) = \sum_{|j|\le N} \frac{\xi_j}{z-\alpha_j}, \qquad \alpha_j = \frac{2\pi j}{L}.$$

Let ν be a real zero of G_N and define $\eta_j = \xi_j/(\alpha_j - \nu)$. Then $\eta \in V_N$ and $\widehat{\eta}(z) = \widehat{\xi}_N(z)/(z-\nu)$. Testing the finite eigenvalue equation against η gives the exact identity

$$\sum_{\gamma\in\pm\Gamma} \frac{|\widehat{\xi}_N(\gamma)|^2}{\gamma-\nu} = \varepsilon_N S_N(\nu), \qquad S_N(\nu) = \sum_{|j|\le N} \frac{\xi_j^2}{\alpha_j - \nu}. \tag{31}$$

For the zero ν_k matched to the positive ordinate γ_k, set

$$B_{k,N} = \frac{\widehat{\xi}_N(\gamma_k)}{\gamma_k - \nu_k}, \qquad R_{k,N}(\nu) = \sum_{m \neq k} \frac{2\nu\,|\widehat{\xi}_N(\gamma_m)|^2}{\gamma_m^2 - \nu^2}.$$

Pairing the ordinates $\pm\gamma_m$ in (31) *yields*

$$\boxed{\gamma_k - \nu_k = \frac{\gamma_k + \nu_k}{2\nu_k}\,\frac{\varepsilon_N S_N(\nu_k) - R_{k,N}(\nu_k)}{|B_{k,N}|^2}.} \tag{32}$$

Proof. The partial-fraction identity

$$\frac{1}{(z-\alpha_j)(z-\nu)} = \frac{1}{\alpha_j - \nu}\left(\frac{1}{z-\alpha_j} - \frac{1}{z-\nu}\right)$$

shows that $G_N(z)/(z-\nu)$ is again a lattice sum; the apparent extra pole vanishes because $G_N(\nu) = 0$. Thus $\eta \in V_N$ and the matrix identity $\langle QW_\lambda \xi, \eta\rangle = \varepsilon_N \langle \xi, \eta\rangle$ becomes (31) by Proposition 9. Isolating the $\pm\gamma_k$ pair gives (32); no remainder is discarded. □

Define the finite eigenvalue depth $D_{\mathrm{eig},N} = -\log_{10}|\varepsilon_N|$ and

$$\widetilde{S}_{k,N} = S_N(\nu_k) - \frac{R_{k,N}(\nu_k)}{\varepsilon_N}.$$

Equation (32) gives the exact relative conversion gap

$$\boxed{D_{\mathrm{eig},N} - D_k = \log_{10}\!\left(|\widetilde{S}_{k,N}|\,\frac{\gamma_k + \nu_k}{2\gamma_k\nu_k|B_{k,N}|^2}\right).} \tag{33}$$

When $\nu_k \approx \gamma_k$ and $B_{k,N} \approx \widehat{\xi}_N'(\nu_k)$, this becomes

$$D_{\mathrm{eig},N} - D_k \approx \log_{10}\!\left(\frac{|\widetilde{S}_{k,N}|}{\gamma_k|\widehat{\xi}_N'(\nu_k)|^2}\right). \tag{34}$$

The remote term requires a truncated zero sum with a controlled tail; any published numerical use of it must state the truncation height, zero source, and tail estimate.

Relation to the doubling conjecture. The exact identity is linear in ε_N and contains two powers of a norming divided difference. It therefore identifies the conditional first-order mechanism for the total displacement $\nu_k(N) - \gamma_k$. Problem 1(iii), however, concerns $\nu_k(N) - \nu_k(\infty)$ at fixed cutoff. Promoting the measured doubled finite-basis exponent still requires subtracting the limiting identity and controlling

$$\varepsilon_N - \varepsilon^\infty, \qquad S_N - S_\infty, \qquad B_{k,N} - B_{k,\infty}, \qquad R_{k,N} - R_{k,\infty}.$$

Thus Proposition 10 supplies the root-transfer layer of the (QG_λ) program; it does not remove the Galerkin layer.

Independent numerical check. On the independent implementation, the two sides of the first-order balance at $(\lambda^2, N, k) = (7, 40, 1)$ agree in ratio 0.999996. The exact conversion formula reproduces predicted/measured gaps 2.4689/2.4689 at $(7, 40)$, 2.69434/2.69435 at $(13, 80)$, and 2.78431/2.78432 at $(20, 144)$. Across $k \le 12$ at $\lambda^2 = 13, 20$, $\gamma_k S_N(\nu_k) \in [-1.07, -1.00]$, while the remote share decreases from 1.4% to 0.15% along the three increasingly saturated configurations. These are T3 measurements from a high-precision numerical implementation,

not interval certificates, and should be replayed from the primary Rust artifacts before final publication.

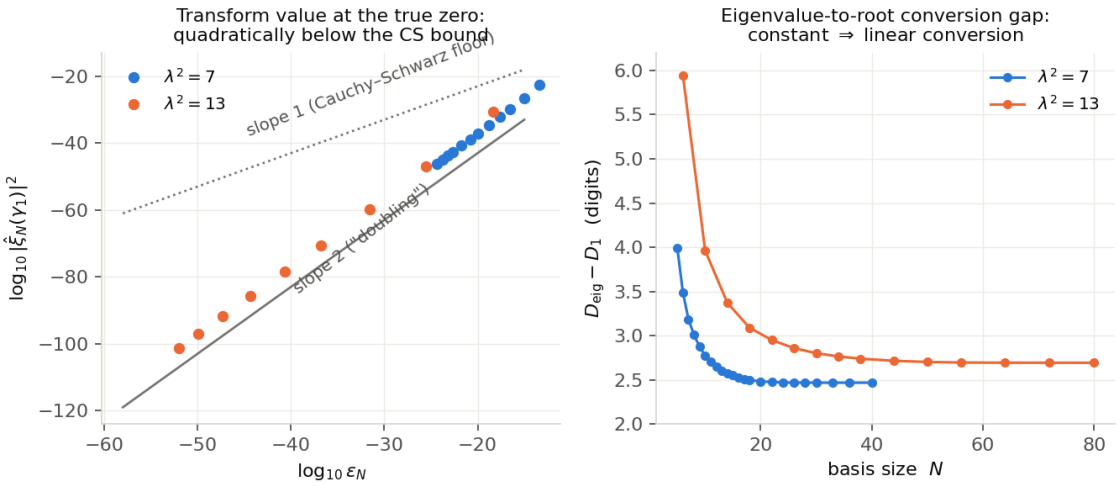


Figure 3: Independent evidence for the conditional root-transfer mechanism. The left panel shows the pre-saturation relation $|\widehat{\xi}_N(\gamma_1)|^2 \asymp \varepsilon_N^2$; it is evidence for the total-displacement mechanism, not by itself a proof of the strict fixed-cutoff doubling. The right panel shows the conversion gap approaching a cutoff-dependent plateau.

Status: Proposition 9, Proposition 10, and (32) *are T1* $|$ $(\mathrm{H_{RH}})$ *in the positive real-zero formulation above. The signed-complex extension used in Proposition 11 removes RH from the transfer identities, while retaining the unresolved remote bound. The observed near-constancy of* $\gamma_k S_N(\nu_k)$*, the remote-term estimates, and the numerical conversion gaps are T3. These identities explain the conversion mechanism; their remainder estimates do not establish convergence or the Riemann Hypothesis.*

8.4 A correction from a second divided form

The same finite eigenstate supplies an error predictor for every simple nonlattice transform root. The first zero is a special case; uniform accuracy in a growing root index requires an additional remainder estimate. The following construction uses finite operator data and no reference-zero errors or fitted conversion constant.

Write $L = \log \lambda^2$, $\alpha_j = 2\pi j/L$, and let a be a real even unit eigenvector of the full finite matrix A, with $Aa = \varepsilon_N a$. Set

$$F(z) = \frac{2}{\sqrt{L}} \sin(zL/2) \sum_{|j| \le N} \frac{a_j}{z - \alpha_j}.$$

At a simple nonlattice root $\nu > 0$ of F, define

$$b_j = \frac{a_j}{\alpha_j - \nu}, \quad S = a^T b, \quad J = b^T A b, \quad f = F'(\nu), \quad h = \frac{F''(\nu)}{F'(\nu)}, \quad \kappa = 1 - \nu h. \tag{35}$$

For $\kappa \neq 0$, the corrected signed displacement estimate is

$$\boxed{\widehat{\delta} = \frac{\nu(J - f^2) - \varepsilon_N S}{\kappa f^2}.} \tag{36}$$

Its relative-error estimate is $|\widehat{\delta}|/\nu$; a known zeta zero enters only the subsequent scoring. The bare local estimate is $-\varepsilon_N S/f^2$. All terms in (36) must use the same normalization, and the subtraction $J - f^2$ needs sufficient source and working precision.

Proposition 11 (Finite corrected-displacement identity)**.** *Let $\gamma > 0$ be a correctly matched simple real zeta ordinate and put $d = \gamma - \nu$ and $B = F(\gamma)/d$, with $B = f$ at $d = 0$. Over the full multiset $z_\rho = (\rho - 1/2)/i$ of nontrivial zeta zeros, define*

$$R(\nu) = \sum_{z_\rho \neq \pm\gamma} \frac{F(z_\rho)^2}{z_\rho - \nu}, \qquad R'(\nu) = \sum_{z_\rho \neq \pm\gamma} \frac{F(z_\rho)^2}{(z_\rho - \nu)^2}.$$

The prime differentiates the rational weights with F and the excluded pair fixed. Set

$$E_{\mathrm{loc}} = \frac{2\nu d B^2}{2\nu + d} - \nu \left\{ B^2 \left(1 + \frac{d^2}{(2\nu + d)^2} \right) - f^2 \right\} - \kappa f^2 d. \tag{37}$$

Then, without assuming RH,

$$\widehat{\delta} - (\nu - \gamma) = \frac{\nu R'(\nu) - R(\nu) - E_{\mathrm{loc}}}{\kappa f^2}. \tag{38}$$

For fixed nonzero f, ν, one has $E_{\mathrm{loc}} = O(d^2)$.

Proof. The secular root condition gives $F_b(z) = F(z)/(z - \nu)$ by partial fractions. The general Weil Hermitian identity [55, Section 1.2], applied to real-type transforms, therefore gives

$$\varepsilon_N S = \frac{2\nu d B^2}{2\nu + d} + R(\nu), \qquad J = B^2 \left(1 + \frac{d^2}{(2\nu + d)^2} \right) + R'(\nu).$$

Both sums converge absolutely for these finite Fourier transforms. At an off-line zero the summand is a square, not an absolute square; conjugate pairs make the total real. The analytic divided transform supplies the displayed value at coincidence. Substitution gives (38). Finally, $B = f + \frac{1}{2} F''(\nu) d + O(d^2)$, so the constant and linear terms in E_{loc} cancel. □

Computing the second form. The rank-two commutator, with $D = \mathrm{diag}(\alpha_j)$, is $[D, A] = g\mathbf{1}^T - \mathbf{1}g^T$. Its generator can be recovered from one column: $g_j = \alpha_j A_{j0}$, $g_0 = 0$. Since $\mathbf{1}^T b = 0$,

$$J = \varepsilon_N \|b\|^2 + (g^T b) r'(\nu), \qquad r(z) = \sum_j \frac{a_j}{z - \alpha_j}. \tag{39}$$

Indeed $(D - \nu I)Ab = \varepsilon_N a - \mathbf{1}(g^T b)$; multiply by $(D - \nu I)^{-1}$ and pair with b. Thus evaluation costs $O(N)$ per root once the generator and eigenstate are available. The full matrix is still needed to determine that eigenstate: the commutator does not fix the diagonal or the eigenvalue decay.

Why the correction helps, and its limit. Under RH, put $E_m = 2F(\gamma_m)^2$. Pairing remote zeros yields

$$\nu R' - R = 2\nu^3 \sum_{m \neq k} \frac{E_m}{(\gamma_m^2 - \nu^2)^2} \geq 0. \tag{40}$$

The contribution with $\gamma_m > H > \nu$ is at most $2\nu^3 E_{>H}/(H^2 - \nu^2)^2$. The bare estimate instead has a remote bound $\nu E_{>H}/(H^2 - \nu^2)$ before division by f^2. The correction cancels the leading behavior $R(\nu) \approx \nu M_2$ of a distant spectral distribution. An unconditional bound needs the absolute size of the signed-complex sum, and finite positive matrices do not supply that bound. The identities above are unconditional; smallness of their remainder uniformly in λ^2, N, k remains open. The denominator κ, local curvature, correct root matching and numerical assembly error must also be controlled.

Retained-data test, 10 September 2026. The first twenty converged stored roots in each of five configurations give Table 4. All 100 signs agree. The reported errors compare error magnitudes, not percentages of large digit depths. The source eigenvectors, root records and matrix ingredients are linked by their artifact identities; the postprocessing was repeated with 100 fewer decimal digits, giving identical exported 32-significant-digit prediction results. This is point arithmetic on common retained sources, not independent assembly or interval certification.

λ^2	N	source bits	maximum relative prediction error
13	120	3386	0.348670%
100	500	3386	0.000275%
1000	800	6708	0.000061%
1000	890	6708	0.000047%
1200	970	6708	0.000035%

Table 4: Unfitted corrected error estimates, first twenty converged roots per configuration. Reference ordinates enter scoring, not the predictor.

At $(\lambda^2, N) = (1000, 890)$, selected higher-root diagnostics reduce the error-estimate discrepancy from 46.10% to 0.00790% at $k = 500$. It is 0.03039% at $k = 575$, but rises to 1.41441% at $k = 600$ and 2.23735% at $k = 610$. Those last two fail a one-percent accuracy gate. These higher roots use reference-seeded Newton refinement of the retained transform, not independent discovery or a root-counting certificate; the selected tests do not prove every intervening index passes. The result advances the root-conversion channel. A formula predicting the eigenvalue and eigenvector from cutoff and resolution alone is still required.

8.5 Local multi-root cancellation and the remaining common-scale problem

Status (10 September 2026). The identities below are exact finite statements under their explicit hypotheses. The reported evaluations use point arithmetic, not outward-rounded numerical certificates. They improve the a posteriori transfer from a finite eigenstate to a root displacement; they do not supply the missing a priori law in $c = \lambda^2, N$ and precision. In particular, no fitted conversion constant or reference-root error enters the new predictor.

Retain the normalization and observables of (35). For a stencil $\mathcal{J}$ of $p+1$ distinct simple positive nonlattice finite roots ν_i, set

$$t_i = \nu_i^2, \qquad a_i^* = \frac{2t_i}{\kappa_i}, \qquad y_i = \frac{J_i - f_i^2 - \varepsilon_N h_i S_i}{\varepsilon_N \kappa_i}.$$

Assume $\varepsilon_N f_i \kappa_i \neq 0$ and correctly match each root to a distinct simple real ordinate γ_i. Put $d_i = \gamma_i - \nu_i$, $B_i = F(\gamma_i)/d_i$, continued by f_i, and define

$$L_i = \frac{2\nu_i d_i B_i^2}{2\nu_i + d_i}, \quad Q_i = B_i^2\left(1 + \frac{d_i^2}{(2\nu_i + d_i)^2}\right), \quad C_i = \frac{Q_i - f_i^2 - h_i L_i}{\varepsilon_N \kappa_i}.$$

Exclude every stencil pair from a *common* remote multiset, write $x_m = z_m^2$, $E_m = 2F(z_m)^2$ for positive-real-part representatives, including multiplicity, and set $M(t) = \sum_{\text{common}} (E_m/\varepsilon_N)/(x_m - t)$. Assume $\gamma_j^2 \neq t_i$ for $i \neq j$ and $x_m \neq t_i$ in the common remote set. Off-line summands retain complex squares and conjugate pairing. The finite Weil identities give exactly

$$y_i = M(t_i) + a_i^* M'(t_i) + C_i + \sum_{j \in \mathcal{J}\setminus\{i\}} \frac{E_j}{\varepsilon_N}\left(\frac{1}{\gamma_j^2 - t_i} + \frac{a_i^*}{(\gamma_j^2 - t_i)^2}\right). \tag{41}$$

The local pair contributes L_i to $\varepsilon_N S_i$ and Q_i to J_i. The respective remote-pair contributions are $\nu_i E_m/(x_m - t_i)$ and $E_m(x_m + t_i)/(x_m - t_i)^2$. Subtraction uses $1 - \nu_i h_i = \kappa_i$.

Proposition 12 (Polynomial-reproducing remote cancellation)**.** *Fix a target* $0 \in \mathcal{J}$*. Suppose weights* w_i *solve*

$$\sum_{i\in\mathcal{J}} w_i\big[P(t_i) + a_i^* P'(t_i)\big] = P(t_0) \quad (\deg P \le p). \tag{42}$$

Then the operator-derived predictor

$$\widehat{\delta}_p = \frac{\varepsilon_N}{f_0^2}\left(\nu_0 \sum_i w_i y_i - S_0\right) \tag{43}$$

has remote error kernel

$$K(x) = \sum_i w_i \left(\frac{1}{x-t_i} + \frac{a_i^*}{(x-t_i)^2}\right) - \frac{1}{x-t_0} = O(x^{-p-2}). \tag{44}$$

For $p = 0$ *this is the previous divided-form correction.*

Proof. Expand each rational term at infinity. Its coefficient of x^{-q-1} is $t_i^q + q a_i^* t_i^{q-1}$. Equation (42) makes this coefficient equal to t_0^q for $q = 0, \dots, p$. This proves the stated cancellation, without assuming that the remote transform is a polynomial. □

For completeness define

$$X = \sum_{j\in\mathcal{J}} \frac{E_j}{\varepsilon_N}\left\{\nu_0 \sum_{i\ne j} w_i \left[\frac{1}{\gamma_j^2 - t_i} + \frac{a_i^*}{(\gamma_j^2 - t_i)^2}\right] - \frac{\mathbf{1}_{j\ne 0}\nu_0}{\gamma_j^2 - t_0}\right\}.$$

Direct substitution in (41) proves the exact remainder

$$\widehat{\delta}_p - (\nu_0 - \gamma_0) = \frac{\varepsilon_N}{f_0^2}\left\{\nu_0 \sum_{\text{common}} \frac{E_m}{\varepsilon_N} K(x_m) + X + \nu_0 \sum_i w_i C_i\right\} + d_0 - \frac{L_0}{f_0^2}. \tag{45}$$

In particular, the cross-stencil and nonlinear local terms cannot be omitted from a proof merely because they are small in a numerical example.

An explicit finite-height bound. Let $\Delta_i = t_i - t_0$, $\max_i \nu_i < H < T$, and $T > \alpha_N$. Require every matched stencil ordinate to lie below H. Define

$$B_K(H) = \sum_i \frac{|w_i|}{(H^2 - t_0)^{p+1}}\left\{\frac{|\Delta_i|^{p+1}}{H^2 - t_i} + |a_i^*|\left[\frac{(p+1)|\Delta_i|^p}{H^2 - t_i} + \frac{|\Delta_i|^{p+1}}{(H^2 - t_i)^2}\right]\right\}.$$

Expanding $1/(x - t_i)$ to order p in Δ_i, with exact remainder $\Delta_i^{p+1}/[(x - t_0)^{p+1}(x - t_i)]$, and differentiating that remainder proves $|K(z^2)| \le B_K(H)$ when $|\Re z| \ge H$. Here $|z^2 - t_i| \ge H^2 - t_i$, including off-line zeros. Platt–Trudgian's finite verification permits $T = 3 \cdot 10^{12}$ [56]. For a complete pair-energy prefix E_H and an absolute complex-tail bound V_T, the uncomputed portion of $\sum_{\text{common}} E_m K(x_m)$ is bounded by

$$(\varepsilon_N - E_H + V_T) B_K(H) + V_T B_K(T). \tag{46}$$

Positivity is used only between H and T; the rest uses absolute variation. A valid conservative choice from the finite rational expansion is

$$V_T \le \frac{16}{L}\cosh^2(L/4) B_M(T)^2 \frac{\log T + 1}{T}, \qquad \mu_{2m} = \sum_j a_j \alpha_j^{2m},$$

$$B_M(T) = \sum_{m<M} \frac{|\mu_{2m}|}{T^{2m}} + \frac{\|a\|_1 (\alpha_N/T)^{2M}}{1 - (\alpha_N/T)^2}.$$

Here $M \geq 1$ is an integer and $\alpha_N = 2\pi N/L$. Corollary 1.2 of Hasanalizade–Shen–Wong [59] implies $\mathcal{N}(t) \leq t \log t$ for $t \geq T$. Integration by parts gives $\sum_{\Im\rho>T} (\Im\rho)^{-2} \leq 2(\log T + 1)/T$; the finite even rational expansion and $|\sin(Lz/2)| \leq \cosh(L/4)$ in the critical strip then give the displayed variation bound. No assumption of global RH is used.

If $|d_i| \leq r_i < 2\nu_i$ and $|F'''| \leq M_{3,i}$ on the corresponding real interval, put $b = M_{3,i}/(6|f_i|)$,

$$C = 2b + (|h_i|/2 + br_i)^2, \qquad G = \frac{|h_i|}{2\nu_i - r_i} + \frac{1}{(2\nu_i - r_i)^2},$$
$$H_i = C + G + (|h_i| + Cr_i)(|h_i| + Gr_i),$$
$$R_i = (1 + |h_i| r_i/2 + br_i^2)^2.$$

Taylor's theorem, with the first-order terms canceled, gives

$$|C_i| \leq \frac{f_i^2 r_i^2 H_i}{|\varepsilon_N \kappa_i|}, \qquad |d_i - L_i/f_i^2| \leq r_i^2 \left(|h_i| + Cr_i + \frac{R_i}{2\nu_i - r_i} \right).$$

Together these formulas give a fully specified conditional finite enclosure. Matching, certified coefficients and directed rounding are separate obligations. If the remainder is enclosed by $[A, B]$, the signed relative displacement error is obtained from $q/(\widehat{\delta}_p - q)$ at the endpoints only when $\widehat{\delta}_p \notin [A, B]$. An interval containing that pole cannot be reported as a finite relative-error enclosure; it must remain unassessed.

Conditioning and falsification. Simplicity alone does not imply $\kappa_i \neq 0$: the even polynomial $F(z) = z^2 - 4$ at $\nu = 2$ is an algebraic counterexample. Nor do nonzero kappas imply an invertible stencil: $F = (z^2 - 1)(z^2 - 4)$ gives $t_i + a_i^* = 5/2$ at both roots, making the $p = 1$ system singular. These are algebraic fixtures, not asserted CCM ground-state examples. A CCM-specific uniform conditioning theorem remains open.

The frozen primary choice was $p = 2$, using the target and the two nearest available computed roots by $|\nu_i - \nu_0|$, ties to lower ν_i. It passed the 1% displacement gate on 110 retained configuration/root pairs. Point substitution into the finite-height remainder covered all 90 pairs with complete energy/Taylor inputs; 20 others lacked those audit inputs. In the original sparse $c = 1000, N = 890$ pool, the signed magnitude errors at $k = 600, 610$ were -0.4876697% and $+0.5583456\%$, with respective point ranges $[-0.4876807, -0.4876566]\%$ and $[0.5583284, 0.5583617]\%$. A separately frozen acquisition of 11 additional indices gave a denser 41-root pool. All 41 primary predictions passed; the errors at $k = 595, 600, 610$ became $+0.0559910\%$, -0.00314985%, $+0.00447358\%$. The previous one-root correction fails at $k = 595$ as well, by -1.8918383%. The new indices use reference-seeded, full-precision Newton validation on the same retained eigenstate, not independent discovery or 41 independent assemblies. There are 121 distinct configuration/root pairs across the two stages, because 30 of the 41 dense-pool cases repeat earlier indices. The fresh dense cases are not included in the earlier 90-case remainder audit. The largest weight 1-norm in the dense primary rule is 4.408; at degree 4 it exceeds 1156. Higher order therefore requires a conditioning audit, even when its point score improves. Rounding the initial predictor inputs y_i, κ_i to 20 significant digits changes its value by at most $1.11 \cdot 10^{-20}$ relatively in 110 tests; this checks postprocessing sensitivity, not eigenstate accuracy. Global Padé and nonlinear rational alternatives failed near the edge and are retained as failures in the research return packet. The previously frozen shifted-erf eigenvalue model also fails the denser $c = 1200, N = 60$–150 window (1.166% RMS and 1.424% maximum depth error) and gives the wrong cutoff-derivative sign at $c = 100, N = 300$. These are dated evidence against that model, not a contradiction of an unrelated coarse value envelope or a new fit.

The exact polynomial window and the common factor. Let $\beta = 2\pi/L$, $x_j = \beta^2 j^2$, $w_0 = 1$, $w_j = \sqrt{2}$ for $j > 0$, $Q_N(x) = \prod_{j=0}^{N} (x - x_j)$ and $P_u(x) = Q_N(x) \sum_{j=0}^{N} w_j u_j/(x - x_j)$.

Then

$$F_u(z) = \frac{(-1)^N 2\pi}{\sqrt{L}\,\beta^{2N+1}} \frac{P_u(z^2)}{\Gamma(N+1+z/\beta)\Gamma(N+1-z/\beta)}. \tag{47}$$

This follows from the finite Gamma product and sine reflection identities, including removable singularities by reciprocal Gamma. Thus at most N positive nonlattice roots are movable polynomial roots; reciprocal Gamma supplies forced lattice zeros beyond the Fourier band. A consecutive prefix of k distinct movable positive roots therefore requires $k \le N$. An isolated root matched to a higher ordinal does not imply that every preceding root has been recovered. This is a counting restriction, not a matching theorem.

For integers $N \ge 1$, real $|\tau| < 1$, and $\Phi_N(s) = \Gamma(N+1)^2/[\Gamma(N+1+s)\Gamma(N+1-s)]$, positive-real Stirling remainders [58] prove

$$\left|\log \Phi_N(N\tau) + NV(\tau) + \tfrac{1}{2}\log(1-\tau^2)\right| \le \frac{1}{6N(1-\tau^2)},$$
$$V(\tau) = (1+\tau)\log(1+\tau) + (1-\tau)\log(1-\tau).$$

In particular $V = \tau^2 + \tau^4/6 + \cdots$; a Gaussian replacement is only central-window information, not a band-wide approximation. The exact extremal quotient uses

$$W_N(z) = \frac{2L}{\Gamma(N+1+z/\beta)^2\Gamma(N+1-z/\beta)^2},$$
$$v_0 = (N!)^{-4}, \qquad v_j = \frac{2}{[(N-j)!(N+j)!]^2} \quad (j>0).$$

These are the numerator weights on positive-real-part zero representatives and the denominator weights on x_j, respectively. Then the precise finite problem is

$$\varepsilon_N = \min_{0 \ne P \in \mathcal{P}_N} \frac{\sum_{\Re z>0} W_N(z) P(z^2)^2}{\sum_{j=0}^{N} v_j P(x_j)^2}.$$

Here the numerator includes multiplicities and complex squares, and defines the positive finite polynomial inner product inherited from the exact even matrix. For its reproducing kernel K_N, the elementary Gram-inverse and Christoffel principles [57] give

$$T_1 = \sum_j v_j K_N(x_j, x_j), \qquad T_2 = \sum_{i,j} v_i v_j K_N(x_i, x_j)^2.$$

The signed-complex numerator must be retained without global RH; finite positivity suffices for this finite-dimensional kernel identity. A uniform relative estimate of this arithmetic weighted kernel is still missing.

Residual and precision obligations. The divided-form algebra specifies errors that an implementation must retain. In full Fourier coordinates write $D = \operatorname{diag}(\alpha_j)$, $[D, A] = ge^T - eg^T$ with fixed boundary vector $e = (1, \ldots, 1)^T$, $b = (D - \nu I)^{-1}a$, $q = Aa - \varepsilon_N a$, $r = -e^T b = \sum_j a_j/(\nu - \alpha_j)$, $G = g^T b$, and $H = g^T(D-\nu I)^{-1}b$. Then $r' = -e^T(D-\nu I)^{-1}b$ and the displacement identity gives

$$b^T A b = \varepsilon_N \|b\|^2 + Gr' - rH + b^T(D-\nu I)^{-1}q.$$

The root-residual term $-rH$ is retained in the new dense postprocessing; the stored eigenvector alone does not certify the last term.

For normalized, sign-matched finite states with $\|u - \widetilde{u}\| \le \eta_u$, the Fourier integral and Cauchy–Schwarz give the real-axis bound

$$|F_u^{(m)} - F_{\widetilde{u}}^{(m)}| \le \frac{\eta_u L^{m+1/2}}{2^m\sqrt{2m+1}}.$$

If $F_u(\nu) = 0$ and $|F_u'| \geq m_0 > 0$ on $[\nu - r, \nu + r]$, $\eta_u\sqrt{L} < m_0 r$, and $\eta_u L^{3/2}/\sqrt{12} < m_0$, the perturbed transform has a unique zero in that interval, displaced by at most $\eta_u\sqrt{L}/m_0$. This is a conditional stability bound. Certified state error, root matching, derivative lower bounds, and stencil conditioning are still needed for an end-to-end precision guarantee.

The arithmetic innovation profile. The equivalent computational route uses the even displacement nodes j^2 and scalar Schur jets: for a pivot $\sigma_N(z)$ of $A_N - zI$,

$$I_N = -\sigma_N'(0)/\sigma_N(0), \qquad T_p = -p[z^p]\sum_{j=0}^{N} \log \sigma_j(z).$$

Fixed-order rank-two generator elimination costs $O(N^2)$ scalar operations. Its exact recurrence was checked on 1,032 rational identities, and an independent analytic assembly reproduced all 366 retained $c = 500$ prefix exports of σ, M, T_1, T_2 to their full 80 significant digits. This does not constitute a derived innovation asymptotic or a stability certificate. At fixed c, if the continuum bottom is positive, logarithmic diagonal growth instead implies $T_p(N) \sim N/(\log N)^p$ for each fixed integer $p \geq 1$; eventual fixed-order ground-state dominance is impossible. The present ramp and the unlimited late-N regime therefore require different uniformity statements.

The boundary-column-only closure was also tested without fitted constants on every $N = 320, \ldots, 1600$ at $c = 5, 9$. Only one of 1281 rows at each cutoff passed a 1% gate. The precise profile remains an open arithmetic problem; it cannot be replaced by that failed local boundary approximation. At a prime-power event $c = q = p^r$, the one-sided cutoff derivatives obey

$$\operatorname{jump} \frac{d[-\log_{10} \varepsilon_N]}{dL} = \frac{2\log p}{\sqrt{q}\log q}\,\frac{(w^T u)^2}{\varepsilon_N \log 10}, \qquad w = (1, \sqrt{2}, \ldots, \sqrt{2})^T.$$

Hellmann–Feynman gives this identity from the rank-one arithmetic jump. It constrains a uniform derivative law; it does not by itself falsify a coarse approximation to eigenvalue values.

8.6 An indexed minimum budget and a mode-resolution criterion

Write $C = \lambda^2$ in this subsection, reserving c for the historical argument coefficient of the scalar ramp model. The target is one law in (C, N, P, k) for $1 \leq k \leq 1000$, with the first-zero law as its $k = 1$ specialization. This is a research target, not a claim of validation over that complete range. A finite root must first be associated with the requested ordinal; an unresolved ordinal is an outcome, not a zero-digit observation.

The indexed minimum. For a specified finite branch $\nu_{N,C,k}$, its fixed-cutoff limit $\nu_{\infty,C,k}$, and the computed root $\widetilde{\nu}_{N,C,P,k}$, use the exact signed decomposition

$$\begin{aligned} e_{\mathrm{num},k} &= (\widetilde{\nu}_{N,C,P,k} - \nu_{N,C,k})/\gamma_k, \\ e_{\mathrm{mode},k} &= (\nu_{N,C,k} - \nu_{\infty,C,k})/\gamma_k, \\ e_{\mathrm{cut},k} &= (\nu_{\infty,C,k} - \gamma_k)/\gamma_k. \end{aligned} \tag{48}$$

Let $B_{j,k}$ be upper bounds on the absolute values of these three channels and $D_{j,k} = -\log_{10} B_{j,k}$. Then

$$D_k \geq -\log_{10}\left(\sum_j 10^{-D_{j,k}}\right) \geq \min\{D_{\mathrm{num},k}(C,N,P), D_{\mathrm{mode},k}(C,N), D_{\mathrm{cut},k}(C)\} - \log_{10} 3. \tag{49}$$

The first expression is the exact logarithm of the summed upper bounds, not a fitted smoothing of the minimum. Equality to a minimum up to a uniformly bounded offset requires two-sided

channel estimates and a quantitative noncancellation condition. Numerical assembly error belongs in $B_{\mathrm{num},k}$ unless it is separately bounded and counted.

Insufficient N enters through the *indexed mode budget* and through branch validity. A fourth "resolution budget" must not count the same mode error twice. The finite-parent identity in Section 8.8 separates eigenvalue, indexed transfer and remainder increments. If its limiting branches and uniform bounds are established, their sum supplies a derived inner minimum for the mode budget. Those estimates have not been proved here.

An exact dual representation of the mode channel. The following specialization of goal-oriented Galerkin duality [60] gives a precise object to estimate. At fixed C, let the real self-adjoint operator A_C have a normalized simple isolated ground state u, eigenvalue ε_∞, and next spectral point $\lambda_2 > \varepsilon_\infty$. Let P_N be the orthogonal Fourier projection and u_N an exact unit Ritz vector, in the operator domain, with eigenvalue $\varepsilon_N < \lambda_2$. Choose its sign so $s_N = \langle u, u_N\rangle > 0$. Define

$$r_N = (A_C - \varepsilon_N)u_N, \qquad P_N r_N = 0,$$
$$Q = I - |u\rangle\langle u|, \qquad R_N = Q(A_C|_{u^\perp} - \varepsilon_N)^{-1}Q.$$

Here $\|R_N\| \le (\lambda_2 - \varepsilon_N)^{-1}$. For real t, let $b(t)$ be the vector representing finite-interval Fourier-transform evaluation, $F_a(t) = \langle b(t), a\rangle$, and set $z_N(t) = R_N b(t)$. For complex t use the analytic coefficient extension and the bilinear transpose pairing. The apparent lattice poles in the coefficient formula for b are removable.

Proposition 13 (Indexed dual mode identity)**.** *Under these hypotheses,*

$$F_{u_N}(t) = s_N F_u(t) + \langle (I - P_N)z_N(t), r_N\rangle. \tag{50}$$

If $\nu_{N,C,k}$ and $\nu_{\infty,C,k}$ are associated real roots, then for some real ξ between them,

$$\nu_{N,C,k} - \nu_{\infty,C,k} = -\frac{\langle (I - P_N)z_N(\nu_{N,C,k}), r_N\rangle}{s_N F_u'(\xi)}, \tag{51}$$

provided the denominator is nonzero.

Proof. Write $u_N = s_N u + e$ with $e \perp u$. Projection of the residual onto $u^\perp$ gives $e = R_N r_N$. Apply the linear evaluation functional and use $P_N r_N = 0$ to prove (50). At the finite root this gives $F_u(\nu_{N,C,k})$; the real mean-value theorem between the associated roots proves (51). □

Proposition 14 (Sufficient mode-resolution condition)**.** *Let a disc of radius r about a finite simple real root $\nu_{N,C,k}$ contain exactly one zero of F_{u_N}, counting multiplicity, and no zero on its boundary. If*

$$\|r_N\| \sup_{|t-\nu_{N,C,k}|=r} \|(I - P_N)z_N(t)\| < \min_{|t-\nu_{N,C,k}|=r} |F_{u_N}(t)|, \tag{52}$$

then F_u also has exactly one zero in that disc. It is real and simple.

Proof. Apply Rouche's theorem to (50); the nonzero constant s_N does not change zeros. Reality under conjugation forces the unique zero in a disc centered on the real line to be real. □

If $|F_{u_N}''| \le M_2$ throughout the disc, a usable lower bound for the right side is $r|F_{u_N}'(\nu_{N,C,k})| - M_2 r^2/2$, when positive. This condition identifies a limiting-transform root, not its zeta ordinal or the cutoff error. Disjoint discs and complete root counts are needed for a prefix.

What would justify doubled convergence. Suppose $\|r_N\| \le a_N$, $\|(I-P_N)z_N(\nu_{N,C,k})\| \le b_{N,k}$, $s_N \ge s_* > 0$, $|F_u'| \ge m_{N,k} > 0$ on the established root interval, and $0 < \Gamma_k \le \gamma_k$. Then

$$B_{\mathrm{mode},k} \le \frac{a_N b_{N,k}}{s_* m_{N,k} \Gamma_k}. \tag{53}$$

In particular, conditional envelopes

$$a_N \le A(C) e^{-a(C)\Phi(N)}, \qquad \frac{b_{N,k}}{s_* m_{N,k} \Gamma_k} \le B(C,k) e^{-b(C,k)\Phi(N)}$$

give

$$D_{\mathrm{mode},k} \ge \frac{[a(C)+b(C,k)]\Phi(N) - \log[A(C)B(C,k)]}{\log 10}. \tag{54}$$

The exponent doubles only if the dual factor supplies the same decay rate with controlled constants and domain. Those CCM rates remain open. This identity does not promote a finite-ramp stretched exponent to an infinite-N coefficient asymptotic.

Minimum required basis size. For requested depth D_*, allocate each channel at most $10^{-D_*}/3$. A sufficient mode threshold is the first N in a proved envelope domain for which the branch condition above holds and

$$[a(C)+b(C,k)]\Phi(N) \ge \log[A(C)B(C,k)] + (D_* + \log_{10} 3)\log 10. \tag{55}$$

This defines a conditional $N_{\mathrm{res}}(C,k,D_*)$; the required functions are not yet an evaluated closed law. A monotone envelope would make it a threshold for all larger N, whereas an isolated measured success can result from signed cancellation.

The count, band and accuracy constraints are different. A consecutive prefix of K movable positive nonlattice roots requires $K \le N$ by the degree-N secular representation. Thus a movable first-1000 prefix requires $N \ge 1000$. Geometric inclusion below $2\pi N/\log C$ neither guarantees accuracy nor is necessary for an isolated accurate match: retained data include successful matches above that band. Neither count nor band placement supplies (55).

Precision also depends on the selected root. Writing $r_u(t) = u \cdot v_N(t)$, where $v_0(t) = 1/t$ and $v_j(t) = \sqrt{2}\,t/(t^2-\alpha_j^2)$, implicit differentiation at a simple nonlattice root gives

$$\mathrm{d}\nu[u](e) = -\frac{e \cdot v_N(\nu)}{r_u'(\nu)}, \qquad \chi_N(\nu) = \frac{\|v_N(\nu)\|}{|\nu r_u'(\nu)|}, \qquad \|v_N(\nu)\|^2 = \frac{1}{\nu^2} + 2\nu^2 \sum_{j=1}^{N} \frac{1}{(\nu^2-\alpha_j^2)^2}. \tag{56}$$

Because $u \cdot v_N(\nu) = 0$, the worst direction is tangent to the unit-state sphere. If a state error were bounded by $C_0 10^{-P}$, the local precision depth would be approximately $P - \log_{10} C_0 - \log_{10} \chi_N$; a finite error bound needs interval derivative and evaluation-vector bounds. Requested digits and matrix bits do not automatically provide that state-error premise. The point sensitivities in the tested high-index stencils have $\log_{10} \chi_N \simeq 672.6$–$676.2$ at $(C,N) = (1000, 890)$ and 734.0–745.8 at $(1200, 970)$. They describe arbitrary coefficient perturbations, not measured corruption or the structured mode error.

Status. The identities and implications above have explicit spectral and branch hypotheses. Twenty-eight exact rational self-adjoint fixture checks verify their finite algebra. Uniform arithmetic residual, dual-tail, derivative and cutoff bounds, with validation across independently qualified source families, remain proof and data targets. Section 8.8 now supplies a complete counted first-1000 test on one stored state.

8.7 Nonlinear local transfer and its exact interpolation defect

The linear moment filter can fail when a neighboring root is outside its local linear regime even though the target is accurate. Retaining the derived nonlinear terms yields an operator-only extension. It remains an a posteriori predictor; it does not replace the indexed budget in Section 8.6.

A division-free system. Retain the finite hypotheses of Section 8.5, but allow an individual κ_i to vanish. Define $Z_i = J_i - f_i^2 - \varepsilon_N h_i S_i$. For proposed displacements d_i, all obtained from the known finite transform, put

$$B_i(d_i) = F(\nu_i + d_i)/d_i, \qquad x_i(d_i) = (\nu_i + d_i)^2,$$
$$L_i(d_i) = \frac{2\nu_i d_i B_i(d_i)^2}{2\nu_i + d_i}, \qquad E_i(d_i) = 2d_i^2 B_i(d_i)^2,$$
$$A_i(d_i) = B_i(d_i)^2\left(1 + \frac{d_i^2}{(2\nu_i + d_i)^2}\right) - f_i^2 - h_i L_i(d_i).$$

Continue B_i by f_i at zero. Let D be the derivative matrix of polynomial interpolation at the $m = p + 1$ distinct nodes $t_i = \nu_i^2$. Eliminating the common polynomial values gives

$$U_i(d) = \frac{\varepsilon_N S_i - L_i(d_i)}{\nu_i} - \sum_{j\neq i} \frac{E_j(d_j)}{x_j(d_j) - t_i},$$
$$H_i(d) = Z_i - \kappa_i U_i(d) - 2t_i(\mathsf{D}U(d))_i - A_i(d_i) - \sum_{j\neq i} E_j(d_j)\left\{\frac{\kappa_i}{x_j(d_j) - t_i} + \frac{2t_i}{(x_j(d_j) - t_i)^2}\right\}. \tag{57}$$

The proposed model solves $H(d) = 0$, approximating only the common remote field by a polynomial of degree p. It retains the local and cross-stencil terms, with no fitted correction coefficient.

Set $\mathsf{K} = \operatorname{diag}(\kappa) + 2\operatorname{diag}(t)\mathsf{D}$. Since $L_i = f_i^2 d_i + O(d_i^2)$ and $A_i, E_i = O(d_i^2)$,

$$H(0) = Z - \mathsf{K}\operatorname{diag}(1/\nu)(\varepsilon_N S), \qquad J_H(0) = \mathsf{K}\operatorname{diag}(f_i^2/\nu_i).$$

If this matrix is nonsingular, the first Newton step gives exactly the division-free linear moment predictor: choose v_i so that $\sum_i v_i[\kappa_i P(t_i) + 2t_i P'(t_i)] = P(t_0)$ for $\deg P \le p$, then $\widehat{\delta} = (\nu_0 \sum_i v_i Z_i - \varepsilon_N S_0)/f_0^2$. Individual zero kappas therefore need not be singularities, although the whole system must be invertible. The current numerical source path uses archived nonzero-kappa y values to reconstruct Z; an actual zero-kappa source requires direct Z acquisition.

Proposition 15 (Exact nonlinear common-field defect)**.** *Let $\Omega(t) = \prod_j (t - t_j)$ and let d^{true} contain the truly matched local displacements. Under the exact finite Weil identities,*

$$H_i(d^{\mathrm{true}}) = 2t_i\Omega'(t_i) \sum_{\mathrm{common}} \frac{E_m}{(x_m - t_i)\Omega(x_m)}. \tag{58}$$

Complex energies retain their squares and conjugate pairing; global RH is not required by this identity.

Proof. For interpolation I_p at the nodes,

$$\frac{1}{x - t} - I_p\left[\frac{1}{x - t}\right] = \frac{\Omega(t)}{(x - t)\Omega(x)}.$$

Differentiate at $t = t_i$. At the true displacements, $U_i = \varepsilon_N M(t_i)$, and all retained local terms cancel from H_i except $2\varepsilon_N t_i[M'(t_i) - (I_p M)'(t_i)]$. Substitution proves the formula. Absolute convergence is as in the preceding finite Weil sums. □

For $|\Re z| \geq H_0 > \max_i \nu_i$, its kernel is bounded by

$$\mathcal{B}_i(H_0) = \frac{2t_i|\Omega'(t_i)|}{(H_0^2 - t_i)\prod_j (H_0^2 - t_j)}.$$

The unknown part is at most $(\varepsilon_N - E_{H_0} + V_T)\mathcal{B}_i(H_0) + V_T\mathcal{B}_i(T)$, under the same verified-height and complete-energy hypotheses as (46). The kernel decays as x^{-p-2}.

Residual is not displacement. For a scaled residual $G(x)$ on an infinity-norm box of radius R about x_0, take fixed invertible Y and suppose

$$q = \sup_x \|I - YJ_G(x)\|_\infty < 1, \qquad \|YG(x_0)\|_\infty + qR \leq R.$$

The map $x \mapsto x - YG(x)$ is then a contraction of the box into itself, so its model root $\widehat{x}$ is unique there. If a truly matched root x^{true} also lies in that box and $G(x^{\text{true}}) = \rho$, then

$$\|\widehat{x} - x^{\text{true}}\|_\infty \leq \|Y\rho\|_\infty/(1-q).$$

This follows by subtracting the fixed-point equations. A point Jacobian and a small Newton residual do not establish the box-wide hypothesis. Source error and Taylor truncation must also enter the error budget.

Measured continuation, with failures retained. The fixed primary model has $p = 2$, a quarter-nearest-spacing displacement domain at every stencil root, and Taylor order 16; order 24 is a numerical crosscheck. Source arithmetic uses 1700 decimal digits and postprocessing 120. All percentages in Table 5 refer to signed error in the predicted displacement $\delta = \nu - \gamma$, not to error in the zero itself. At $k = 700$ the solver finds no descending step within the local

(C, N)	k	Linear error	Nonlinear error	Outcome
$(1000, 890)$	610	-0.00745679%	-0.00745681%	within 1%
	616	-0.0270812%	-0.0266569%	within 1%
	618	-86.8459%	-2.22210%	fails 1%
	619	$+71.9606\%$	-0.0444023%	within 1%
	620	$+35.8510\%$	-1.04454%	fails 1%
$(1200, 970)$	650	-0.0000251935%	-0.0000251935%	qualified point pass
	675	$+2.56032\%$	-0.0248694%	qualified point pass
	700	-15.7757%	$+25.2621\%$	unqualified
	725	$+49.5347\%$	-10.7295%	unqualified

Table 5: Five exploratory revisits and four frozen new-index tests in an existing source family. The latter give two qualified passes and two unqualified targets, not four successes.

domain, and a neighbor seed has a different nearest-reference ordinal. At $k = 725$ the solver converges but a neighbor ordinal drifts. The two qualified targets have measured relative depths 23.8950 and 4.97824 digits. Reference-seeded acquisition and nearest-reference checks do not certify independent ordinal discovery.

The exact kernel also identifies the remaining nearby-field obstruction. At target 618, the observed fractional predictor error is -0.0222210; the known common sum propagated with a point Jacobian gives -0.0222233, of which the excluded pair 621 supplies -0.0219676. The point linearized unknown-tail width is about $1.03 \cdot 10^{-10}$. The seed for 621 had instead reached the same finite root as 622; adding that duplicate does not represent the important omitted pair. A diagnostic 64-subinterval scan in its reference-midpoint window finds no sign-changing rational-secular bracket. This finite scan does not prove absence of a root.

Thirty-five exact rational/complex fixture checks verify the nonlinear algebra. Twenty-seven direct evaluations of the retained transform at predicted points give maximum relative discrepancies from order 24 of $3.26 \cdot 10^{-41}$ and $7.08 \cdot 10^{-24}$ in the two families. These checks, the 28 dual-identity checks, and all unresolved outcomes are recorded with input and script hashes in the research checkpoint. They establish neither outward-rounded source certificates nor a uniform convergence law for $1 \le k \le 1000$.

8.8 A complete thousand-root test and a proved mode breaker

The continuation of 10–11 September 2026 uses two authenticated retained even states at $C = 250$, $N = 500, 1500$, with source precision 5047 bits. They were produced by Toolkit v0.13.2; they are not new v0.15.0 eigenpair solves. Source manifest, payload and transport hashes were checked. The following interval statements concern the exact stored decimal vectors. They do not bound their distance to the exact assembled or infinite CCM eigenstates. Roots were seeded from 18 significant reference digits and refined with Arb at 1650 decimal digits. The known reference table was used for scoring. A subsequent independent Arb check at 1300 decimal working digits encloses all 1000 reference values; see Section 8.9. This does not validate all 2500 stored reference digits.

A counted first-1000 prefix. At $N = 1500$, all 1000 requested roots have disjoint positive interval-Newton enclosures inside their respective reference-midpoint windows. Their relative depths are listed in Table 6. An exact polynomial count additionally rules out extra interleaved roots. In $x = (z/\beta)^2$, $\beta = 2\pi/\log C$, define

$$P_N(x) = a_0 \prod_{j=1}^{N} (x - j^2) + 2x \sum_{j=1}^{N} a_j \prod_{\substack{1 \le \ell \le N \\ \ell \ne j}} (x - \ell^2). \tag{59}$$

Here a_j are full even-vector coefficients, $a_{-j} = a_j$. Clearing their common power-of-ten denominator gives an integer polynomial. For any rational $A < B$ at which P_N is nonzero,

$$H_{A,B}(y) = (1+y)^N P_N\left(\frac{A + By}{1+y}\right)$$

has as many positive roots as P_N has roots in (A, B), counting multiplicity. Descartes' sign rule gives an exact upper count from the coefficient sign variations of $H_{A,B}$.

For $N = 1500$, $A = 0$, $B = 1555864$, that upper count is exactly 1000. The 1000 disjoint enclosures below $\beta\sqrt{B} = 1419.422489383493\ldots$ give the matching lower count. Also $P_N(0) \ne 0$, and this threshold is below $\beta(N+1)$, the first fixed reciprocal-Gamma carrier zero. Thus these are exactly the first 1000 positive zeros of the stored transform, all simple. This is a finite stored-state ordinal certificate; it is not an infinite-operator theorem.

The fixed local predictor across the complete prefix. The degree-two operator moment rule uses each target and its two nearest computed roots, with lower ordinate breaking a distance tie. No stencil is selected using the reference error. For the same scaled moment matrix as in the preceding derivation, its weights satisfy $B^T v = e_0$, and

$$\widehat{\delta}_k = \frac{\varepsilon}{f_k^2}\left(\nu_k \sum_{i \in I_k} v_i \frac{Z_i}{\varepsilon} - S_k\right), \qquad \delta_k = \nu_k - \gamma_k.$$

All 1000 signed predictions pass the fixed one-percent displacement-error gate. The largest $|\widehat{\delta}_k/\delta_k - 1|$ is $1.6911472 \cdot 10^{-7}$, at $k = 833$. This is about 0.00001692% of the displacement,

k	D_k, $N = 500$	D_k, $N = 1500$	D_k, $N = 1500$, 500-digit coefficients
1	667.3144	1175.3987	499.1423
19	628.6777	1138.4084	481.4055
100	508.8755	1034.6344	430.3573
250	283.2394	884.9406	355.5659
500	unmatched	674.2352	250.6994
750	unmatched	487.4204	157.9075
1000	unmatched	305.8987	66.6005

Table 6: Retained-state relative depths. "Unmatched" denotes failure of the requested movable-root association, not a zero-digit measurement. The final column isolates decimal coefficient rounding, not matrix assembly.

not that percentage of the ordinate. The archimedean source chain was authenticated and the commutator generator reconstructed without loading the dense matrix. The calculation retains the finite root-residual term in the rank-two identity. Outward postprocessing of retained point inputs does not certify their assembly error. The predictor is a posteriori: it uses the computed eigenstate, eigenvalue and neighboring roots. Its success does not close an a priori formula in C, N, P, k.

A genuine insufficient-mode example. At the same C and source bits, the $N = 500$ state yields 454 requested window matches in the 1000-target scan; the contiguous matched movable prefix ends at 362. Exact sign variations below $x = 304874$ give 362 polynomial roots, matching the disjoint enclosures. More strongly, the enlarged interval $305915 < x < 307123$, which contains the entire reference window for $k = 363$, has zero sign variations after the above transformation. There is *no movable polynomial root* in that window. This certifies the failure independently of the Newton acquisition path.

The same fixed degree-two operator rule, using the nearest two roots within the counted 362-root movable prefix, passes all 362 targets. Its worst relative displacement error is 0.00752314 (0.752314%), at $k = 362$. Thus the two complete prefix tests contain 1362 successful configuration/root pairs. The missing $k = 363$ branch is retained as a resolution failure, not supplied with an extrapolated error score.

The fixed carrier zeros must be distinguished from these movable roots. There are 52 such carrier zeros below the $N = 500$ prefix threshold, and the carrier root with lattice index 554 lies in the enlarged $k = 363$ window. The complete transform therefore has a different ordinal count from its movable polynomial roots in this regime. In contrast, no carrier root lies below the counted first-1000 prefix at $N = 1500$. The condition $\beta(N + 1) > \gamma_K$ excludes carrier zeros through the reference height. Equivalently, $N \geq \lfloor \gamma_K \log C/(2\pi) \rfloor$ suffices for that particular geometric exclusion. It does not guarantee movable-root existence or accuracy. Isolated accurate movable matches above the band remain possible: at $N = 500$, $k = 350$ has 18.7161 digits and $\nu/(\beta N) = 1.07521$.

An accuracy threshold gives a stricter measured prefix. At $N = 500$, the movable prefixes meeting 32, 100 and 300 relative digits have lengths 341, 320 and 240. At $N = 1500$, all 1000 meet each of these thresholds. These are finite observations; they neither locate a sharp minimum N between 500 and 1500 nor prove monotonicity in N.

A separately tested precision channel. Round the retained even coefficients independently to P_d significant decimal digits, without changing C, N or recomputing the matrix. For $r_a(\nu) = 0$, the numerical root shift has first variation

$$\Delta\nu_{\mathrm{num}} = -\frac{\Delta r_a(\nu)}{r_a'(\nu)}.$$

The associated componentwise relative condition is

$$\chi_{\mathrm{cw}}(\nu) = \frac{|a_0/\nu| + \sum_{j=1}^{N} |2a_j\nu/(\nu^2 - \alpha_j^2)|}{|\nu r_a'(\nu)|}. \tag{60}$$

Decimal nearest rounding has componentwise error at most $5 \cdot 10^{-P_d}|a_j|$, giving the local budget $P_d - \log_{10}(5\chi_{\mathrm{cw}})$. A finite bound additionally needs a derivative/branch enclosure. The measured $\log_{10}\chi_{\mathrm{cw}}$ grows from 0.85338 at $k = 1$ to 433.39080 at $k = 1000$.

At eight fixed indices and $P_d = 500, 900, 1300$, all 24 rounded-state roots were enclosed independently. The signed first variation, added to the measured base displacement, predicts total displacement with worst relative error below $6.39 \cdot 10^{-64}$. Combining it instead with the operator prediction above, without using the measured base displacement, gives 24 of 24 one-percent passes and worst relative error $4.602 \cdot 10^{-9}$. At $k = 1000$, 500-digit rounding yields 66.6005 relative digits; 900-digit rounding already retains the 305.8987-digit base-state result. These experiments support a root-dependent precision budget but do not identify requested toolkit precision with a proved coefficient-error bound.

A falsified height extrapolation. The historical first-fifty-root fit at $(C, N) = (250, 1500)$ has rounded coefficients $a_0 = 1178.828$, $a_2 = 5.0432 \cdot 10^{-5}$ in $2\log_{10}|\xi'(\rho_k)| + 2\log_{10}\gamma_k + a_0 + a_2\gamma_k^2$. Keeping those coefficients fixed over all 1000 roots gives maximum digit error 24.5486, attained by overprediction at $k = 1000$, and RMS error 7.7863 digits. Coefficient rounding can contribute at most about 0.00151 digits there, far below this failure.

Retaining only the higher-order Gamma-carrier correction does not repair the extrapolation. With $s = \gamma_k/\beta$ and $\Phi_N(s) = \Gamma(N+1)^2/[\Gamma(N+1+s)\Gamma(N+1-s)]$, adding $2[\log\Phi_N(s) + \psi_1(N+1)s^2]/\log 10$ to the historical fit increases the maximum digit error to 127.5348. This is a failed candidate, not a contradiction of the exact Gamma factorization: the extremal polynomial contributes additional height dependence. It rules out this specific correction as the complete indexed law.

A derived refinement of the mode budget. Write the operator-predicted displacement as $\widehat{\delta}_{N,k} = \varepsilon_N \mathcal{T}_{N,k}$, where

$$\mathcal{T}_{N,k} = \frac{\nu_{N,k}\sum_{i\in I_k} v_i Z_i/\varepsilon_N - S_{N,k}}{f_{N,k}^2}, \qquad \mathcal{R}_{N,k} = \delta_{N,k} - \varepsilon_N \mathcal{T}_{N,k}.$$

For associated roots at two basis sizes $N < M$, the exact finite increment is

$$\begin{aligned}\nu_{N,k} - \nu_{M,k} = {} & (\varepsilon_N - \varepsilon_M)\mathcal{T}_{M,k} \\ & + \varepsilon_N(\mathcal{T}_{N,k} - \mathcal{T}_{M,k}) + (\mathcal{R}_{N,k} - \mathcal{R}_{M,k}).\end{aligned} \tag{61}$$

This separates change in the eigenvalue with the parent transfer held fixed, change in the indexed transfer, and the predictor remainder. It is an algebraic identity, not a unique causal attribution: reversing the order of the two changes produces another valid decomposition. The three displayed terms sum to the finite increment and therefore do not duplicate a fourth copy of the same mode error.

The transfer factor itself varies strongly with N and k. At $C = 250$, write

$$\ell_k = \log_{10}|\mathcal{T}_{500,k}/\mathcal{T}_{1500,k}|.$$

The computed values are

k	1	19	100	250	350
ℓ_k	0.05565	1.70202	17.73021	93.67247	269.55472

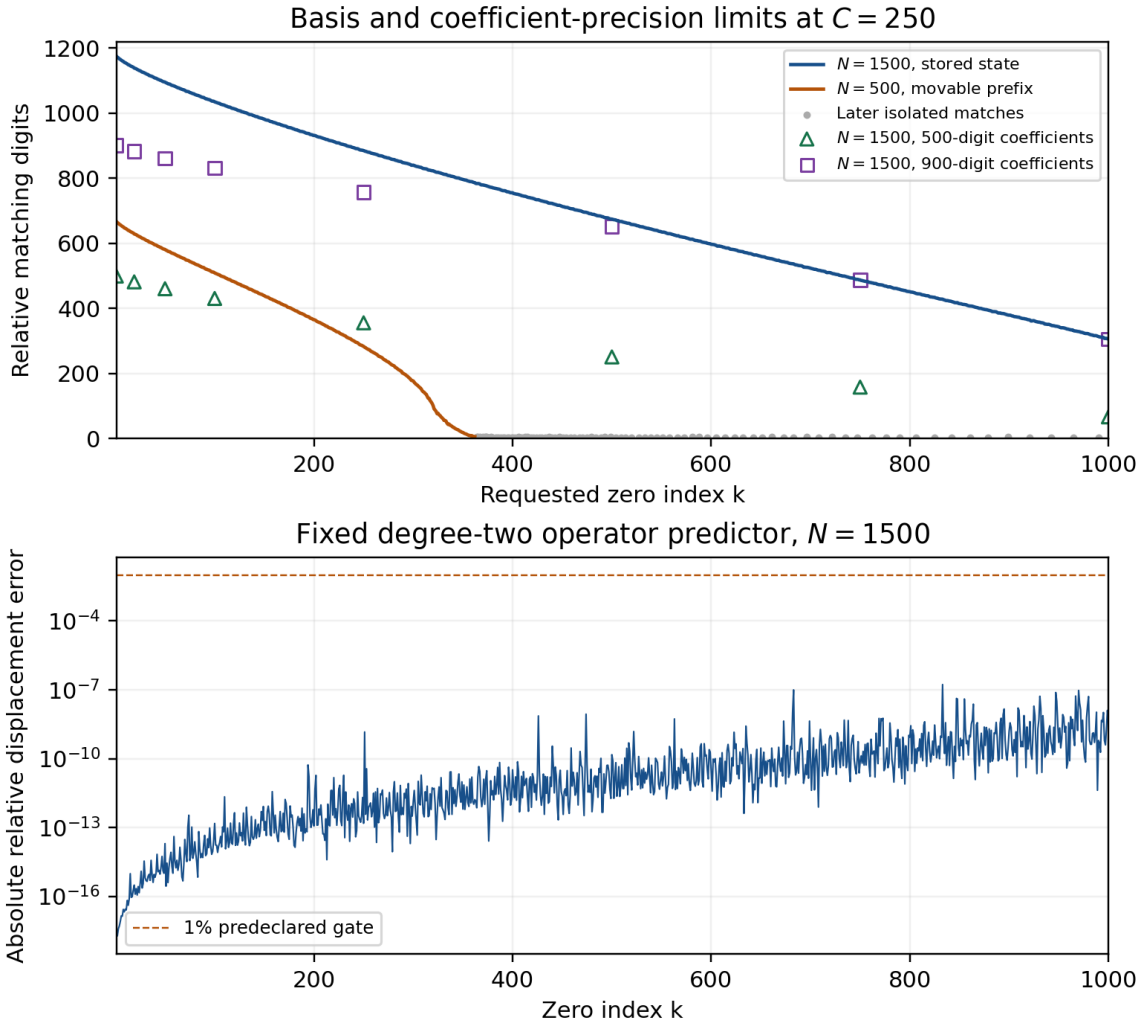


Figure 4: The same cutoff with different basis sizes and controlled coefficient precision, followed by the fixed operator-predictor error on the counted first-1000 prefix. Later isolated $N = 500$ matches do not extend its qualified movable prefix. All data concern retained point sources.

In the parent-anchored split, the eigenvalue-change term supplies 87.97% of the increment at $k = 1$, but only $1.8612 \cdot 10^{-18}$ of it at $k = 100$. A model that varies the eigenvalue while holding this parent transfer factor fixed misses most of the high-index change. These finite data rule out treating the transfer factor as a function of C, k alone over these two basis sizes.

If fixed-cutoff limits and branch correspondence are established, passing $M \to \infty$ in (61) supplies an actual mode decomposition. Relative upper bounds B_E, B_T, B_R on its three terms then give

$$D_{\mathrm{mode},k} \geq -\log_{10}(B_E + B_T + B_R) \geq \min\{D_E, D_T, D_R\} - \log_{10} 3.$$

This is a precise possible source of the additional indexed minimum: the transfer term can limit accuracy even when the eigenvalue budget is large. Its uniform CCM rate has not been derived. The finite comparison uses independently assembled retained sources and supplies no bound on the infinite remainder or assembly difference.

Remaining analytic task. The indexed minimum in (49), combined with a valid branch gate, survives these tests. A closed law still requires uniform CCM estimates for the primal residual, indexed dual tail, root derivative, arithmetic cutoff contribution and numerical state error. The exact count supplies a practical finite branch test, and the operator moment rule supplies a precise correction target. Neither replaces those missing estimates.

8.9 Global transfer germs and finite positive bounds

The successful operator predictor still uses an indexed root derivative. The following continuation addresses its global shape and gives a bounded finite reconstruction. It does not yet replace the state by a closed C, N, P, k rate. The tests reuse the complete prefixes from Section 8.8; they are not new independent data.

8.9.1 Keep the carrier exact

Put $L = \log C$, $\beta = 2\pi/L$, $t = z^2$, and $\Xi(z) = \xi(1/2 + iz)$. The exact polynomial representation gives

$$\frac{F(z)}{F(0)} = \Phi_N(z/\beta)\frac{P_N((z/\beta)^2)}{P_N(0)}, \qquad \Phi_N(s) = \frac{\Gamma(N+1)^2}{\Gamma(N+1+s)\Gamma(N+1-s)}.$$

Let g_m be the origin coefficients of $\log(F(\sqrt{t})/F(0)) - \log(\Xi(\sqrt{t})/\Xi(0))$. Removing the carrier gives polynomial-quotient coefficients

$$h_m = g_m + \frac{\zeta(2m, N+1)}{m\beta^{2m}}, \qquad m \geq 1. \tag{62}$$

All coefficients are obtained from the retained Fourier state by Taylor arithmetic, with no fit to root errors. The quotient has distinct poles and zeros when the two root sets differ. Even exponentially small displacements do not enlarge the true Taylor convergence radius.

At $C = 250, N = 1500$, the 64-term raw germ gives maximum squared-slope digit error $1.6432 \cdot 10^{-11}$ across all 1000 targets. At $N = 500$ it fails severely above the carrier band. Keeping Φ_N exact and using a [31/32] Padé approximant to $\sum_{m\geq 1} h_m t^{m-1}$ gives 1000/1000 and 362/362 passes for the combined displacement predictor. The largest relative errors are respectively $1.69115 \cdot 10^{-7}$ and 0.784842%. In the lower-mode case some operator and slope errors cancel; the slope-only digit error can reach 0.005299. Accordingly, the combined score is not an independent one-percent slope bound. Reference heights and ξ' remain inputs to this shape test.

8.9.2 A positive finite counting measure

Write the movable roots in physical coordinates as $0 < \nu_1 < \cdots < \nu_N$. Suppose $\nu_i \geq \gamma_i$ for $i \leq N$, choose $K < N \leq J$, and assume $\nu_K < \gamma_{K+1}$. Define

$$H_{K,J}(t) = \sum_{i=K+1}^{N} \log(1 - t/\nu_i^2) - \sum_{j=K+1}^{J} \log(1 - t/\gamma_j^2),$$
$$\Delta(x) = \#\{K < j \leq J : \gamma_j^2 \leq x\} - \#\{K < i \leq N : \nu_i^2 \leq x\}.$$

The ordering implies $\Delta \geq 0$. Set $s_0 = \gamma_{K+1}^2$ and $v = t/s_0$.

Proposition 16 (Finite Stieltjes enclosure)**.** *For $0 < v < 1$,*

$$S(v) := \frac{H_{K,J}(s_0 v)}{v} = \int_0^1 \frac{d\sigma(w)}{1 - vw}, \qquad d\sigma(w) = \Delta(s_0/w)\, dw \geq 0.$$

Its moments are

$$\mu_\ell = \frac{1}{\ell + 1} \left\{ \sum_{j=K+1}^{J} (s_0/\gamma_j^2)^{\ell+1} - \sum_{i=K+1}^{N} (s_0/\nu_i^2)^{\ell+1} \right\}.$$

Let G_n be its n-node Gaussian quadrature rational function, and $\widetilde{G}_n$ the corresponding function for $(1 - w)d\sigma(w)$. If the moment systems are nondegenerate, then

$$G_n(v) \leq S(v) \leq \frac{\mu_0 - v\widetilde{G}_n(v)}{1 - v}. \tag{63}$$

Proof. Integration by parts gives $H_{K,J}(t) = \int \Delta(x) t/[x(x - t)]\, dx$; the boundary term at infinity vanishes. Substitute $w = s_0/x$ to obtain the measure and its moments. The Hermite interpolation remainder for $w \mapsto (1 - vw)^{-1}$ at the Gaussian nodes is nonnegative, since the $2n$-th derivative is positive and the node polynomial is squared. Integrating proves $G_n \leq S$. For the modified measure, $\widetilde{S}(v) = \mu_0/v - (1 - v)S(v)/v$. Its Gaussian lower bound gives the upper inequality in (63). □

This is the classical Gaussian/right-endpoint-Radau mechanism, with its sign and normalization made explicit. It uses $2n + 1$ original moments. Deflation is essential: without removing the first K displaced pairs, the small early displacements still place support near γ_1^2. Finite Hankel positivity alone does not justify evaluation through that support. Deflation here uses measured roots and therefore supplies an enclosure of a reconstruction, not an independent prediction of them.

For completeness, if π_n is monic orthogonal for σ, $q_n(v) = v^n \pi_n(1/v)$ and $h_n^{\mathrm{orth}} = \int \pi_n^2 d\sigma$, the orthogonality identity gives

$$S - G_n = \frac{v^{2n}}{q_n(v)^2} \int \frac{\pi_n(w)^2}{1 - vw}\, d\sigma(w).$$

It lies between $v^{2n} h_n^{\mathrm{orth}}/q_n(v)^2$ and that quantity divided by $1 - v$. The loss as $v \to 1$ is explicit; deriving the associated index K from C, N, P remains a separate problem.

8.9.3 Full xi tail without global RH

The finite reference product is extended to the full Ξ by its analytic Taylor coefficients. Write positive-height complex zeros as $z_r = \gamma_r + i\eta_r$, $|\eta_r| < 1/2$. The even product uses z_r^2, not $|z_r|^2$. After the first J certified zeros, its tail coefficients are

$$c_m = -[t^m] \log \frac{\Xi(\sqrt{t})}{\Xi(0)} - \frac{1}{m} \sum_{j=1}^{J} \gamma_j^{-2m}.$$

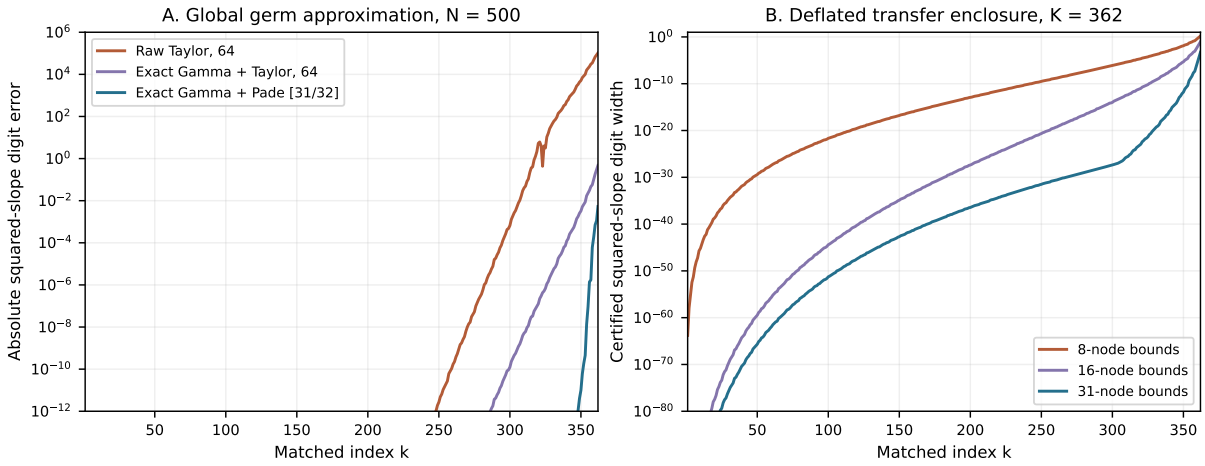


Figure 5: At $C = 250, N = 500$, treating the Gamma carrier exactly repairs the raw global-germ failure (left). Finite positive-measure bounds tighten with quadrature order and widen near the unresolved edge (right). The right panel uses measured deflation; it is not a forward prediction. Squared-slope digit error and interval width are different quantities.

For $2\pi e < T < \gamma_{J+1}$ and $0 \le t < T^2$, the remainder after order M satisfies

$$|R_M(t)| \le \frac{t^{M+1} B_{2M+2}(T)}{(M+1)(1-t/T^2)}, \tag{64}$$
$$B_p(T) = \frac{T^{1-p}}{2\pi}\left\{\frac{\log(T/(2\pi))}{p-1} + \frac{1}{(p-1)^2}\right\} + 0.28T^{-p}\left(2\log T + \frac{1}{p}\right).$$

Indeed, $|z_r| \ge \gamma_r$ bounds each complex logarithmic remainder. The Lehman estimate stated in [61, Lemma 1], with its permitted constant 0.28, bounds the resulting sum of inverse powers. Multiplicity is included. This argument makes no global RH assumption.

At $C = 250, N = 500$, all 500 polynomial roots are certified positive, real and simple. Independent Arb reference enclosures at 1300 decimal working digits certify their strict ordering above the corresponding zeta ordinates, and contain all 1000 stored reference values at that precision. This does not validate all 2500 digits in the reference file. All 1362 previously reported displacement intervals also contain the differences formed with these independent reference enclosures. For $K = 362, J = 1000$, 64 moments and $n = 31$, all 362 direct finite-product checks lie strictly inside (63). Adding the order-32 tail with $T = 1419.4$ gives a maximum tail remainder $3.78881 \cdot 10^{-24}$. The worst full logarithmic width, at $k = 362$, is 0.000618556. Its multiplicative spread in a squared-slope contribution is at most 0.123788%. This bounds that contribution only, with measured deflation and the stored-state qualifications retained.

8.9.4 Input-dependent resolvent quantities

The analytic route back to C, N is explicit, although its uniform rate has not been derived. Partition the exact even-sector matrix at the zero-frequency coordinate as

$$A(C,N) = \begin{pmatrix} a & c^T \\ c & B \end{pmatrix}, \qquad B \ge bI, \quad b > 0.$$

For a simple $0 < \varepsilon < b$, with nonzero central eigenvector component,

$$\frac{u_{\text{tail}}}{u_0} = -(B-\varepsilon I)^{-1}c = -w(\varepsilon), \qquad \varepsilon = a - c^T(B-\varepsilon I)^{-1}c.$$

The full even coefficients obey $a_j/a_0 = -w_j/\sqrt{2}$. Consequently

$$\frac{F(z)}{F(0)} = \operatorname{sinc}(zL/2)\left[1 + \sqrt{2}\sum_{m\geq 1} z^{2m}\sum_{j=1}^{N}\frac{w_j(\varepsilon)}{\alpha_j^{2m}}\right]. \tag{65}$$

The series is an origin identity within the frequency-pole radius. It derives the origin germ from weighted inverse-block moments, rather than supplying an already closed convergence formula.

Set $w_0 = B^{-1}c$, $\eta = \|w_0\|^2$, $s = a - c^T w_0$ and $U = s/(1+\eta)$. If $s > 0$ and $U < b$, the Schur equation has one root in $(0, U]$, and

$$\frac{s}{1+\eta+U\eta/(b-U)} \leq \varepsilon \leq U, \qquad \|w(\varepsilon) - w_0\| \leq \frac{U\sqrt{\eta}}{b-U}. \tag{66}$$

To see this, expand the resolvent exactly: $s = \varepsilon(1+\eta) + \varepsilon^2 w_0^T (B - \varepsilon I)^{-1} w_0$. The last quadratic form lies in $[0, \eta/(b-\varepsilon)]$. This proves the upper bound and, after substituting $\varepsilon \leq U$ in the remainder, the lower bound. Strict decrease of the Schur function gives uniqueness, and the vector bound is the resolvent identity.

Uniform CCM estimates for b, s, w_0 and its weighted moments are now a specific input-only target. Source assembly error and cancellation in s must be controlled before these become numerical certificates. The local numerator, branch gate and cutoff comparison also remain necessary. The indexed minimum over numerical, mode/transfer and cutoff budgets is retained; no fitted minimum term or universal resolution threshold is claimed from these finite results.

8.10 A verified finite input route and an indexed dual reduction

The next computation strengthens the insufficient-mode example to the ground state of the finite analytic matrix. At $C = 250, N = 500$, the matrix is assembled from digamma and trigamma formulas with finite endpoints and prime powers, with Arb rounding and explicit geometric tail bounds. No retained eigenstate or zeta ordinate enters this matrix/eigenpair construction. The interval implementation encloses 1830 values from an independent high-precision assembly control. These checks validate the implementation; the analytic identities and remainder bounds remain its mathematical basis.

Enclosed scalar branch. At a real shift z, positive pivots of $B - zI$ certify its positivity. The central Schur jet yields

$$\sigma(z) = a - z - c^T(B - zI)^{-1}c, \quad -\sigma'(z) = 1 + \eta(z), \quad b(z) = \frac{1}{\operatorname{tr}(B - zI)^{-1}} \leq \lambda_{\min}(B - zI).$$

Here $\eta(z) = \|(B - zI)^{-1}c\|^2$. If $0 < a_* < b(z)$, then throughout $|t| \leq a_*$ the magnitude of the Schur derivative lies between

$$M_- = 1 + \frac{\eta(z)}{(1 + a_*/b(z))^2}, \qquad M_+ = 1 + \frac{\eta(z)}{(1 - a_*/b(z))^2}.$$

Verifying $a_* M_- > |\sigma(z)|$ gives opposite endpoint signs and the enclosure $z + \sigma(z)/[M_-, M_+]$ for its unique root. The positive noncentral block identifies this root as the lowest finite eigenvalue.

The direct interval elimination fails to prove positive pivots at 1000 and 3000 working digits, and succeeds at 5000. Seven scalar refinements give relative eigenvalue radius below $7.47 \cdot 10^{-1415}$. Direct interval vector back-substitution remains too wide. Instead, a point central-normalized vector $v = (1, v_{\text{tail}})$ is checked by recomputing

$$r_v = (B - \varepsilon I)v_{\text{tail}} + c$$

in interval arithmetic. A verified $g \leq \lambda_{\min}(B - \varepsilon I)$ gives $\|v_* - v\| \leq \|r_v\|/g =: E$. If $E < \|v\|$, the unit-vector error is at most $2E/(\|v\| - E)$. The resulting bound is $2.271 \cdot 10^{-1424}$. The large guard precision is a feature of this certification algorithm, not a necessary physical mode or precision budget.

A finite-operator mode breaker. Five hundred disjoint positive interval-Newton root enclosures, together with a nonzero leading coefficient of the degree-500 numerator, exhaust its movable roots for the enclosed ground vector. Previous roots are warm seeds only. The first 362 movable roots match their corresponding zeta midpoint windows, and the $k = 363$ window contains no movable root. Every one of the 362 earlier displacement intervals contains the new bound. Thus this breaker now concerns the independently enclosed finite analytic operator. Carrier roots are still counted separately. The later regularization in Section 8.21 supplies an analytic certificate for the $C = 250, N = 1500$ thousand-root ground source.

The historical 5047-bit unit vector differs from the new finite ground state by approximately $1.36603 \cdot 10^{-858}$; its eigenvalue has relative difference $6.51756 \cdot 10^{-849}$. These are finite source errors consistent with the retained root measurements, not a storage-corruption finding.

A failed larger-cutoff extrapolation. The fixed global-germ rule from Section 8.9 does not give uniform higher-cutoff edge accuracy. The following counts use a one-percent *inverse-squared-slope* gate, rather than the earlier combined-predictor gate:

(C, N)	64 coefficients, [31/32]	128 coefficients, [63/64]
$(1000, 890)$	33/58	34/58
$(1200, 970)$	4/10	3/10

These are all eligible previously acquired points, not complete certified prefixes or blind data. Mismatched and unsuccessful acquisitions remain recorded. The higher-order system is first unresolved at 1100 working digits; all targets are scored at 1700 digits and the model failures remain. More coefficients do not repair it uniformly, and some misses occur at roots that still have several accurate digits.

Two global vectors determine the indexed dual. Write $B_\varepsilon = B - \varepsilon I$, $D = \operatorname{diag}(1^2, \dots, N^2)$, and the inherited displacement identity as

$$DB_\varepsilon - B_\varepsilon D = rw^T - wr^T, \qquad w_j = \sqrt{2}, \quad r_j = \sqrt{2}\, j b_j.$$

Set $X = B_\varepsilon^{-1}$, $p = Xr$, $q = Xw$, and $R_t = (D - tI)^{-1}$ away from the finite frequency poles. Then

$$XR_t w = R_t\{q(1 + p^T R_t w) - p(q^T R_t w)\}. \tag{67}$$

Indeed, $DX - XD = qp^T - pq^T$ and $XR_t - R_t X = R_t(DX - XD)R_t$. Applying this identity to w proves the formula. Once p, q are known, each indexed dual needs two Cauchy sums and linear work in N, rather than another dense inverse solve. Numerical cancellation in this identity still requires control.

For $t = (z/\beta)^2$, put $K(z) = 2z \sin(zL/2)/(\sqrt{L}\,\beta^2)$. The tail transform functional is $-K(z)R_t w$. Thus the exact central-normalized finite transform obeys

$$F_*(z) - F_v(z) = K(z)(XR_t w)^T r_v. \tag{68}$$

This follows directly from $(v_* - v)_{\text{tail}} = -Xr_v$. Both the inverse-displacement identity and this sign convention pass 156 exact rational component checks. A derivative lower bound and a valid root neighborhood turn (68) into an indexed displacement bound. The specific norm $\|XR_t w\|$ can be much smaller than the generic inverse norm. Its uniform dependence on C, N, k, together with the structured residual's dependence on P, is a concrete remaining target for the convergence law. The finite computation and exact reduction do not yet establish that rate.

8.11 An exact root-specific dual and the precision budget

The inverse response in (67) simplifies further at a movable transform root. This separates the conditioning of a zero from the conditioning of the eigenvector as a whole.

Proposition 17 (Root-specific inverse reduction)**.** *Use the notation of Section 8.10, and write the central-normalized ground vector as* $(1, v)$*. Suppose* B_ε *is invertible, so that* $B_\varepsilon v = -D^{-1}r$*. At a nonzero movable nonlattice transform root* ν*, set* $t = (\nu/\beta)^2$ *and* $H(t) = r^T R_t v$*. Then* $H(t) \neq 0$ *and*

$$(B - \varepsilon I)^{-1} R_t w = -\frac{DR_t v}{tH(t)}. \tag{69}$$

Proof. The root equation is $tw^T R_t v = 1$. The displacement identity and $DR_t = I + tR_t$ give

$$B_\varepsilon DR_t v = R_t r\,(tw^T R_t v - 1) - tR_t w\,(r^T R_t v).$$

The first term vanishes. If $H(t) = 0$, invertibility of B_ε and DR_t would force $v = 0$, contradicting the root equation. Multiplication by B_ε^{-1} proves the assertion. □

Thus the complete dual field at a root needs only a weighted ground-vector profile and one scalar Cauchy sum. The result does not assert that the matrix inverse has small norm. For a candidate vector with residual $e = B_\varepsilon v + D^{-1}r$, the exact off-root version, whenever its denominator is nonzero, is

$$XR_t w = \frac{\psi XR_t r + XDR_t e - DR_t v}{tH(t)}, \qquad \psi = tw^T R_t v - 1, \quad X = B_\varepsilon^{-1}. \tag{70}$$

The residual term remains even at a root of the candidate transform. If $B_\varepsilon \succeq gI > 0$, its contribution has norm at most $\|DR_t e\|/(g|tH(t)|)$. Authenticating stored coefficients does not justify dropping this term.

A finite precision channel. At a simple true root, the local relative residual condition is

$$\chi_k = \frac{|K(\nu_k)|\,\|DR_t v\|}{|\nu_k F'(\nu_k)|\,|tH(t)|}. \tag{71}$$

This concerns a Euclidean residual for a central-normalized vector. For a finite error bound, let x be a candidate transform root and let $I = [x - a, x + a]$ have $|F'| \geq m > 0$ throughout. Equation (68) implies that, if

$$E = \frac{|K(x)|\,\|XR_{(x/\beta)^2} w\|\,\|e\|}{m} < a,$$

then F has exactly one root in I, at distance at most E from x. Indeed, F is strictly monotone and has opposite endpoint signs. An indexed reference window and the preceding root count must also be qualified; this local statement alone does not assign an ordinal.

If an implementation separately establishes $\|e\| \leq R(C, N)2^{-P}$, and χ_I denotes the preceding finite bound divided by the reference ordinate and by $\|e\|$, then

$$D_{\text{num}} \geq P \log_{10} 2 - \log_{10}\big(\chi_I R(C, N)\big). \tag{72}$$

Combining a qualified physical-error bound with this numerical bound gives the minimum of the two depths minus $\log_{10} 2$. Resolving physical error into mode and cutoff channels gives the three-channel minimum minus $\log_{10} 3$. These are sufficient bounds; they are not equality laws in the presence of cancellation. An absent branch is not assigned an invented zero-digit result. Uniform bounds for the slope, H, the weighted profile and $R(C, N)$ remain necessary for a closed C, N, P, k law.

Independent finite verification. At $C = 250, N = 500$, point structured solves for $p = Xr, q = Xw$ were certified by interval residuals and the positive noncentral gap. Their norm-error bounds are $1.294 \cdot 10^{-1087}$ and $4.631 \cdot 10^{-1093}$, respectively. All 500 dual norms and condition enclosures computed from (69) agree with that independent two-field route. The generic comparison replaces $\|XR_t w\|$ by $\|R_t w\|/g$ with the same normalization and derivative:

k	$\log_{10} \chi_k$	decimal gain over generic bound
1	1.626068	660.023099
19	38.681006	642.296202
100	157.461113	583.402455
250	382.353647	471.255218
350	646.208731	338.173011
362	658.531742	331.647159

The inverse-field implementation passes 324 exact rational component checks; the root reduction and its residual remainder pass a separate 486 checks. The certified field construction took 28.0 seconds locally, and direct root-specific evaluation took 5.8 seconds. These are recorded run times, not portable performance guarantees.

The historical vector error does not spoil the zeros. The retained 5047-bit vector has noncentral residual norm at most $6.390 \cdot 10^{-1518}$ against the enclosed analytic matrix. Evaluating the signed identity (68) at its roots, with an enclosed derivative of the true transform, produces 500 disjoint interval-Newton inclusions for the true finite roots. Every inclusion overlaps the earlier finite-root certificate. All historical-to-analytic root shifts are now resolved with sign, including the first 127 shifts that the direct component-box comparison could not resolve.

The relative motion is approximately $10^{-1516.284}$ at the first root and $10^{-859.183}$ at index 362, despite unit-vector error near 10^{-858}. Across all 362 matched zeta windows, the fractional change in the reported zeta displacement is below $1.073 \cdot 10^{-849}$. This quantifies the historical data's stability; it does not certify the larger $N = 1500$ source or an infinite-operator convergence rate.

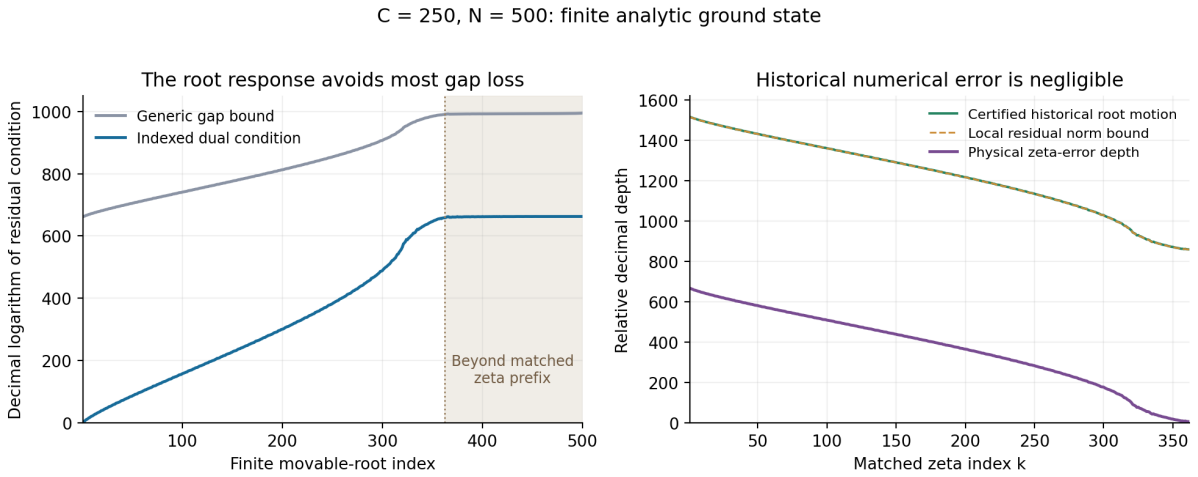


Figure 6: Finite indexed conditioning and historical source error at $C = 250, N = 500$. Left: the root-specific dual avoids hundreds of decimal digits of the generic gap penalty. Shading marks movable indices beyond the 362-root matched zeta prefix. Right: the certified historical root motion and the local residual norm estimate are far smaller than the physical zeta error. Higher depth means smaller error.

8.12 A mode ladder, edge failures, and a symmetric root pencil

The next experiment fixes $C = 250$ and independently assembles the finite matrix before computing states at $N = 750, 1000, 1250, 1500$. It therefore tests mode resolution without varying the cutoff. Matrix coefficients use 2500 decimal digits with interval remainder bounds, and the point Schur iteration uses 2200 digits with a fixed relative-correction stop. The first 1000 reference windows are all requested. Reference ordinates enter root acquisition and scoring, but not matrix assembly or the eigenvalue computation. Every failed acquisition and every isolated match beyond the complete prefix remains recorded.

Counted point-state results. Exact integer-polynomial sign variations bound the number of roots below each prefix endpoint; disjoint interval-Newton enclosures supply the matching lower count. These counts concern the newly serialized decimal state. The first failed window contains no movable polynomial root at $N = 750$ or 1000, as proved by a separate exact count. Carrier zeros are counted separately. The following scores use only the complete matched prefix and the previously fixed nearest-three-root predictor.

N	Complete prefix	Within 1%	Worst relative prediction error
750	578	571/578	2.43679
1000	797	792/797	0.707276
1250	1000	1000/1000	$1.91481 \cdot 10^{-5}$
1500	1000	1000/1000	$1.69115 \cdot 10^{-7}$

The $N = 1500$ run independently reproduces the earlier retained-state maximum prediction error. At $N = 1250$, the first and thousandth zeros have respectively 1091.30879 and 70.32801 relative decimal digits. At $N = 1500$ they have 1175.3987 and 305.8987. Thus the extra 250 modes gain about 84.09 digits at the first zero but 235.57 at the thousandth, while $-\log_{10} \varepsilon_N$ gains only 84.09. A common eigenvalue scale with an N-independent height correction cannot explain this ladder. The mode and index dependence must interact.

Existence does not ensure predictor accuracy. At $N = 750$, the first failed prediction is $k = 572$, although its zero has 6.59871 relative digits and lies in its proper window. The linear predictor has the wrong sign. A nonlinear local solve improves the relative error to -1.95771%, still outside the gate. At $N = 1000$, the first failed prediction is $k = 793$; the nonlinear error is $+3.17251\%$. Taylor orders 16 and 24 give the same conclusions. Other failures include wrong signs and two unsuccessful local-domain solves. Convergence of the model solver is not convergence to the true displacement.

An exploratory, predeclared degree ladder $p = 2, 3, 4, 6, 8$ uses the same polynomial-reproduction identity, with no fitted coefficients. For nearest-root stencils, degree eight reduces the worst error to $2.842 \cdot 10^{-12}$ at $N = 1250$ and $1.196 \cdot 10^{-20}$ at $N = 1500$. Every target in those two prefixes passes at every tested degree. It does not repair the lower-mode edges: degree eight passes only 568/578 and 791/797 targets at $N = 750, 1000$. Predecessor-only stencils also leave failures. These exploratory successes do not replace the original primary rule or establish a uniform remainder estimate.

Finite source qualification. A backward-error construction now certifies positivity of each noncentral block. Exact rounded final generators and positive pivots define a surrogate $\widehat{B} = L \operatorname{diag}(p) L^T > 0$. Structured interval recurrences enclose its generators, diagonal and inverse trace. For the actual analytic block at point shift z,

$$E = \max_i \sum_j |\widehat{B}_{ij} - (B - zI)_{ij}|, \qquad g = 1/\overline{\operatorname{tr}(\widehat{B}^{-1})} - E > 0$$

proves $B - zI \succeq gI$. This follows from the inverse-trace bound and the symmetric row-norm perturbation bound. The exact construction, including perturbed-generator fixtures, passes 1413 rational component checks jointly with the pencil identities below. An initial 3200-digit verification at $N = 1500$ is too wide and is retained as a failure; 4500 verification digits succeed without changing its point state.

The independently enclosed full residual, divided by the vector norm, is η. If its eigenvalue inclusion about z lies below $z + g$, principal-block interlacing identifies a unique simple ground eigenvalue. The new ladder has certified relative eigenvalue radii below $5.04 \cdot 10^{-166}$, $1.81 \cdot 10^{-172}$, $5.31 \cdot 10^{-178}$ and $2.45 \cdot 10^{-182}$, respectively. The generic unit-vector error bounds range from $2.60 \cdot 10^{-176}$ to $1.03 \cdot 10^{-193}$. They do not by themselves transfer every tiny point-state displacement to the exact ground transform; indexed dual or local root enclosures remain necessary. Increasing the $N = 1000$ point arithmetic from 2200 to 2600 digits changes its eigenvalue by less than $7.13 \cdot 10^{-1217}$ relatively, but leaves the shared finite Newton stopping error. Arithmetic stability is a separate check.

Proposition 18 (Symmetric pencil for movable squared frequencies)**.** *Write the finite even matrix as* $A = \left(\begin{smallmatrix} a & c^T \\ c & B \end{smallmatrix}\right)$*, with* $D = \operatorname{diag}(j^2)$*,* $c = D^{-1}r$ *and* $[D, B] = rw^T - wr^T$*. Let its exact ground vector be central-normalized as* $(1, v)$ *with eigenvalue* ε*, and suppose* $B_\varepsilon = B - \varepsilon I > 0$ *and* $1 + w^T v \neq 0$*. Define*

$$H = B_\varepsilon - cw^T - wc^T + (a - \varepsilon)ww^T, \tag{73}$$

$$T = B_\varepsilon D - wr^T = DB_\varepsilon - rw^T. \tag{74}$$

Then $H > 0$*,* T *is symmetric, and their generalized eigenvalues are the movable squared-frequency roots* $t = (\nu/\beta)^2$*, counted with multiplicity. Hence all such squared roots are real. They are positive if the odd parity block satisfies* $A_{\mathrm{odd}} - \varepsilon I > 0$*.*

Proof. Set $U = I + vw^T$. The eigen-equations give $B_\varepsilon v = -c$ and $v^T B_\varepsilon v = a - \varepsilon$, so $H = U^T B_\varepsilon U > 0$. Moreover $HU^{-1}D = B_\varepsilon D - wr^T = T$, and the displacement identity makes T symmetric. The determinant lemma gives

$$\det(D - tU) = \det(D - tI)[1 - tw^T(D - tI)^{-1}v].$$

This is the movable polynomial up to a nonzero scalar, including the lattice by polynomial continuation. The matrix $U^{-1}D$ is similar to the real symmetric $H^{-1/2}TH^{-1/2}$. Finally, the CCM parity formulas give $T = D^{1/2}(A_{\mathrm{odd}} - \varepsilon I)D^{1/2}$, proving the last assertion. □

The pencil depends explicitly on the finite matrix and its ground eigenvalue; its vector cancels. Under the stated positivity hypotheses, its ordered roots satisfy the exact min–max formula

$$t_k = \min_{\dim V = k} \max_{0 \neq x \in V} \frac{x^T T x}{x^T H x}.$$

This supplies a route to parameter-dependent resolution estimates. Evaluating the full pencil is still a spectral computation, not the desired input-only law. The exceptional case $1 + w^T v = 0$, root multiplicity, and matching to zeta windows require separate treatment. For approximate states, residual corrections cannot be omitted.

The requested minimum budget must therefore be accompanied by an indexed admissibility condition controlling the target branch, its neighbors, and the transfer remainder. A missing branch is not a zero-digit prediction. Uniform estimates in C, N, k, together with a verified implementation residual as a function of P, remain the missing steps toward a full convergence law for indices 1–1000.

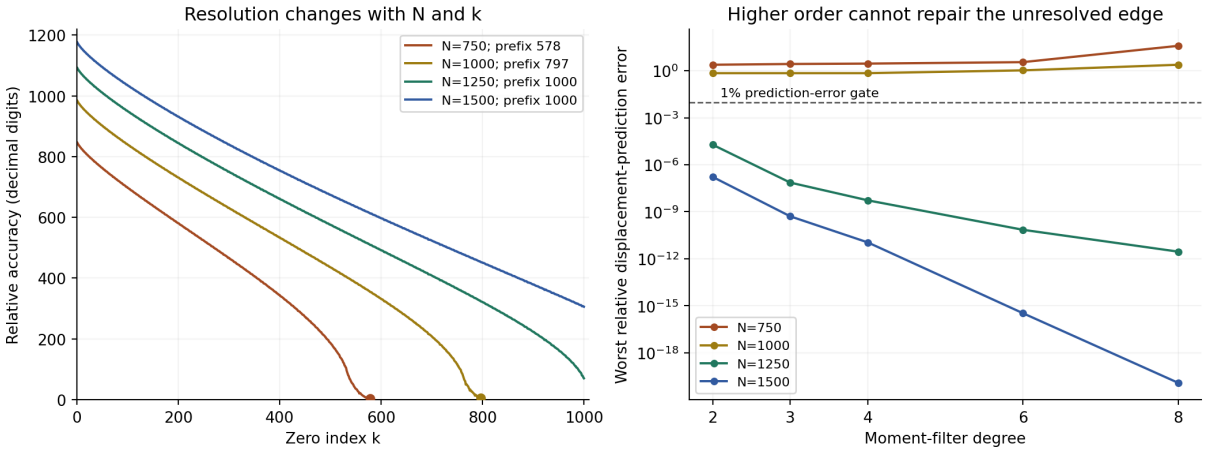


Figure 7: The fixed-$C = 250$ mode ladder. Left: all counted matched-prefix point-state depths, with no continuation through absent windows. Right: the full-prefix worst prediction error at each predeclared filter degree. Higher order sharply improves the two well-resolved thousand-root runs but leaves the lower-mode edge failures. Finite ground-eigenvalue certificates and point-state root certificates have distinct scopes.

8.13 A zero-shift pencil and two indexed error channels

Real-root localization through a self-adjoint quotient is already established in Connes–van Suijlekom, Theorem 5.6 and Proposition 5.10 [2]. The following parity-squared pencil is an explicit finite form useful for indexed error analysis; it does not claim a new localization theorem. It separates two contributions that a uniform convergence law must estimate.

Proposition 19 (The arbitrary-shift virtual transform)**.** *Use the notation of Proposition 18. For a real shift z with $B_z = B - zI$ invertible, define*

$$v_z = -B_z^{-1}c, \quad q_z = B_z^{-1}w, \quad \sigma_z = a - z + c^T v_z, \quad W_z = 1 + w^T v_z.$$

Let $H_z = B_z - cw^T - wc^T + (a-z)ww^T$ and $T_z = B_z D - wr^T$. For $R_t = (D - tI)^{-1}$,

$$\begin{aligned}\det(T_z - tH_z) = \det(B_z)\det(D - tI) \\ \cdot \{W_z[1 - tw^T R_t v_z] - t\sigma_z w^T R_t q_z\}.\end{aligned} \tag{75}$$

If $W_z \neq 0$, its roots are the movable squared-frequency roots of the transform with central-normalized tail $\widetilde{v}_z = v_z + (\sigma_z/W_z)q_z$.

Proof. Apply the two-by-two determinant lemma to $B_z^{-1}(T_z - tH_z)$, written as $D - tI$ plus two rank-one matrices, and use $c = D^{-1}r$ and $[D, B] = rw^T - wr^T$. Factoring out W_z gives $1 - tw^T R_t \widetilde{v}_z$. Polynomial continuation includes the diagonal nodes. The movable-polynomial identity then proves the claim. □

The virtual transform is generally not a ground eigenstate. At the exact ground shift $z = \varepsilon$, $\sigma_z = 0$, recovering the ground transform. At $z = 0$ the pencil requires only the finite input matrix. Its evaluation remains a spectral computation, rather than a closed convergence formula. The arbitrary-shift determinant and congruence identities pass 1044 exact rational checks, including nonzero Schur complements.

Fixed-shift mode resolution. With the rectangular injection $\mathcal{J} = (-w, I)^T$, one has $H_z = \mathcal{J}^T(A_{\text{even}} - zI)\mathcal{J}$ and $T_z = D^{1/2}(A_{\text{odd}} - zI)D^{1/2}$. The even trial functions vanish at the

endpoints, and differentiation maps them to the odd trial space. At fixed C, z, increasing N therefore gives principal matrix pairs. Where the metric is positive, min–max yields

$$t_k^{(N+1)}(z) \le t_k^{(N)}(z) \le t_{k+1}^{(N+1)}(z).$$

The ground shift changes with N, so this does not automatically prove monotonicity for ground-transform roots. Nor does it identify the limiting Ritz spectrum with all zeta zeros: a global space and completeness argument are still required. Since the movable polynomial has degree at most N, $N \ge k$ is necessary to resolve a complete prefix of k movable roots. It is not sufficient; the preceding mode ladder shows substantial additional resolution requirements. Carrier zeros cannot supply missing movable branches.

Exact shift sensitivity and the minimum budget. For a simple branch with eigenvector x,

$$\frac{dt}{dz} = \frac{x^T[t(I + ww^T) - D]x}{x^T H_z x}.$$

At a nonlattice ground root, take $x = R_t v$ and put $\Psi(t) = 1 - tw^T R_t v$ and $h(t) = r^T R_t v$. Then

$$x^T[t(I + ww^T) - D]x = 1/t - v^T R_t v, \qquad x^T H_\varepsilon x = h(t)\Psi'(t)/t.$$

Equivalently, the previous divided-vector observables give $\nu'(\varepsilon) = -S/(J(\nu) - \varepsilon\|b_\nu\|^2)$. The full residual term is necessary for approximate states. This endpoint derivative is not the integrated shift and is not the displacement from a reference zero. Along a continuously matched simple branch, the exact decomposition is

$$\nu_k(\varepsilon) - \gamma_k = \underbrace{\nu_k(0) - \gamma_k}_{\text{finite zero-shift error}} + \underbrace{\int_0^\varepsilon \nu_k'(z)\, dz}_{\text{energy-shift error}}.$$

If both relative channel errors are positive and have depths D_R, D_E, their total depth is exactly

$$D = \min(D_R, D_E) - \log_{10}(1 + 10^{-|D_R - D_E|}).$$

For bounds on the magnitudes of the relative channel errors, adding an arithmetic channel 10^{-D_P} instead gives

$$D_{\text{total}} \ge -\log_{10}(10^{-D_R} + 10^{-D_E} + 10^{-D_P}) \ge \min(D_R, D_E, D_P) - \log_{10} 3.$$

Thus a minimum budget has an exact error-sum basis, conditional on quantitative channel bounds and indexed admissibility. It cannot assign a meaningful convergence depth to an unresolved branch. The earlier indexed residual estimate supplies a conditional P-dependent arithmetic channel. Uniform C, N, k estimates for the other channels, their slopes and branch resolution remain open.

A certified thousand-root zero-shift prefix. At $C = 250, N = 1500$, residual-aware scalar identities certify the virtual tail of the analytic zero-shift pencil. In particular, for point solves v_p, q_p and residuals $e = Bv_p + c$, $e_q = Bq_p - w$,

$$\sigma = a + c^T v_p + v_p^T e - e^T B^{-1} e, \qquad W = 1 + w^T v_p - q_p^T e + e_q^T B^{-1} e.$$

The previously certified lower bound $B \succeq gI > 0$ bounds the quadratic remainders. These identities and the indexed response identities pass 135 additional exact rational checks. The first certificate encloses and counts all 1000 roots but cannot resolve displacement signs at indices 1–25. Increasing only point-solve precision leaves that limitation unchanged. The analytic

assembly, requested at 2500 decimal digits, has some interval radii of order 10^{-2320}; requested precision is not a certified error bound.

Reassembling the same analytic matrix at 3000 decimal digits gives 6004 refined generator enclosures contained in their earlier bounds. The resulting norm error bound for the saved virtual coefficients is $1.087 \cdot 10^{-1321}$. Interval-Newton root inclusions and an interval polynomial Descartes upper count now certify exactly 1000 distinct positive movable roots below $t = 1555864$, one in each of the first 1000 reference windows. All 1000 displacements are strictly positive. For indices $1, 19, 100, 250, 500, 750, 1000$, their relative depths are, respectively,

$$1180.20046,\ 1141.72691,\ 1036.95900,\ 886.65473,\ 675.44086,\ 488.28489,\ 306.46160.$$

This certifies the finite analytic zero-shift pencil, not a global limit or every corresponding analytic ground-transform displacement. The initial unresolved signs and all precision refinements are retained. This limitation concerns verification widths in the independent analytic assembly.

The remaining fraction depends strongly on the index. Comparing the saved point transforms, the ratio $(\nu_k(0) - \gamma_k)/(\nu_k(\varepsilon) - \gamma_k)$ at $N = 1500$ rises strictly at all 999 successive steps, from $1.57863 \cdot 10^{-5}$ to 0.27359962. At $N = 500$, the fixed sparse probe gives 0.00012010 at $k = 1$ and 0.87358620 at $k = 362$. The point comparisons therefore show that discarding the zero-shift channel becomes increasingly inaccurate near the resolution edge. The positive zero-shift certificate above does not silently upgrade these two-source comparisons to analytic ground-root certificates.

A predeclared exploratory closure tests the simpler candidate

$$\widehat{\delta}_k = \frac{\varepsilon \nu_k'(\varepsilon)}{2A_k - 1}, \qquad A_k = \frac{F_0'(\nu_k(\varepsilon))}{F_\varepsilon'(\nu_k(\varepsilon))},$$

with both transforms central-normalized. The closure is unproved; the exact pencil derivative does not prove it. No measured reference displacement enters its construction, and no constants are fitted. At $N = 1500$ it passes the correct-sign and 1% gate for all 1000 targets, with worst relative error 0.00132420 (about 0.13242%). At $N = 500$ it passes only 302/362 targets: every index 303–362 fails, and the maximum error is 7.94672%. It is thus falsified as a uniform replacement for the stronger neighbor-based predictor. The failed closure remains useful evidence about the missing boundary response, not a valid uniform convergence formula.

8.14 The monic pencil polynomial and a cutoff discriminator

The zero-shift pencil supplies a monic extremal problem that is different from the ground-state Rayleigh quotient and the bordering evaluation constraint. In the squared coordinate $t = (\nu/\beta)^2$, write

$$Q_N(t) = \prod_{j=1}^{N} (t - j^2), \qquad P_a(t) = a_0 Q_N(t) + \sum_{j=1}^{N} w_j v_j \frac{t Q_N(t)}{t - j^2}, \qquad w_j = \sqrt{2}.$$

This polynomial map from the even coefficient vector $a = (a_0, v)$ is invertible. Its leading coefficient is $\ell^T a$, where $\ell = (1, w)$. Thus degree at most $N-1$ is exactly the endpoint-vanishing space $a = \mathcal{J}x$ used above.

Proposition 20 (Monic characterization of the zero-shift pencil)**.** *Suppose the finite even matrix A is positive definite. In the induced polynomial inner product $\langle P_a, P_b \rangle = a^T A b$, the degree-$N$ monic orthogonal polynomial has coefficient vector*

$$a_{\mathrm{OP}} = \frac{A^{-1}\ell}{\ell^T A^{-1} \ell}.$$

Where the central-normalized virtual vector in Proposition 19 exists at $z = 0$, its polynomial is a nonzero multiple of this monic polynomial.

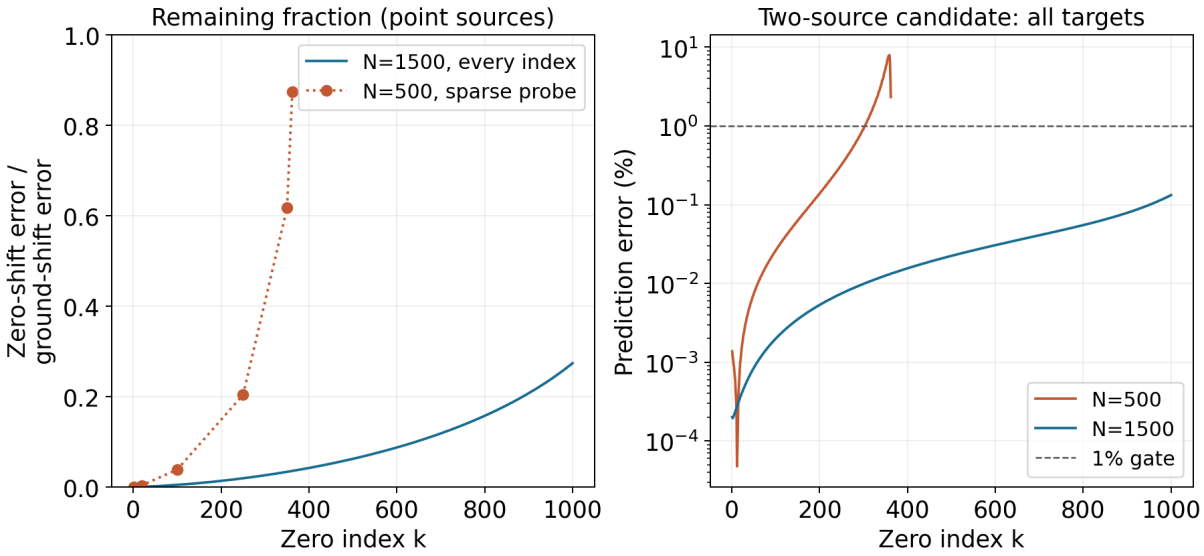


Figure 8: Two contributions and an unsuccessful simplification, at $C = 250$. Left: the remaining zero-shift fraction of the point ground error, including the complete thousand-root comparison and the fixed sparse lower-mode probe. Right: all targets of the unfitted two-source candidate, including its lower-mode failures. Analytic zero-shift root certification and point ground comparisons have distinct scopes.

Proof. The leading-coefficient constraint is $\ell^T a = 1$. Strict positivity makes its quadratic minimizer unique, with Euler equation $Aa = \eta\ell$. Every lower-degree polynomial has coefficient vector in $\ker \ell^T$, proving orthogonality. For the zero-shift variables, $Bv = -c$, $Bq = w$ and $c^T q = 1 - W$ give exactly

$$A\begin{pmatrix} 1 \\ v + (\sigma/W)q \end{pmatrix} = \frac{\sigma}{W}\ell.$$

After leading-coefficient normalization this is the same minimizer. □

The expression through $A^{-1}\ell$ also avoids dividing by a possibly vanishing central coordinate. This is a finite algebraic statement; it does not require RH. It gives the correct monic object for studying the zero-shift channel, without reinterpreting the ground polynomial as an ordinary monic minimizer.

The varying measure and its limitation. The Gamma-transform identity gives, under a positive real atomic Weil representation, the explicit measure

$$d\mu_{C,N}(t) = \sum_m \frac{2L\, m_m\, \delta_{(\gamma_m/\beta)^2}(dt)}{\Gamma(N + 1 + \gamma_m/\beta)^2 \Gamma(N + 1 - \gamma_m/\beta)^2},$$

where m_m is the multiplicity and reciprocal Gamma factors are interpreted by entire continuation. It represents the polynomial inner product above. Without the positive real hypothesis, the full signed-complex Weil functional remains the moment source; finite matrix positivity alone does not identify it with this positive zeta-atomic measure. Moments through degree $2N$ converge, which is all that is needed here. No all-degree finite-moment assumption is made.

In the degree-$(N-1)$ space the pair H_0, T_0 is precisely the moment pair for multiplication by t. Its eigenvalues are therefore the zeros of the degree-N orthogonal polynomial. With distinct ordered positive-weight atoms t_k, min–max gives $\theta_k \geq t_k$; infinite nondegenerate support gives strictness. Coincident carrier zeros can remove weights and must not be silently counted. Repeated ordinates contribute weights rather than independent scalar coordinates. Thus comparison with a zero list counted with multiplicity needs its own matching convention.

Factoring $\pi_N(t) = (t - \theta_k)q_k(t)$ and using orthogonality gives the exact indexed identity

$$\theta_k - t_k = \frac{\int (t - t_k)q_k(t)^2 \, d\mu_{C,N}(t)}{\int q_k(t)^2 \, d\mu_{C,N}(t)}.$$

Multiplication by $\beta^2/(\nu_k(0) + \gamma_k)$ converts it to ordinate error. It exposes the weighted contribution away from the target atom, but estimating that contribution uniformly is still necessary. The Gamma weight has the previously derived interior potential $2N[(1+\tau)\log(1+\tau) + (1-\tau)\log(1-\tau)]$, $\tau = \gamma \log C/(2\pi N)$. There is now a legitimate monic object to analyze with that potential. Its varying discrete measure cannot be replaced by a fixed-measure or Gaussian model without proof. Rational Ritz convergence through constrained logarithmic potentials provides relevant methods [62]; its asymptotic hypotheses and individual-index implications remain to be checked here. The finite polynomial, moment-pencil and virtual-vector identities pass 283 exact rational checks on positive atomic fixtures.

A new cutoff at the same mode count. An independently assembled $C = 125, N = 1500$ case uses 2300 decimal assembly digits and 2000 point-solve digits. A 3000-digit prefix control lies inside all 84 corresponding generator intervals. All first 1000 reference windows are requested before the results are seen, using the unchanged nearest-three-root predictor and its 1% gate. Interval roots of the serialized point state and an exact polynomial upper count give a complete matched prefix of 1000, with no carriers below its counting threshold. Every prediction passes.

C $(N = 1500)$	D_1	D_{1000}	Worst relative prediction error
125	662.16446	21.94445	$2.49960 \cdot 10^{-6}$
250	1175.39874	305.89872	$1.69115 \cdot 10^{-7}$

The $C = 125$ thousandth zero has more Fourier-band headroom ($\tau \simeq 0.72717$, against 0.83156 at $C = 250$), yet far fewer accurate digits. Mode headroom alone therefore does not determine accuracy. The cutoff and indexed mode budgets must be distinguished. The new state has an enclosed matrix residual, but this experiment does not certify every tiny displacement of the analytic ground transform. Neither the fixed-parameter successes nor the finite monic identity close the uniform C, N, P, k law.

8.15 Constructive resolution and a continuum action

The preceding monic identification makes the unresolved mode budget a weighted polynomial approximation problem. It also permits a sufficient indexed bound that does not mistake a signed orthogonality identity for a positive error estimate.

Proposition 21 (Cardinal-subspace resolution bound)**.** *Assume the positive real atomic measure of the preceding subsection, with distinct ordered positive-weight atoms t_i and weights μ_i. Let θ_k be its degree-N zero-shift Ritz root. Choose $k \le N$ polynomials q_i of degree at most $N-1$ with $q_i(t_j) = \delta_{ij}$ for $1 \le i, j \le k$. Put*

$$D_k = \operatorname{diag}(\mu_1, \dots, \mu_k), \qquad E_k = \sum_{m>k} (t_m - t_k)\mu_m q(t_m)q(t_m)^T, \qquad B_k = \operatorname{tr}(D_k^{-1}E_k).$$

Then $0 \le \theta_k - t_k \le B_k$. In particular, d relative digits for the zero-shift ordinate are guaranteed if

$$N \ge k, \qquad B_k \le t_k\big(2\,10^{-d} + 10^{-2d}\big).$$

Proof. The cardinal evaluations make the chosen trial space k-dimensional. For $q_c = \sum_i c_i q_i$, its norm squared is at least $c^T D_k c$, while its energy relative to t_k is at most $c^T E_k c$: the first k atoms contribute $\sum_i \mu_i(t_i - t_k)c_i^2 \le 0$. Min–max therefore bounds $\theta_k - t_k$ by the largest eigenvalue of $D_k^{-1/2} E_k D_k^{-1/2}$, hence by its trace since it is positive semidefinite. The lower bound is the full multiplication-operator min–max bound. Finally the relative ordinate error is at most $\sqrt{1 + B_k/t_k} - 1$, which gives the stated digit threshold. □

This bound needs only finite moments once the cardinals are supplied. If U contains their coefficients in the polynomial basis of the moment pair (T, H), then exactly

$$E_k = U^T(T - t_k H)U - \operatorname{diag}\big(\mu_i(t_i - t_k)\big).$$

For example, cardinal polynomials on the first M atoms, $k \le M \le N$, also remove atoms $k+1, \dots, M$. All 3311 exact rational checks on 119 finite positive-measure fixtures pass. The bound is conditional on the stated measure, and evaluating it currently uses reference atoms; it is not yet an explicit C, N, k estimate. A failed upper bound is inconclusive, whereas $N < k$ cannot represent a complete movable prefix. The ground-energy shift and the P-dependent arithmetic channel must still be added before applying the earlier soft minimum.

A model with an explicit mode-dependent turning point. The rational-Ritz potential approach[62] suggests retaining the finite Fourier lattice in a continuous approximation. Write $L = \log C$ and $A = 2\pi N/L$. Replace its pole density by $L\mathbf{1}_{|x|<A}/(2\pi)$ and the zeta atoms by the smooth density $\log(|x|/(2\pi))/(2\pi)$. These replacements discard discreteness and the exact sine carrier. The latter density is even negative for $|x| < 2\pi$, so this is a formal logarithmic-defect model, not a proved positive equilibrium measure at the origin.

Solving the corresponding finite Hilbert-transform problem with potential zero outside $[-a, a]$ and no square-root endpoint term gives

$$\log \frac{a}{4\pi} = \frac{2L}{\pi} \arcsin\left(\min\left\{ \frac{A}{a}, 1 \right\} \right). \tag{76}$$

Thus $a = 4\pi C$ when $N \ge 2C \log C$; otherwise there is a unique solution $a > A$. This is a model transition, not an established CCM resolution threshold. For $4\pi < x < 4\pi C$, inversion gives

$$N_{\mathrm{turn}}(C, x) = \frac{x \log C}{2\pi} \sin\left(\frac{\pi \log(x/(4\pi))}{2 \log C} \right). \tag{77}$$

Setting $x = \gamma_k$ exposes the indexed requirement; the separate degree bound $N \ge k$ remains necessary for a complete movable prefix.

For $0 \le x < a$, let $s = \sqrt{a^2 - x^2}$. In the mode-limited case $A < a$, put $b = \sqrt{a^2 - A^2}$. The candidate leading exponent is

$$\begin{aligned} \mathcal{S}_{C,N}(x) &= s - x \arccos(x/a) + \frac{L}{\pi} I(x), \\ I(x) &= A \log \left| \frac{s+b}{s-b} \right| - x \log \left| \frac{As + bx}{As - bx} \right|. \end{aligned} \tag{78}$$

The apparent singularities cancel at $x = A$, where $I(A) = 2A \log(a/A)$. When $A \ge a$, omit I. In particular, $\mathcal{S}_{C,N}(0) = a + 4N \operatorname{arcosh}(a/A)$ in the mode-limited regime and $4\pi C$ otherwise. Independent differentiation, quadrature, Hilbert-transform and edge-limit evaluations pass 38 high-precision point checks; these are formula checks, not interval certificates of a CCM approximation.

An unfitted test and its limitations. The exploratory prediction $D_k = \mathcal{S}_{C,N}(\gamma_k)/\log 10$ uses no fitted constants. Table 7 compares it only on the counted, reference-window-matched point-state prefixes inside its domain. All 5000 requested rows are retained, including the two valid $N = 750$ targets outside the modeled turning point and every unresolved source target. The model's extra two $N = 1000$ windows are not valid indexed measurements and do not enter its score.

The mode-dependent endpoint lies within two ordinates of both observed prefix boundaries. At $C = 250, N = 1500$ it reduces the bare cutoff action's 189.33-digit RMS discrepancy to 2.08

C	N	Counted prefix	Model prefix*	RMS digits	1% passes
125	1500	≥ 1000	≥ 1000	2.1482	4/1000
250	750	578	576	2.4308	0/576
250	1000	797	799	2.4457	0/797
250	1250	≥ 1000	≥ 1000	2.3131	0/1000
250	1500	≥ 1000	≥ 1000	2.0765	1/1000

Table 7: Leading continuum action against stored point-state root errors. The RMS measures error in relative decimal depth; the stricter final column tests relative displacement error below 1%. *Only the first 1000 reference ordinates are counted for the model.

digits. Nevertheless, only 5 of the 4373 eligible comparisons pass the displacement gate: the action is *not* an accurate final error formula. Its root-specific prefactor, discrete carrier effects, transfer from the zero-shift measure to the ground source, uniform remainder and arithmetic budget remain open. Neither a fitted correction nor a clipped zero-digit claim is added.

8.16 Curvature, amplitude, and exact source transfer

The leading action has now been tested at the level of its local curvature and its root-specific prefactor. These tests distinguish mechanism evidence from an accurate displacement formula. With the notation of (76), direct differentiation on the positive side gives

$$\mathcal{S}''(0+) = \frac{1 - 2L\sqrt{a^2 - A^2}/(\pi A)}{a} \quad (A < a), \qquad \mathcal{S}''(0+) = \frac{1}{a} \quad (A \geq a).$$

After restoring the exact ξ' height feature, the linear Gamma envelope cancels the action's $-\pi|x|/2$ term. The resulting local quadratic coefficient is $a_{2,\mathrm{model}} = \mathcal{S}''(0+)/(2\log 10)$. It has no fitted parameters. Its comparison with the earlier first-50 fits is a comparison of a local coefficient with a finite-range fit, not an assertion that the two definitions coincide:

C	N	Earlier fitted a_2	Model a_2	Relative difference
1000	890	$-4.2777\,10^{-4}$	$-4.25028\,10^{-4}$	-0.6410%
1000	800	$-5.0343\,10^{-4}$	$-4.99223\,10^{-4}$	-0.8356%
1200	970	$-4.3028\,10^{-4}$	$-4.27892\,10^{-4}$	-0.5550%
100	500	$1.2519\,10^{-4}$	$1.25192\,10^{-4}$	0.00155%
250	1500	$5.0432\,10^{-5}$	$4.99851\,10^{-5}$	-0.8861%
250	500	$-2.5214\,10^{-4}$	$-2.47408\,10^{-4}$	-1.8769%
13	120	$1.4732\,10^{-3}$	$1.32923\,10^{-3}$	-9.7724%

Every sign agrees; the previously identified low-C outlier is retained. This supports the finite-mode mechanism but does not settle its remainder.

A trial amplitude and a remote-energy discriminator. Let $\Xi(x) = \xi(\frac{1}{2} + ix)$ and define, inside the model endpoint,

$$F_m(x) = \Xi(x)e^{G(x)}, \qquad 2G(x) = \mathcal{S}(|x|) - \mathcal{S}(0) + \pi|x|/2.$$

For $\beta J < a$, the finite lattice sums

$$Z_J = \frac{1}{L}\sum_{|j|\leq J} F_m(\beta j)^2, \qquad W_J(\gamma) = -\frac{\gamma}{L}\sum_{|j|\leq J} \frac{F_m(\beta j)^2}{\beta j - \gamma}$$

specify the trial normalization and Stieltjes factor without fitting. If the source approximation and the omission of remote energy were valid, the bare identity would give

$$\frac{\delta_k}{\gamma_k} \approx \frac{\varepsilon W_J(\gamma_k)}{\gamma_k^2 |\xi'(\frac{1}{2} + i\gamma_k)|^2 e^{\mathcal{S}(\gamma_k) - \mathcal{S}(0) + \pi\gamma_k/2}}.$$

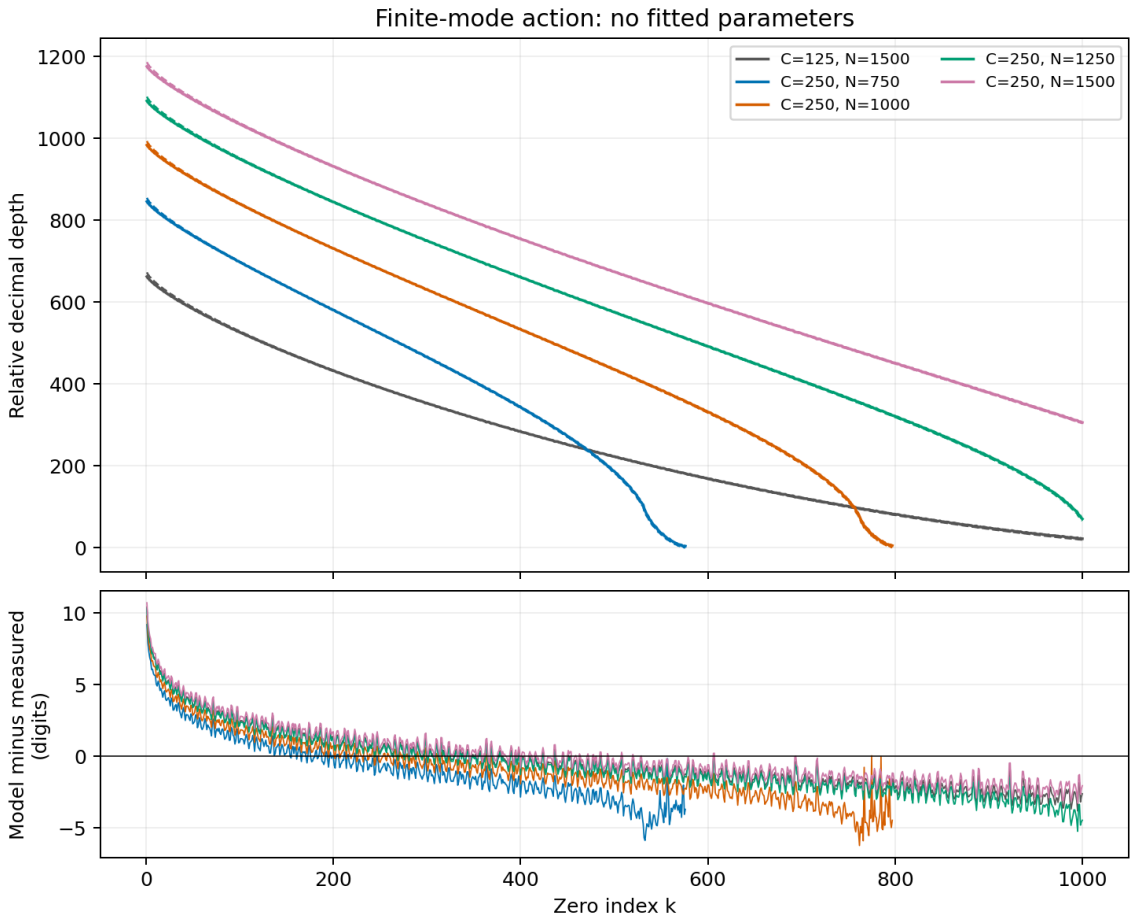


Figure 9: Measured depths (solid) and the derived continuum-model action (dashed), with their difference below. Close agreement on a thousand-digit scale does not establish accurate displacement prediction.

The normalization Z_J cancels. This diagnostic uses reference ordinates and reference ξ' values, as well as the computed eigenvalue; it is not a parameter-only prediction. Lattice sums through $|j| = 128$ and 192 agree closely, but the finite added-shell bounds are not infinite tail certificates.

A separate exploratory hypothesis assigns paired energy the density

$$dE_m(y) = \frac{2a}{\pi y\sqrt{y^2 - a^2}}\, dy, \qquad y > a.$$

Equivalently a^2/y^2 has the arcsine distribution. In the unscaled squared ordinate $t = \gamma^2$, its exact resolvent identity is

$$R_m(t) = 1 + t\int \frac{dE_m(y)}{y^2 - t} = \frac{1}{\sqrt{1 - t/a^2}}, \qquad 0 \le t < a^2.$$

The proposed density is not derived for CCM. The comparison below uses the measured ratio of displacement to the bare operator estimate, with the same counted-prefix and model-domain gates as before. It tests only the remote multiplier, not the whole error law.

C	N	Trial RMS digits	Trial 1% passes	Remote 1% passes	Worst remote error
125	1500	0.3580	43/1000	682/1000	4.0053%
250	750	0.8270	17/576	538/576	218.292%
250	1000	0.7975	26/797	763/797	82.0279%
250	1250	0.5893	38/1000	781/1000	2.3615%
250	1500	0.2670	50/1000	1000/1000	0.5128%

The full trial passes only 174/4373 eligible targets. Its modeled Stieltjes numerator closely matches the stored operator quantity, while its derivative amplitude increasingly fails at high index. For $C = 250, N = 1500$, the modeled-to-measured derivative-squared ratio falls from 0.99988 at $k = 1$ to 0.13147 at $k = 1000$; the Stieltjes ratio stays close to one. This directs the next derivation to amplitude and source transfer, rather than another fitted depth curve. The remote model passes 3764/4373 but fails strongly near underresolved edges. The exploratory central-normalized closure $F_0'/F_\varepsilon' \approx (1 + \sqrt{1 - \gamma_k^2/a^2})/2$ also improves with resolution: its resulting displacement predictor passes 1000/1000 at $N = 1500$, but only 250/362 at $N = 500$, both at $C = 250$. It is not a uniform replacement for the earlier operator predictor. All these comparisons are exploratory on previously inspected data; no constants are fitted. Twenty-one independent 100-digit point checks verify the model curvature and distribution identities only.

An exact transfer bound that stays regular at the ground. The following result supplies a rigorous finite alternative to assuming the two source amplitudes agree. Here $\mathbf{A}$ denotes the even matrix, to distinguish it from the Fourier endpoint $A = 2\pi N/L$.

Proposition 22 (Monic source resolvent)**.** *Suppose $\mathbf{A} > 0$ has simple lowest eigenvalue ε and ground vector g with $\ell^T g \neq 0$. Put*

$$a_0 = \frac{\mathbf{A}^{-1}\ell}{\ell^T\mathbf{A}^{-1}\ell}, \quad \mathcal{J} = \begin{pmatrix} -w^T \\ I \end{pmatrix}, \quad H_z = \mathcal{J}^T(\mathbf{A} - zI)\mathcal{J}, \quad b = \mathcal{J}^T a_0.$$

For $0 \le z \le \varepsilon$, $H_z > 0$ and the unique monic stationary vector is

$$a_z = a_0 + z\mathcal{J}H_z^{-1}b, \qquad a_\varepsilon = \frac{g}{\ell^T g}.$$

For every fixed real linear functional d,

$$|d^T(a_z - a_0)| \le z\sqrt{(\mathcal{J}^T d)^T H_z^{-1}(\mathcal{J}^T d)}\sqrt{b^T H_z^{-1} b}.$$

Proof. The only possible kernel of $\mathbf{A} - zI$ is the ground line at $z = \varepsilon$, which is not contained in $\ker \ell^T$; its compression is therefore positive. Write $a_z = a_0 + \mathcal{J}r$. Stationarity gives $H_z r = zb$, proving the formula and its endpoint. Cauchy–Schwarz in the H_z^{-1} inner product proves the bound. □

Taking d to be transform differentiation at a fixed ordinate bounds the amplitude change; a bound below $|d^T a_0|$ excludes a vanishing derivative there. Comparisons at different roots and changes from monic to unit normalization still require their own factors. The constrained energy $h(z) = a_z^T(\mathbf{A} - zI)a_z$ also obeys

$$h'(z) = -\|a_z\|^2, \qquad h''(z) = -2(\mathcal{J}^T a_z)^T H_z^{-1}(\mathcal{J}^T a_z) \le 0.$$

Consequently $h(0) = \int_0^\varepsilon \|a_z\|^2 dz$ and $\varepsilon \le h(0)/\|a_0\|^2$. These statements pass 3332 exact rational checks, including the singular full-matrix ground endpoint.

Returning to $t = (\nu/\beta)^2$, Parseval and the Gamma transform give a useful interpretation:

$$d\eta_N(t) = \frac{L\beta\, dt}{2\pi\sqrt{t}\,\Gamma(N+1+\sqrt{t})^2\Gamma(N+1-\sqrt{t})^2}, \qquad d\mu_z = d\mu_0 - z\, d\eta_N.$$

The measure representation is conditional on the earlier positive real Weil hypothesis; the matrix proposition is not. For $z > 0$ the displayed measure is signed, even though its lower-degree form remains positive. Thus a positive-measure zero-shift asymptotic cannot simply be transferred to the ground source. The degree-N ground polynomial has zero shifted norm. Uniform estimates for this resolvent, the discrete prefactor and the precision channel remain necessary to close the C, N, P, k law.

8.17 The exact Fourier boundary and a saturation control

The next discriminator retains the exact Gamma carrier instead of its leading interior approximation. Let $y = x/\beta$, $\tau = y/N$, and

$$V_N(y) = (N+y)\log\frac{N+y}{N} + (N-y)\log\frac{|N-y|}{N},$$
$$J_N(y) = \left|\frac{\Gamma(N+1)^2}{\Gamma(N+1+y)\Gamma(N+1-y)}\right| e^{V_N(y)}.$$

Here $0\log 0 = 0$ and reciprocal Gamma is continued through its zeros. Inside $y < N$, Stirling's leading factor is $(1-\tau^2)^{-1/2}$, which loses uniformity at the last retained Fourier mode. The exact factor has a finite transition there.

Proposition 23 (Uniform factorization of the known carrier)**.** *Define the exact positive-real Stirling remainder by*

$$R(u) = \log\Gamma(u+1) - (u+\tfrac{1}{2})\log u + u - \tfrac{1}{2}\log(2\pi).$$

For $N \ge 1$, $y \ge 0$, $h = y - N$, put $\Phi(h) = e^{h - h\log|h|}|1/\Gamma(1-h)|$, with $\Phi(0) = 1$. Then, including carrier zeros by continuity,

$$J_N(y) = \sqrt{\pi N}\,\Phi(h)\left(1 + \frac{h}{2N}\right)^{-1/2} e^{2R(N) - R(N+y)}.$$

Away from those exact zeros, omitting the final exponential incurs relative error below $1/(6N)$, uniformly for $y \ge 0$.

Proof. Substitute the defining remainder identity at the two positive arguments N and $N+y$. All large logarithmic terms cancel to give the displayed factorization. Binet's formula[63] implies that R is positive and decreasing, with $R(u) < 1/(12u)$. Therefore $0 < 2R(N) - R(N+y) < 1/(6N)$, and the relative error is $1 - e^{-[2R(N)-R(N+y)]} < 1/(6N)$. At a carrier zero both expressions vanish, so no relative error is assigned. □

At $y = N$ the leading size is $\sqrt{\pi N}$; at $y = N+1, N+2, \ldots$ the forced carrier zeros remain exact. For the interior remainder $E = 2R(N) - R(N+y) - R(N-y)$, Binet's integral also gives

$$-\frac{y^2}{6N(N^2-y^2)} \le E \le 0, \qquad 0 \le y < N.$$

These are estimates of a known factor, not a CCM error bound. The factorization and signed bounds pass 693 Arb checks at 150 digits, including both sides of the boundary and exact carrier zeros.

Improved exploratory curves, followed by a failed control. Multiplying the previous trial transform by the exact signed Gamma factor and recomputing its finite lattice sums reduces the five-case RMS depth errors to $0.216, 0.183, 0.188, 0.114, 0.059$ digits, in the same configuration order as the preceding subsection. Nevertheless, only 386/4373 displacements pass the unchanged 1% gate. A separate imaginary-order Bessel transport hypothesis raises this to 700/4373, still insufficient. It replaces a leading quarter-power amplitude by

$$Q_B(a,x) = \frac{e^{-\pi x/2}\,\mathrm{Re}\, I_{ix}(a)}{I_0(a)} e^{-S_a(x)+a}, \qquad S_a(x) = \sqrt{a^2-x^2} - x\arccos(x/a).$$

The candidate divides the Gamma-corrected bare displacement by Q_B. The Bessel definition is standard[64]; no CCM differential equation identifying that counterpart is established. Its evaluation required adaptive precision, reaching 1600 digits. Earlier 90-digit results with inadequate ball widths are retained as unqualified attempts.

A new control holds $C = 125$ fixed and raises N from 1500 to 2000, keeping the same 2300-digit assembly, 2000-digit point solve, reference targets and operator-predictor rule. All 1000 point-state roots have disjoint enclosures in their requested reference windows and a matching exact polynomial upper count. No carrier lies below the counting threshold. All 1000 unchanged operator predictions pass; their worst relative error is $2.51079 \cdot 10^{-6}$.

N at $C = 125$	Eigenvalue depth	D_1	D_{1000}
1500	665.08700647	662.16445877	21.94444762
2000	665.19096739	662.26841961	22.01817255

The isolated Gamma correction predicts a *loss* of 0.06942 digits at $k = 1000$, whereas the measured depth *gains* 0.07372. Its mode-change prediction passes only 133/1000 ratio tests. The full Gamma and Gamma–Bessel trials pass only 53/1000 and 64/1000 on the new state. Thus the improved earlier curves do not justify transferring that isolated factor into a uniform convergence law. The polynomial amplitude must also be controlled. The new state has an interval matrix residual, with residual/eigenvalue below $3.252 \cdot 10^{-179}$; this is not a certificate for every tiny analytic ground-transform displacement.

A sharper model statement about the minimum mode budget. For $x > 4\pi$ and $C > x/(4\pi)$, write $\ell_x = \log(x/(4\pi))$. Equation (77) becomes

$$N_{\mathrm{turn}}(C,x) = \frac{x\ell_x}{4}\frac{\sin u}{u}, \qquad u = \frac{\pi\ell_x}{2\log C}.$$

It increases with C, from $x\ell_x/(2\pi)$ at the cutoff boundary to $x\ell_x/4$ as $C \to \infty$. At fixed N, the largest model endpoint occurs at

$$C_* = \frac{N}{2W(N/2)}, \qquad a_{\max} = \frac{2\pi N}{W(N/2)},$$

where W is the positive Lambert function. These follow from the explicit endpoint equation; they are not CCM capacity theorems. The degree requirement $N \geq k$ remains separate. In the mode-limited branch, putting $s = \sqrt{a^2 - x^2}$ and $b = \sqrt{a^2 - A^2}$ gives

$$\partial_N \mathcal{S} = 2 \log \left| \frac{s+b}{s-b} \right| > 0, \qquad x < a, \quad x \neq A.$$

Yet its C-derivative need not be positive. Close to the model endpoint, with $\delta = a - x \ll a - A$,

$$\mathcal{S}(a - \delta) \sim \frac{2\sqrt{2}}{3\sqrt{a}} \left(1 + \frac{2LA}{\pi b} \right) \delta^{3/2}.$$

The cutoff-limited coefficient has bracket one; this expansion is not uniform when the two boundaries coalesce. Fifty-one independent point checks verify these model identities. They explain why a useful indexed minimum must account for distance from the resolution boundary, rather than merely clip an inaccurate depth to zero. Uniform prefactor, source-transfer and arithmetic estimates are still required.

8.18 Exact joint transport under a mode increment

The failed saturation control points to an exact finite replacement for an isolated carrier correction. Let the next even matrix be $M = \left(\begin{smallmatrix} A & b \\ b^T & d \end{smallmatrix}\right)$, with $M > 0$, and extend the old leading-coefficient functional ℓ by $w = \sqrt{2}$. Define

$$u = A^{-1}\ell, \quad h = (\ell^T u)^{-1}, \quad a = hu, \quad v = A^{-1}b, \quad \sigma = d - b^T v, \quad \eta = \ell^T v - w.$$

Here a is the old monic minimizer, and h is its form norm; it is not a ground eigenvalue.

Proposition 24 (Joint monic mode update)**.** *Put $\mathcal{D} = \sigma + \eta^2 h$, $q = \eta a - v$ and $\tau_* = -\eta h/\mathcal{D}$. The new monic minimizer and norm are*

$$a_{\mathrm{new}} = \begin{pmatrix} a \\ 0 \end{pmatrix} + \tau_* \begin{pmatrix} q \\ 1 \end{pmatrix}, \qquad h_{\mathrm{new}} = \frac{h\sigma}{\mathcal{D}}, \qquad \log_{10} \frac{h}{h_{\mathrm{new}}} = \log_{10} \left(1 + \frac{\eta^2 h}{\sigma} \right).$$

Proof. For a fixed last coefficient τ, completing the square under $\ell^T x + w\tau = 1$ gives $x = a(1 + \eta\tau) - \tau v$ and the energy $h(1 + \eta\tau)^2 + \sigma\tau^2$. Its minimum gives the formulas. Also $\ell^T q + w = 0$, so the update has one lower polynomial degree. □

Set $s = (N+1)^2$ and $\kappa = \sqrt{2}\, s Q_N(s)$. In the new form let K_{new} reproduce polynomials of degree at most N, and put

$$r(t) = \frac{K_{\mathrm{new}}(t, s)}{K_{\mathrm{new}}(s, s)}, \qquad h_r = \frac{1}{K_{\mathrm{new}}(s, s)}.$$

The vector $(q, 1)/\kappa$ represents r: it has degree at most N, value one at s, and is orthogonal to $(t - s)$ times every old polynomial of degree less than N. Consequently

$$h_r = \frac{\mathcal{D}}{\kappa^2}, \qquad P_{N+1}(t) = (t - s)P_N(t) - \xi_N r(t), \qquad \xi_N = \frac{\eta h \kappa}{\mathcal{D}}.$$

This identity holds at $t = s$ as well. If K_{old} reproduces old degrees at most $N-1$, the orthogonal direct sum also gives

$$K_{\text{new}}(t,u) = (t-s)(u-s)K_{\text{old}}(t,u) + \frac{r(t)r(u)}{h_r}.$$

No global positive zeta measure is needed for these finite-form identities. Under the earlier conditional measure interpretation, $d\mu_{C,N} = (t-s)^2 d\mu_{C,N+1}$ is consistent with the same embedding; division alone could omit a mass at the removed carrier.

Write $g_N(t) = [\Gamma(N+1+\sqrt{t})\Gamma(N+1-\sqrt{t})]^{-1}$. Since $g_N = (s-t)g_{N+1}$, the corresponding monic transforms satisfy

$$F_{N+1}(x) = F_N(x) + I_N(x), \qquad I_N(x) = (-1)^N \sqrt{L}\,\xi_N r((x/\beta)^2) g_{N+1}((x/\beta)^2).$$

The carrier change cancels the embedded part of the polynomial exactly. The remaining update depends on η, σ and r, not merely on the Gamma factor. These identities pass 2275 exact rational checks on 35 positive fixtures and 585 interval checks on five existing $C = 125$ prefixes, $N = 10, 20, 40, 80, 120$. The latter use 2100 working digits and verified positive interval pivots. An initial 700-digit direct-inverse check failed arithmetic qualification at $N = 120$ and is retained separately.

At an old zero x_N, a constant-sign derivative bound $|F'_{N+1}| \geq m_N > 0$ on $[x_N - \rho, x_N + \rho]$, together with $|I_N(x_N)| < m_N \rho$, gives a unique next zero and $|x_{N+1} - x_N| \leq |I_N(x_N)|/m_N$. A summable indexed majorant would control the mode tail. Uniform parameter estimates, identification of its limit, transfer to the changing ground shift, and the finite precision remainder remain open. The exact monic norm formula must not be substituted for the ground-eigenvalue convergence rate.

8.19 Certified prefix transfer and a finite resolution threshold

The joint update can be evaluated for every prefix without solving a new eigenproblem at each N. Write $x_j = j^2$ and $A_{ij} = (r_i w_j - w_i r_j)/(x_i - x_j)$, with original leading functional $w = (1, \sqrt{2}, \ldots)^T$. Let $\widetilde{r}_j, \widetilde{w}_j, d_j$ be the generators and positive pivot when natural-order LDL elimination reaches index j. Starting with $R_{-1} = 1, W_{-1} = 0$, define, away from the Fourier nodes,

$$c_j(t) = \frac{R_{j-1}(t)\widetilde{w}_j - W_{j-1}(t)\widetilde{r}_j}{t - x_j}, \qquad \begin{pmatrix} R_j \\ W_j \end{pmatrix} = \begin{pmatrix} R_{j-1} \\ W_{j-1} \end{pmatrix} - \frac{c_j}{d_j}\begin{pmatrix} \widetilde{r}_j \\ \widetilde{w}_j \end{pmatrix}.$$

If $c(t)_j = w_j/(t - x_j)$ uses the original generators, elimination gives

$$R_N = 1 - c(t)^T A_N^{-1} r, \quad W_N = -c(t)^T A_N^{-1} w, \quad H_N = w^T A_N^{-1} w = \sum_{j=0}^{N} \frac{\widetilde{w}_j^2}{d_j}.$$

Thus the monic transform is

$$F_N(\nu) = -\frac{2\sin(\nu L/2)}{\sqrt{L}} \frac{\nu}{\beta^2 H_N} W_N((\nu/\beta)^2).$$

Node limits must be taken separately. One $O(M^2)$ factorization and one $O(M)$ transfer per evaluation point provide all $N \leq M$. Direct expansion of the determinant-one update also proves

$$\frac{R_N(t)W_N(u) - W_N(t)R_N(u)}{u - t} = \sum_{j=0}^{N} \frac{c_j(t)c_j(u)}{d_j}, \qquad R_N W'_N - W_N R'_N = \sum_{j=0}^{N} \frac{c_j(t)^2}{d_j} > 0.$$

This positive derivative kernel supplies a local root tool, but poles of R_N must still be checked; positivity does not justify Newton iteration across one of them.

Proposition 25 (Ordinal counts and nested zero-shift interlacing)**.** *Let $A_M > 0$ be a consistent finite source, $D = \operatorname{diag}(x_j)$, and $S = A_M D + rw^T = DA_M + wr^T$. The degree-$N$ monic squared roots are the generalized eigenvalues of $(J_N^T S_N J_N, J_N^T A_N J_N)$, where $J_N = [-w_{1:N}^T; I_N]$. Away from nodes and zero leading pivots,*

$$Z_N(t) = \frac{W_N(t)}{R_N(t)} = w^T (S_N - tA_N)^{-1} w, \qquad \#\{\theta_{j,N} < t\} = n_-(S_N - tA_N) + \mathbf{1}_{\{Z_N(t)>0\}} - 1.$$

The full-pencil negative count is obtained from the signs of $d_j(x_j - t)R_j/R_{j-1}$. Moreover,

$$\theta_{k,N+1} \le \theta_{k,N} \le \theta_{k+1,N+1}, \qquad 1 \le k \le N.$$

Proof. The compressed-pencil representation is the finite monic variational identity. The determinant identity $\det(S_j - tA_j) = \det(A_j) \prod_{i=0}^{j} (x_i - t) R_j(t)$ gives the pivot signs. The bordered matrix with blocks $S_N - tA_N, w, w^T, 0$ has the inertia of $S_N - tA_N$ plus the sign of $-Z_N$. Compression to $\ker w^T$ instead leaves one positive and one negative direction; equating negative counts proves the count formula. Finally, the spaces

$$V_N = \{v \in \mathbb{R}^{M+1} : v_j = 0 \ (j > N),\ w^T v = 0\}$$

are nested and have dimension N. All their Ritz values concern the same pair (S, A_M). The min–max principle proves interlacing. □

An unresolved interval sign is an unresolved count, not a rounded decision. The transfer and inertia identities pass 7290 and 5292 exact rational checks on 30 positive fixtures each. A separate streaming and nested-compression check passes 3328 checks on 52 fixtures.

Transfer to the analytic source. Positive rounded factors define a precise surrogate, denoted by hats. Its roots alone do not certify the analytic matrix. Put $H = J^T A J$, $T = J^T S J$ and suppose

$$\widehat{H} \succeq gI, \quad \|H - \widehat{H}\| \le E_H, \quad \|T - \widehat{T}\| \le E_T, \quad e = E_H/g < 1, \quad f = E_T/g.$$

If the lowest surrogate squared root exceeds f, Loewner bounds and min–max give, with the same index,

$$\frac{\widehat{\theta}_k - f}{1 + e} \le \theta_k \le \frac{\widehat{\theta}_k + f}{1 - e}.$$

Here the bounds include both the matrix generators and the perturbed leading functional. Interval reconstruction of the natural factors, an inverse-trace lower bound, and the existing analytic generator enclosures establish these hypotheses for the $C = 125, M = 2000$ source and all its prefixes. A baseline count at $\nu = 0.1$ verifies positivity for every prefix. For $\widehat{\theta}_k \le 2 \times 10^6$, the resulting frequency motion is at most 3.184×10^{-1458}. Consequently the following error intervals concern the finite *analytic zero-shift* pencil, including the reference-zero uncertainty.

N	matched prefix	$D_1^{(0)}$	$D_{1000}^{(0)}$
1000	902	$[656.2556, 656.2565]$	unmatched
1250	1000	$[666.7057, 666.7067]$	$[22.1208, 22.1218]$
1500	1000	$[666.8629, 666.8639]$	$[22.3172, 22.3182]$
1750	1000	$[666.9341, 666.9350]$	$[22.3718, 22.3728]$
2000	1000	$[666.9774, 666.9784]$	$[22.3993, 22.4003]$

Here $D_k^{(0)} = -\log_{10}|(\nu_{k,N}^{(0)} - \gamma_k)/\gamma_k|$. All 5000 requested pairs are retained: 4902 have certified error brackets; the other 98 are proved above their reference windows. Three initial linearization brackets at $N = 1000$, $k = 900, 901, 902$ failed despite window membership and were refined by count bisection. The initial arithmetic attempt with full reference balls was also unresolved and is retained. Qualified point evaluations separate the reference radius, use 2300 or 3500 decimal working digits, and are followed by ordinal and source checks. The source certificate uses 6000 decimal working digits. These are working precisions, not a minimum-bit-P theorem.

A measured minimum mode budget. Guarded counts at the reference midpoint boundaries classify every integer $N = 1, \ldots, 2000$ and $k = 1, \ldots, 1000$: two million dependent target pairs from the same source. Both guard endpoints use a fixed 10^{-6} frequency margin; the pencil inequalities certify that their equal counts persist at the analytic midpoint. No count is unresolved. The first complete 1000-target window match occurs at $N = 1091$; $N = 1090$ matches only 999. All individual matches persist through $N = 2000$. This is a window-resolution threshold, not a specified digit-accuracy threshold or a statement about larger N.

The previously derived, unfitted continuum turning-point budget is

$$N_{\mathrm{turn}}(C, x) = \frac{x\ell_x}{4}\frac{\sin u_x}{u_x}, \qquad \ell_x = \log\frac{x}{4\pi}, \qquad u_x = \frac{\pi\ell_x}{2\log C}, \qquad 4\pi < x < 4\pi C.$$

At $x = \gamma_k$ it has 1.653 modes RMS discrepancy from the 1000 individual entry thresholds. It predicts 1090.163 for $k = 1000$, whose actual threshold is 1091. Yet $\max\{k, \lceil N_{\mathrm{turn}}\rceil\}$ is insufficient at 249 indices, by as many as six modes. Its closeness cannot be promoted to a sufficient CCM budget; neither can an empirical six-mode padding be declared uniform. Reference heights here assess a model, not a reference-free zero predictor.

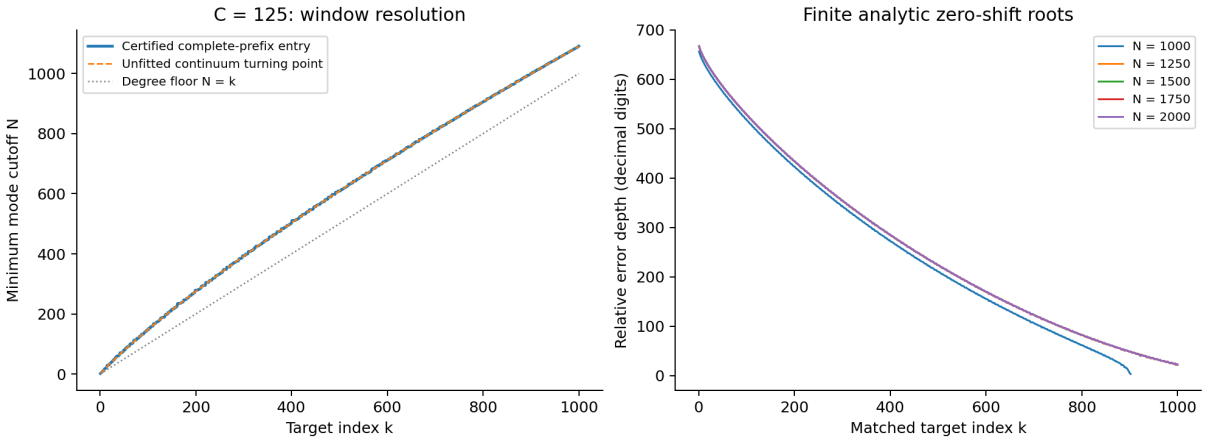


Figure 10: Certified zero-shift resolution and errors at $C = 125$. The continuum threshold closely tracks the finite entry curve but fails as a sufficient integer budget. The $N = 1000$ error curve ends at its last matched target, $k = 902$; no unmatched value is scored.

For a fixed upper boundary b_k, interlacing makes $N_{\mathrm{entry}} = \min\{N \geq k : \#(\nu_{j,N}^{(0)} < b_k) \geq k\}$ a monotone entry event. A lower-boundary count is also needed: a root could later leave its window downward. Replacing the window by $\gamma_k(1 \pm 10^{-d})$ provides a count-based finite d-digit test. The desired law therefore needs a resolution gate before combining physical, ground-source and arithmetic error budgets. For established upper bounds E_R, E_G, E_P, the triangle inequality gives

$$D_{\mathrm{total}} \geq -\log_{10}(E_R + E_G + E_P) \geq \min\{D_R, D_G, D_P\} - \log_{10} 3.$$

This is a conditional bound, not an equality or a fitted rate. Uniform parameter bounds for its channels remain open. In particular, zero-shift interlacing and source certification do not

certify the changing ground-transform root errors or identify the infinite-cutoff limit with the zeta zeros.

8.20 Certified accuracy budgets beyond window resolution

Window membership and digit accuracy require different mode budgets. The preceding finite interlacing yields an exact accuracy threshold when a terminal analytic root lies strictly above its reference. Throughout this subsection the source is the *zero-shift* pencil; the ground correction is treated separately below.

Proposition 26 (Finite one-sided accuracy threshold)**.** *Let a common positive analytic source and its prefixes satisfy Proposition 25, with all generalized squared roots positive. Suppose* $\nu_{k,M}^{(0)} > \gamma_k$*. For* $k \le N \le M$ *and* $d > 0$*, set* $b_{k,d} = \gamma_k(1 + 10^{-d})$*. Then*

$$0 < \frac{\nu_{k,N}^{(0)} - \gamma_k}{\gamma_k} < 10^{-d} \quad \Longleftrightarrow \quad \#\{j : \nu_{j,N}^{(0)} < b_{k,d}\} \ge k.$$

The accuracy event is monotone in N *over this finite family. Its first occurrence, if any, is certified by counts at two adjacent mode sizes.*

Proof. Interlacing gives $\nu_{k,N}^{(0)} \ge \nu_{k,M}^{(0)} > \gamma_k$. The remaining strict upper inequality is equivalent to the stated ordinal count, which is monotone in N by the same interlacing. □

The analytic terminal-root certificates establish this positivity premise for every $k = 1, \dots, 1000$ at $(C, M) = (125, 2000)$ and $(250, 1500)$. Two exact decimal endpoints guard each reference-based threshold by 10^{-6} times its requested absolute tolerance. The indexed source inequalities in Section 8.19 verify that the lower endpoint remains below the threshold and the upper endpoint above it after analytic source perturbation. Equal entry indices at the two guards therefore certify the exact integer minimum. Equality, unresolved arithmetic and a non-reached tolerance are distinct outcomes; none is rounded into a pass.

	$C = 125,\ M = 2000$		$C = 250,\ M = 1500$	
Depth d	prefix K	minimum N	prefix K	minimum N
6	1000	1092	1000	1220
12	1000	1103	1000	1226
25	989	1239	1000	1235
50	896	1212	1000	1246
100	753	1201	1000	1262
250	456	1323	1000	1406
500	127	1179	734	1500
750	0	–	407	1500
1000	0	–	133	1496

Here K is the complete prefix attaining the requested relative accuracy within the tested family, and N is the smallest common budget for that prefix. All 18 000 targets are retained: 12 495 have certified exact integer minima and 5505 do not reach the tolerance through their tested maximum. No final target is unresolved. At $C = 250$, 13 371 endpoint attempts require refinement from 3500 to 5000 decimal working digits; the failed first attempts are retained. These are evaluation precisions for this certificate, not necessary bit-P limits.

The full unfitted action inversion remains imperfect. At $C = 125$, the 25-digit model misses 14 reachable targets; at 500 digits it underbudgets 92 of the 127 mutually reachable targets, by as many as 19 modes, and predicts one additional target not reached by $N = 2000$. At $C = 250$,

the six-digit budget is insufficient at three indices; at 1000 digits it underbudgets 112 of 133 mutually reachable targets, by as many as 11 modes. All non-reach outcomes are included in the comparison. These discrepancies concern the required number of modes, not the relative error of a displacement predictor.

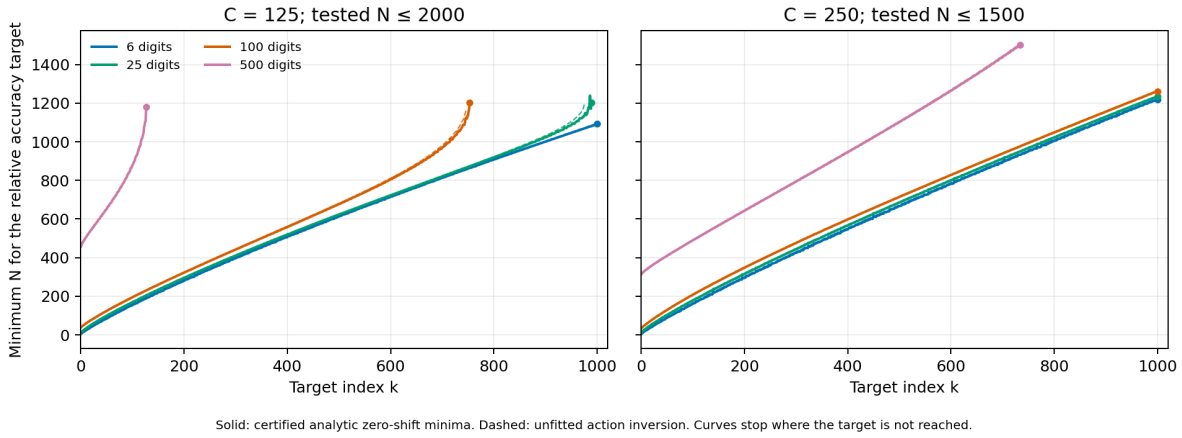


Figure 11: Specified-accuracy mode budgets for the finite analytic zero-shift pencils. The cutoff and requested depth change both the reachable prefix and its mode requirement. No fixed integer padding has been fitted.

For a complete prefix through K, the required budget is the maximum of the individual minima through K, not automatically the minimum for index K alone. A target not reached at the tested maximum M fails at all smaller mode sizes in this family. This does not establish failure for $N > M$ or in the infinite-mode limit. The count benchmark uses reference ordinates to assess accuracy; it is not itself a reference-free prediction law.

An unfitted additional-mode term. There is a useful local inversion of the previously derived continuum action. For fixed C, $L = \log C$ and $4\pi < x < 4\pi C$, put

$$u = \frac{\pi \log(x/(4\pi))}{2L}, \quad N_0 = \frac{xL \sin u}{2\pi}, \quad b_0 = x \cos u, \quad K_0 = 1 + \frac{2L}{\pi} \tan u.$$

Implicit differentiation of the model endpoint equation at $a = x$ gives $a'(N_0) = 4x/(b_0 K_0)$. The model action consequently obeys

$$S(C, N_0 + \Delta, x) = \frac{16\sqrt{2}\, x}{3 b_0^{3/2} \sqrt{K_0}} \Delta^{3/2} + o(\Delta^{3/2}).$$

Its local inverse at small depth is

$$N_{\mathrm{model}}(C, x, d) = N_0 + x^{1/3} \cos u \, K_0^{1/3} \left(\frac{3 \log 10}{16\sqrt{2}} \right)^{2/3} d^{2/3} + o(d^{2/3}).$$

This coefficient is derived, not fitted. The expansion is nonuniform as $u \to \pi/2$ and is not presumed accurate even at six digits. The comparison above inverts the full implicit action, with the degree floor $N \geq k$; it does not substitute this local expansion. There are 160 numerical model-identity and limiting checks on 20 fixtures, separate from the analytic CCM count certificates.

If a bound $|D_N - S_N / \log 10| \leq B$ were established, the actual integer budget would lie between the model inverse budgets at $d - B$ and $d + B$, with endpoint and degree conventions retained. Locally,

$$\Delta N \simeq \frac{\log 10 \, \Delta D}{\partial_N S}.$$

Thus a modest missing amplitude in digit space can produce a substantial budget error when the action slope is small. Beyond its formal fixed-C cap the model slope is zero while the finite pencil still changes with N. Neither an empirical padding nor a minimum term alone supplies the missing amplitude and finite-mode tail. The ground correction, indexed numerical error and limiting identification remain necessary for the requested C, N, P, k convergence law.

8.21 A positive pencil for the exact finite ground source

The full matrix $A - \varepsilon I$ is singular at the ground eigenvalue. A positive rank-one lifting removes this obstacle without changing the monic ground transform or its root pencil. Write $D = \operatorname{diag}(j^2)$ and $[D, A] = rw^T - wr^T$, with $w = (1, \sqrt{2}, \ldots)^T$.

Proposition 27 (Ground-pencil preserving regularization)**.** *For $\rho > 0$, set*

$$B_\rho(z) = A - zI + \rho ww^T, \qquad r_\rho = r + \rho Dw, \qquad S_\rho(z) = B_\rho(z)D + r_\rho w^T.$$

If J spans $\ker w^T$, then

$$J^T B_\rho(z) J = J^T(A - zI)J, \qquad J^T S_\rho(z) J = J^T((A - zI)D + rw^T)J.$$

If ε is a simple ground eigenvalue and $w^T v_g \neq 0$, then $B_\rho(\varepsilon) > 0$. Its monic minimizer is $v_g/(w^T v_g)$, and its compressed pencil is exactly the ground-root pencil, independently of ρ.

Proof. The displaced commutator has generator r_ρ. The added term in S_ρ is $\rho(ww^T D + Dww^T)$, which vanishes after compression, as does the added term in B_ρ. The form $A - \varepsilon I$ is positive semidefinite with nullspace spanned by v_g. Its rank-one addition is strictly positive because $w^T v_g \neq 0$. Under the monic constraint $w^T a = 1$ it adds the constant ρ to the energy, so the minimizer is unchanged and equals the normalized null vector. □

Put $y(z) = B_\rho(z)^{-1} w$ and $H(z) = w^T y(z)$. Where $B_\rho(z) > 0$,

$$H'(z) = \|y(z)\|^2 > 0, \qquad H(\varepsilon) = \rho^{-1}.$$

The leading-functional solve and the natural positive factors therefore give a scalar refinement step $(\rho^{-1} - H)/\|y\|^2$. This point step is followed by an analytic certificate, rather than interpreted as one.

Specifically, an interval residual for an explicitly serialized vector and shift z gives some analytic eigenvalue in $[z - R, z + R]$. If $B_\rho(z) \succeq gI$ with $g > R$, rank-one interlacing gives $\lambda_2(A - zI) \geq g$. The residual interval then contains the unique ground eigenvalue. Including R as a scalar diagonal operator error transfers the positive surrogate to $B_\rho(\varepsilon)$ and hence to the unchanged analytic ground-root pencil. A scalar shift error has operator norm R; it does not acquire a factor $\sqrt{N+1}$.

The lifting, compression and leading solve pass 3241 exact rational checks on 240 fixtures, including singular sources. A multiple ground eigenvalue, or a ground vector orthogonal to w, is not repaired by this rank-one update and must fail the positivity condition explicitly.

The two independently enclosed finite sources now support all 1000 analytic ground-root ordinals. Point refinement uses 4000 decimal working digits, the serialized-vector residual uses 4500, and the positive source certificate uses 6000. These are certification settings, not necessary precision requirements for a claim run. No reference zero enters source assembly or the ground refinement.

C	N	D_1 (approx.)	D_{1000} (approx.)	predictor error bound
125	2000	662.268420	22.018173	2.521×10^{-6}
250	1500	1175.398737	305.898719	1.792×10^{-7}

The depths are approximate summaries of analytic intervals. The last column bounds the worst relative displacement-prediction error over all 1000 indices, using the previously frozen three-root operator predictor without refitting. Its inputs remain the earlier computed eigenstates; the comparison denominator is now an enclosed error of the true finite analytic ground root. All 2000 one-percent prediction checks pass, including the source and reference uncertainties.

The ground-eigenvalue relative radii are below 2.432×10^{-1477} and 3.554×10^{-1614} respectively. Exact decimal brackets, seeded from the earlier point displacements with relative half-width 10^{-8}, receive counts $k-1$ and k before analytic source inflation and window checks. All requested targets and arithmetic attempts are retained. The regularization certifies the roots at the stated complete N values; smaller prefixes of the same shifted matrix must not be misidentified as their own ground-state sources.

This strengthens the certification scope of the earlier thousand-root comparisons. It establishes neither a uniform error bound for the operator predictor nor a closed parameter-only convergence law. In particular, good performance at these two resolved sources does not remove the recorded low-mode edge failures.

Exact transport of the ground correction. For fixed N, let $a(z) = Jq(z)$ be a simple positive generalized-root branch of $(T(z), H(z))$, where here $T(z) = J^T((A-zI)D+rw^T)J$ and $H(z) = J^T(A-zI)J$. Differentiation and symmetry cancel the eigenvector derivatives:

$$\theta_k'(z) = \frac{\theta_k(z)\|a(z)\|^2 - a(z)^T D a(z)}{a(z)^T(A-zI)a(z)}, \qquad \nu_k(\varepsilon) - \nu_k(0) = \int_0^\varepsilon \frac{\beta^2 \theta_k'(z)}{2\nu_k(z)}\, dz.$$

Positivity and simple-branch continuation must hold throughout this interval. Positivity alone does not fix the numerator's sign: for $D = \mathrm{diag}(0,1)$, $w = (1,1)^T$, $r = 0$ and $A = \mathrm{diag}(a,d)$, the scalar root is $\theta(z) = (d-z)/(a+d-2z)$ and $\theta'(z) = (d-a)/(a+d-2z)^2$, which admits both signs. There are 302 exact scalar derivative and sign checks. This is not a CCM monotonicity counterexample; it explains why the zero-shift nested-pencil proof cannot simply be applied with a changing ground shift ε_N. A uniform indexed estimate of this integral, and of the mode, cutoff and finite-P channels, is still required.

8.22 Quantitative transfer from eigenvalue error to mode-root error

This result concerns the genuine fixed-cutoff mode channel. It sharpens the generic dual identity of Section 8.6 using the CCM divided-difference structure. The use of adjoint-weighted Galerkin residuals is classical [60]; the estimate below exploits the two inverse-frequency powers in the even adjoint tail. The finite transform is that of [1, Prop. 5.9]. Qualitative ground-state convergence under the hypotheses of [2, Thm. 6.1] does not itself supply this quantitative rate.

Fix $c = \lambda^2 > 1$, $L = \log c$, and $\alpha_j = 2\pi j/L$. Assume the exact real self-adjoint realization on $\ell^2(\mathbb{Z})$ is $A = D_L + K$, with D_L diagonal, even, increasing with $|j|$, $d_j \sim \log|j|$, $\|K\| \le K_*$, and $\mathrm{Dom}\, A = \mathrm{Dom}\, D_L$. Assume reflection commutes with A, and the off-diagonal entries satisfy

$$(\alpha_i - \alpha_j)A_{ij} = g_i - g_j, \qquad g_j = \alpha_j A_{j0}.$$

Let a be a real even unit ground state *in the even sector*, with eigenvalue ℓ, simple and isolated there by a gap $g > 0$. Write $B = A - \ell I$. Thus B is nonnegative on even vectors; no positivity of its odd restriction is assumed. Let v_N be a unit even ground Ritz vector on $|j| \le N$, phase aligned with a, and set

$$\Delta_N = \ell_N - \ell, \qquad Q_N = I - P_N, \qquad r_N = (A-\ell_N)v_N = Q_N K v_N.$$

Fix a positive simple nonlattice real root $\nu = \nu_{c,\infty,k}$ of F_a, and define

$$b_j = a_j/(\alpha_j - \nu), \qquad b_j^+ = \nu a_j/(\alpha_j^2 - \nu^2),$$
$$S = \langle a, b\rangle = \langle a, b^+\rangle, \qquad G = \sum_j g_j b_j, \qquad p = 2L^{-1/2}\sin(\nu L/2).$$

Proposition 28 (Exact even adjoint and its positive denominator)**.** *Under these hypotheses $b \in \operatorname{Dom} A$, $G \neq 0$, and*

$$F_{v_N}(\nu) = \frac{p}{G}\left\{\Delta_N\langle b^+, v_N\rangle + \langle Q_N b^+, r_N\rangle\right\}. \tag{79}$$

Moreover, with $f = F_a'(\nu)$,

$$\langle b, Bb\rangle = \frac{Gf}{p} = 2\langle b^+, Bb^+\rangle > 0. \tag{80}$$

In particular the coefficient

$$\mathcal{C}_{c,k} = -\frac{pS}{Gf} = -\frac{S}{\langle b, Bb\rangle} \tag{81}$$

is well defined. No nonvanishing assumption on its numerator is made.

Proof. The multiplier $(\alpha_j - \nu)^{-1}$ is bounded and commutes with D_L, so it preserves its operator domain. The series $\sum b_j$ is absolutely convergent by Cauchy–Schwarz. So is G, since $(A_{j0})_j \in \ell^2$ and $\alpha_j/(\alpha_j - \nu)$ is bounded. Neither g nor the all-ones sequence is being treated as an ℓ^2 vector. The root condition is $\sum_j b_j = 0$. Subtracting the ith eigenvector equation after division by $\alpha_i - \nu$, and using the off-diagonal identity, gives the convergent component identity

$$(Bb)_i = \sum_j A_{ij} a_j \left(\frac{1}{\alpha_j - \nu} - \frac{1}{\alpha_i - \nu}\right) = -\frac{G}{\alpha_i - \nu}.$$

If $\phi_j = p/(\nu - \alpha_j)$ is the evaluation representer, this says $Bb = (G/p)\phi$. Taking the even part and pairing with v_N proves (79) whenever $G \neq 0$.

For completeness, nonvanishing and positivity follow in the even sector itself. Differentiating the secular sum at its root yields $f/p = -\sum_j a_j/(\alpha_j - \nu)^2$. Evenness and the root condition give $\sum_j a_j/(\alpha_j^2 - \nu^2) = 0$. Subtracting the two energies then gives

$$\langle b, Bb\rangle - 2\langle b^+, Bb^+\rangle = -G\sum_j \frac{a_j}{\alpha_j^2 - \nu^2} = 0.$$

The even vector b^+ is not proportional to a: proportionality would confine the support of a to one value of $|j|$, whose even transform has no positive nonlattice zero in its secular factor. The simple even-ground hypothesis therefore gives $\langle b^+, Bb^+\rangle > 0$. The component identity now proves $G \neq 0$ and (80). This positivity argument concerns this divided vector; it does not assert positivity of the full operator. □

Proposition 29 (Projection energy and the omitted adjoint term)**.** *Put $t_N = Q_N a$, $\tau_N^2 = \|t_N\|^2$, $q_N = \langle t_N, Bt_N\rangle$,*

$$m_N = d_{N+1} - \ell - K_*, \qquad \chi_N = K_*^2/(gm_N).$$

If $\tau_N^2 < 1$, $m_N > 0$ and $\chi_N < 1$, then

$$q_N(1 - \chi_N) \le \Delta_N \le q_N/(1 - \tau_N^2). \tag{82}$$

If also $\alpha_{N+1} > \nu$, then

$$|\langle Q_N b^+, r_N\rangle| \le \Delta_N \beta_N, \tag{83}$$

$$\beta_N = \frac{\nu}{\alpha_{N+1}^2 - \nu^2}\sqrt{\frac{1 + 2K_*/m_N}{1 - K_*^2/(gm_N)}}.$$

At fixed c, k, $\beta_N = O_{c,k}(N^{-2})$.

Proof. Let $w = P_N a$. Since $Ba = 0$, one has $\langle w, Bw\rangle = q_N$ and $P_N B P_N w = -P_N K t_N$. Expand w in an orthonormal eigenbasis of $P_N B P_N$: its first eigenvalue is Δ_N, and every other one is at least g by min–max. Consequently

$$q_N \le \Delta_N + \|P_N K t_N\|^2/g \le \Delta_N + K_*^2 \tau_N^2/g.$$

On the omitted subspace $B \ge m_N I$, so $\tau_N^2 \le q_N/m_N$. This proves the lower bound; testing $w/\|w\|$ gives the upper bound.

On the omitted subspace write $Q_N b^+ = M t_N$, where M is the diagonal multiplier $\nu/(\alpha_j^2 - \nu^2)$ and $\|M\| \le \mu_N := \nu/(\alpha_{N+1}^2 - \nu^2)$. Since $D_L - \ell I$ is positive there and commutes with M,

$$\langle M t_N, B M t_N\rangle \le \mu_N^2 (q_N + 2K_* \tau_N^2) \le \mu_N^2 \Delta_N \frac{1 + 2K_*/m_N}{1 - \chi_N}.$$

Both vectors are even. Cauchy–Schwarz in the nonnegative even form B therefore gives

$$|\langle Q_N b^+, B v_N\rangle|^2 \le \langle Q_N b^+, B Q_N b^+\rangle \Delta_N.$$

The $\Delta_N v_N$ term vanishes in this pairing, proving the assertion. □

An explicit root remainder. Choose $I = [\nu - r, \nu + r]$ on which $h = \inf_I |F_a'| > 0$, and put $M_1 = L^{3/2}/\sqrt{12}$, $M_2 = L^{5/2}/\sqrt{80}$. These bound the first and second Fourier-evaluation derivatives of unit vectors on the real axis, by the second and fourth moments of the centered interval. The gap gives $e_N := \sqrt{2\Delta_N/g} \ge \|v_N - a\|$. Define

$$m = h - M_1 e_N, \qquad X_N = |p/G|\Delta_N\big(|S| + \|b^+\| e_N + \beta_N\big).$$

If $m > 0$ and $X_N < mr$, F_{v_N} has exactly one root ν_N in I, with $|\nu_N - \nu| \le X_N/m$. Taylor's theorem gives

$$\begin{aligned} |\nu_N - \nu - \mathcal{C}_{c,k}\Delta_N| \le{}& \frac{|p/G|\Delta_N(\|b^+\| e_N + \beta_N)}{|f|} \\ &+ \frac{M_1 e_N X_N}{m|f|} + \frac{M_2 X_N^2}{2m^2|f|}. \end{aligned} \tag{84}$$

Indeed, the derivative of F_{v_N} has fixed sign and magnitude at least m on I, while $|F_{v_N}(\nu)| \le X_N$. The endpoint signs prove existence and uniqueness. Expand at ν, use $|F_{v_N}'(\nu) - f| \le M_1 e_N$, and apply (79) to obtain the displayed remainder. Thus, as $N \to \infty$ with c, k fixed,

$$\boxed{\nu_{c,N,k} - \nu_{c,\infty,k} = \mathcal{C}_{c,k}\Delta_N + O_{c,k}(\Delta_N N^{-2} + \Delta_N^{3/2}).} \tag{85}$$

If $S = 0$, this formula does not imply a nonzero leading term or equality of the two convergence rates. The onset $m_N > K_*^2/g$, pole separation, and the root-derivative margin cannot be suppressed in growing-c, k use. For example, $A = 3I - \mathbf{1}\mathbf{1}^T$ on frequencies $-1, 0, 1$ has a uniform simple ground state. Its secular root $\nu = 1/\sqrt{3}$ has $S = 0$. This is an exact Loewner example, not a CCM state.

Residuals and arithmetic conditioning. For an approximate parent let $r_a = (A - \ell)a$ and $s = \sum_j a_j/(\alpha_j - \nu)$. The component identity becomes

$$B b = -G(\alpha - \nu)^{-1} + e_b, \qquad (e_b)_j = \frac{(r_a)_j + s g_j}{\alpha_j - \nu}.$$

For any even child, with $r_v = (A - \ell_N)v$, its exact finite algebra gives

$$F_v(\nu) = \frac{p}{G}\left\{(\ell_N - \ell)\langle b, v\rangle + \langle b, r_v\rangle - \langle e_b, v\rangle\right\}.$$

Using only the omitted block of r_v requires an additional bound on its retained block. For two scalar terms $Y = q + t \neq 0$, changing only q by a relative η changes Y by exactly ηq; the amplification $|q/(q+t)|$ can be large. It is one component of the precision budget, not a complete prescription for P or evidence that an input artifact is corrupt.

Resolution and the remaining parameter law. For certified upper bounds on Δ_N and the explicit remainder,

$$B_{\mathrm{mode},k} = \frac{|\mathcal{C}_{c,k}|\overline{\Delta}_N + \overline{\mathcal{R}}_{c,k,N}}{\gamma_k}$$

feeds the three-channel sufficient budget in Section 8.6. This still depends on the continuum ground state, its tail, gap and root conditioning; no closed uniform function of c, N, P, k has been established. In particular Δ_N is not the finite eigenvalue ℓ_N, and $\nu_{c,\infty,k}$ is not being identified with the Riemann ordinate γ_k.

The intrinsic first attainment at depth D_* is the minimum of $\{N \in \mathcal{A}_{c,k} : |\nu_{c,N,k} - \gamma_k|/\gamma_k \leq 10^{-D_*}\}$, with value $+\infty$ for an empty set. A persistent threshold instead requires every later integer to be admissible and accurate. They may differ under cancellation. The monotone zero-shift family in Section 8.20 supplies its own special ordering theorem; that theorem is not automatically transferred to the changing-ground family considered here.

The admissibility set must specify the root observable. For positive nonlattice T, the finite transform has exactly

$$J_{c,N}(T) = \max\{0, \lfloor TL/(2\pi)\rfloor - N\}$$

forced positive lattice locations below T outside the retained band. This counts locations, not multiplicity or all transform zeros. A reference-matched movable root need not have the same ordinal in the full transform; other roots and separation must also be counted.

Verification scope. The displayed arguments were independently reconstructed and checked with 3,900 exact rational identities and 414 separate 80-digit Ritz checks on 18 finite Loewner fixtures. These fixtures test the algebra, including the parity energy identity and retained-residual defects. They are not CCM assemblies and do not prove a uniform arithmetic tail estimate. Finite-parent numerical comparisons do not replace the continuum hypotheses or the explicit omitted-mode term in this theorem.

Reproduced finite-parent comparisons. A separate retained-data replay uses the stored $c = 1000$, $N = 800,890$ states and the same 1700-digit analytic generator. The exact two-term identity agrees at all 13 fixed indices to relative discrepancy below $2 \cdot 10^{-397}$. Nine pairs meet the frozen accuracy and root-residual gate. The full two-term Newton displacement passes all nine one-percent tests; deleting the omitted-mode term or using the parent coefficient alone passes none. At $k = 1$ omission already causes -1.60335% error. This is consistent with a fixed-cutoff late theorem whose onset is not yet attained in the large-cutoff ramp.

The measured relative depths make the indexed resolution dependence clear:

Target label k	$N = 800$	$N = 890$
1	1260.364036	1358.789387
100	1088.105848	1190.938797
300	729.949771	862.073328
500	73.905037	299.440729
525	29.817503	180.934808

These are reference-assisted point matches, not certified full-spectrum ordinals or minima over untested integers. At the parent target heights labeled $500, 600, 610$, respectively $1, 142, 155$ forced lattice locations already lie below them outside the retained band. The labels therefore require their explicit movable-root convention.

Four late comparisons at $c = 5, 9$, using $320 \to 640$ and $640 \to 1280$, instead pass the one-percent parent-coefficient test. Their overlapping states and 45-significant-digit coefficient summaries are retained with summary-rounding allowances. The original interval-center eigenvalue calculation, which fails the high-index identity, is also retained; Rayleigh products on the same vectors and generator repair it. Replaying these computations, including both versions, reproduces all eight comparison tables byte for byte and the selected exact-decimal identity fields. No source assembly or reference-zero certification is inferred from this replay. A separate contraction audit has two failures among 11 retained late rows, so a universal one-half gain-ratio shortcut is not supported by that corpus.

8.23 Local rational root bounds near the indexed resolution edge

The global Fourier derivative bound in Section 8.22 is deliberately conservative. It can fail to establish a useful neighborhood when the transform and its derivatives are small together. The following local bound retains that cancellation in the Taylor coefficients. It is elementary rational-function analysis, not a new asymptotic formula for the arithmetic ground state.

For an even finite state, set $F_u(t) = p(t)R_u(t)$, with $p(t) = 2L^{-1/2}\sin(tL/2)$. Away from the Fourier lattice,

$$R_u(t) = \frac{u_0}{t} + \frac{1}{\sqrt{2}}\sum_{j=1}^{N} u_j \left\{ \frac{1}{t-\alpha_j} + \frac{1}{t+\alpha_j} \right\}, \qquad \alpha_j = 2\pi j/L.$$

Here u_j are orthonormal even-sector coefficients. Let $R(t) = \sum_i w_i/(t-p_i)$, fix a real center x off its poles, and put

$$c_m = (-1)^m \sum_i \frac{w_i}{(x-p_i)^{m+1}}, \qquad d_i = |x-p_i|, \qquad q_i = r/d_i < 1.$$

For an integer $M \geq 1$ define

$$V_M(r) = \sum_{m=2}^{M} m|c_m| r^{m-1} + T_M(r),$$
$$T_M(r) = \sum_i \frac{|w_i|}{d_i^2} \frac{q_i^M\{(M+1) - Mq_i\}}{(1-q_i)^2}.$$

Proposition 30 (Local derivative margin and displacement enclosure)**.** *For $|h| \leq r < \min_i d_i$, $|R'(x+h) - c_1| \leq V_M(r)$. If*

$$a := |c_1| > V_M(r), \qquad |c_0| < \{a - V_M(r)\}r,$$

then R has exactly one real root $x + h_*$ in $(x-r, x+r)$, and

$$\left| h_* + \frac{c_0}{c_1} \right| \leq \frac{V_M(r)}{a}\, r.$$

If this interval also excludes every zero of $\sin(tL/2)$*, the same existence and uniqueness hold for the finite transform in that interval.*

Proof. Expand each simple fraction geometrically. The differentiated tail is bounded by

$$\frac{|w_i|}{d_i^2}\sum_{m=M+1}^{\infty} m q_i^{m-1} = \frac{|w_i|}{d_i^2}\frac{q_i^M\{(M+1)-Mq_i\}}{(1-q_i)^2}.$$

Keep the cancellations in the finitely many c_m before taking absolute values. The derivative has constant sign and magnitude at least $a - V_M(r)$ throughout the interval. The strict residual inequality gives opposite endpoint signs and therefore one root. Integrating $R'(x+h) - c_1$ from zero to that root gives the displacement bound. The last assertion follows because the sine factor never vanishes there. □

All quantities admit outward-rounded interval evaluation. An unresolved derivative, a radius reaching a pole or a carrier zero, or an unsuccessful Taylor-order limit is an unresolved certificate, not a claimed absent root. Arithmetic precision used to verify this finite rational function is distinct from accuracy of its source eigenvector.

A stronger finite check on the retained mode pair. The unchanged $c = 1000$, $N = 800 \to 890$ pair from Section 8.22 was evaluated at all thirteen previously frozen indices. The parent and child vectors are the exact decimal vectors in the retained files; no eigensolve or matrix assembly was repeated. Interval evaluation uses 2100 decimal digits, and an independent direct-power evaluation at 2600 digits rechecks every certified derivative margin and both endpoint signs. Nine hundred separate exact rational fixture checks verify the derivative-tail implementation.

All thirteen parent neighborhoods and nine child neighborhoods certify. The nine originally eligible pairs all retain the one-percent pass for the frozen full two-term Newton prediction. The largest certified relative comparison bound is below $7.36 \cdot 10^{-10}$, including the deliberate enclosure padding; it is not an estimate of the actual model error. The old global energy/Fourier-radius calculation established neighborhoods at only four of these indices. The new child orders are

k	1	20	100	300	400	450	475	500	525
M	1	1	1	1	1	3	5	9	21

The four originally ineligible child cases remain unresolved: $k = 550$ reaches the fixed order limit 96, while the proposed neighborhoods for $k = 575, 600, 610$ meet a Fourier carrier zero. They are retained rather than removed or relabeled as converged. These are local certificates for serialized states, not analytic-ground source certificates, full-spectrum ordinal counts, or thirteen independent matrix experiments.

Three different notions of sufficient resolution. The condition $\alpha_{N+1} > \nu$ in the earlier adjoint-tail bound is a condition for that particular estimate, not a necessary condition for an accurate movable root. In this retained example, $N = 800, k = 500$ has about 73.905 relative digits while $\nu L/(2\pi) \simeq 891.819 > N + 1$. A hard rule discarding every target beyond the retained Fourier band would therefore discard a successful reproduction.

There is nevertheless a degree obstruction: after multiplication by $t\prod_{j=1}^{N}(t^2 - \alpha_j^2)$, the secular numerator is a polynomial of degree at most N in t^2. A nonzero even state therefore has at most N distinct positive nonlattice secular roots. Resolving an entire prefix of K such roots requires $N \geq K$; it does not guarantee the required accuracy, nor identify a reference label with a spectral ordinal.

Finally, writing $\rho = \nu/\alpha_{N+1} < 1$, the earlier multiplier is

$$\frac{\nu}{\alpha_{N+1}^2 - \nu^2} = \frac{1}{\alpha_{N+1}}\frac{\rho}{1-\rho^2}.$$

Its fixed-k N^{-2} behavior becomes only N^{-1} when ρ stays strictly between zero and one, and loses that suppression as $\alpha_{N+1} - \nu$ remains bounded. These statements concern this explicit factor at fixed cutoff, with the other theorem constants controlled; they are not lower bounds for the actual error. An accuracy-dependent minimum-N law must distinguish the degree floor, an estimate's admissibility conditions, and the actual indexed accuracy threshold.

What this removes, and what remains. This local certificate closes the root-nonlinearity qualification for the nine retained eligible comparisons. It does not determine the arithmetic mode innovation, cutoff displacement, or source uncertainty from c, N, P, k. Those remain the work needed for the parameter law.

9 Discussion

Formula first. The empirical and analytic questions are deliberately separated. The first question is whether a caller who supplies $(\lambda^2, N, P_{\mathrm{req}})$ receives an accurate digit prediction. Tables 2 and 3 answer that question directly: the fixed published submodel is accurate to about one digit on independent ramps through $\lambda^2 = 20$, remains useful with a residual of about six digits at the edge of the calibrated domain, and fails decisively when extrapolated unchanged to $\lambda^2 = 1000$. The second question is why the working formula has its observed form. The Gram, Galerkin, prolate, determinant, and prime-event programs address that second question; none is allowed to redefine the caller-facing law without a better whole-cutoff validation result.

Relation to Groskin [7]. Groskin independently implemented the Connes–van Suijlekom matrix and measured a Galerkin convergence exponent $s(c)$ growing with the cutoff, proposing a Paley–Wiener analyticity-strip mechanism. The present work builds on that mechanism: Prop. 2(ii) gives the precise local-slope coordinate in which his $s(c)$ can be compared with our measured stretched-exponential. Establishing that the eigenvalue and root experiments sample the same asymptotic tail remains part of the transfer problem. The candidate strip rate is the archimedean $\Gamma_{\mathbb{R}}$ value $\delta = \pi/4$. What is new here is the digit-space formulation, the Shannon coordinate, the calibrated closed form $q_\gamma(\lambda^2)$, and the embedding of the rate in the full three-budget minimum. A second independent reimplementation reproduced the fixed-form ramps at $\lambda^2 = 7, 13, 20$ and supplied the large-cutoff constraint in Section 5.7; this corroborates the law on its stated domain while preventing an unqualified extrapolation.

Groskin's revised analysis, extended to cutoff $\lambda^2 = 100$, is independent corroboration of this two-regime structure. He finds that no single smooth-convergence law fits his eigenvalue-space data and explicitly declines to assert a convergence rate, reporting instead an unresolved tension between a *drifting* effective exponent and an apparent *finite* convergence limit. Both halves have a common empirical interpretation here: Prop. 2(ii) explains how a stretched exponential produces the drift (its local exponent $2s_{\mathrm{loc}}(N) = aqN^q$ grows with N), while the prime ceiling of Section 6 supplies the finite limit $D_{\mathrm{Primes}}(\lambda^2)$. What looks like an irreconcilable choice within a single-rate framework is simply the pre-saturation ramp and the ceiling of the minimum law (2), seen in one dataset.

Relation to Śliwiński [15] and Suzuki [5]. Śliwiński proves an inverse-logarithmic lower bound on the *mean* spectral error over the whole spectrum (Thm. 3.1), a different regime from the first-zero convergence here, and consistent with it, since the mean is dominated by the unconverged high-index eigenvalues while $\nu_1 \to \gamma_1$ converges super-exponentially. Suzuki's screw-function analysis supplies the archimedean $\ln|z|$ multiplier and the prime-frequency support that underlie Sections 5.2–5.5.

Origin of the leading rate: arithmetic, not archimedean. The prime ceiling is *modeled* by an archimedean object, the prolate eigenvalue of Section 6, yet the exponential rate is not carried by the archimedean part of the Weil form itself. Diagonalizing the screw-function split [5] at high precision, the archimedean-only localized form is $O(1)$-indefinite, with no exponentially small eigenvalue. Restricting it to the Sonin subspace, the orthogonal complement of the band-concentrated prolate modes, where Connes–Consani establish archimedean Weil positivity [18] (surveyed in [4]), restores positivity, but only as an $O(1)$ gap: the smallest Sonin eigenvalue is 2.26, 2.28, 2.57 at $\lambda^2 = 5, 8, 13$, essentially flat in λ^2, against a floor $K\lambda^2 = 27, 44, 71$. A direct diagonalization of the archimedean-only form at $\lambda^2 = 13$–50 finds the same structure from the eigenvector side: the lowest even mode is a delocalized Sonin-type state with an $O(1)$ eigenvalue, so the band-concentrated plunge eigenvector exists only for the full form. Archimedean positivity and the plunge are therefore *decoupled*: the place at infinity supplies an $O(1)$ spectral gap, while the exponentially small $\varepsilon^\infty \sim e^{-4\pi\lambda^2}$ is produced by the cancellation between the archimedean and prime parts of the full form. This locates one of the analytic steps behind the leading coefficient: a quantitative archimedean–prime cancellation estimate on the even ground eigenvector. The preceding constrained CCM-to-prolate quotient comparison is a separate open step, which is why the full CCM identification of K is T3.

Boundary spectral mass as a proof route. For a unit finite ground state let $w_0 = |\langle (2N+1)^{-1/2}\mathbf{1}, u_0\rangle|^2$. Across 56 independent configurations at $\lambda^2 \in \{7, 13, 20\}$, the ratio ε_N/w_0 remains between 0.398 and 1.762 while ε_N changes by more than 25 decimal orders. Because $\sum_j \xi_j$ is proportional to the boundary value of the finite Fourier state, this gives a Bonami–Karoui-type boundary-action interpretation of the depth [25]. The identity and its normalization should be derived explicitly before it carries theorem-level weight; at present it is one promising analytic attack on the mode law, not a replacement for it.

Next formula-selection experiment. The large-cutoff data do not yet choose among a corrected exponent, a changing onset coefficient, or a new coordinate. The next validation pass will compare five fixed candidate surfaces by leaving out entire cutoffs: the current q_γ model; the fitted linear relation for $1/q$; a saturated exponent; a logarithmically corrected pre-plunge coordinate; and a two-stage prediction $D_{\mathrm{eig},N} - C_{\mathrm{conv}}$. The selected replacement, if any, must improve the $\lambda^2 = 150, 200, 250, 500, 1000$ holdouts without degrading the existing $\lambda^2 \le 100$ surface.

Practical use. On the calibrated domain, the law is a parameter-selection guide: to target D^* digits, pick $\lambda^2 \gtrsim D^* \ln 10/(4\pi) = D^*/5.46$ so the ceiling clears D^*; use equation (10) to choose N so the ramp clears D^*; and choose P_{req} so the precision budget is nonbinding. Increasing any one parameter past the binding minimum yields nothing. At cutoffs beyond 100, the minimum and cutoff budgets remain useful, but the closed q_γ mode map must not be used for planning until the large-cutoff transition model is replaced or directly calibrated.

10 Rigor ledger

Table 8 collects the rigor status of every component of the law, together with the open item that would promote each to the next level. A fourth column types each open item: *named lemma* means the missing step is a precisely posed statement inside an existing classical theory (variational principle, Rouché, Szegő, Gevrey characterization), so that closing it is a matter of executing known machinery; *new analysis* means genuinely new estimates are required. The table preserves both categories as an explicit program for strengthening the paper.

Table 8: Rigor status of each component. T1: rigorous. T2: derived modulo explicitly named intermediate estimates. T3: derived in structure and validated against data, with a map not yet rigorous. T4: empirical. Open-item type: *named lemma* = precisely posed inside existing theory; *new analysis* = new estimates required.

Component	Status	Open item	Type
Minimum structure (Prop. 1), lower bound	**T1**	—	—
Minimum structure, matching upper bound	**T1** | (θ)	uniform quantitative noncancellation $E \geq \theta \max_i \varepsilon_i$	empirical check
$D_{\mathrm{Wprec}} = P_{\mathrm{req}} + O(1)$, slope 1	**T1** | stability	end-to-end backward error and conditioning audit	numerical analysis
Assembly-validity premise for empirical states	**T2**/validated	independent cutoff or closed-form check; residuals; structural and provenance gates	verified computation
Retained Fourier edge $T_{\max} = 2\pi N/\ln\lambda^2$	**T1**	—	—
Geometric capacity $K^{(0)}_{\max}$ (RvM estimate)	**T1**	oscillatory $S(T_{\max})$ and remainder retained by the exact count	—
Finite prime-power path: continuous/piecewise analytic with rank-one derivative jumps	**T1**	propagate each event through eigenstate and root response	perturbation theory
Stretched-exp rate unification (Prop. 2)	**T1**	—	—
General simple-zero displacement (Lem. 2)	**T1**	—	—
Exact logarithmic decomposition (Prop. 3)	**T1**	—	—
Endpoint flatness and nonanalyticity (Prop. 6)	**T1**	determine the actual boundary regime and tail	new analysis
Tail stability and late energy (Props. 4–5)	**T1** | simple gapped ground	cutoff-uniform gap and coefficient-tail law	new analysis
Indexed linear mode transfer (Section 8.22)	**T1** | stated hypotheses	uniform tail, root and gap bounds; onset	spectral analysis
Local rational root bounds (Section 8.23)	**T1** (finite)	analytic source uncertainty; uniform parameter estimates	interval analysis
Tail-to-transform transfer (Lem. 3, Cor. 1)	**T1** | stated hypotheses	establish coefficient decay and simple limiting zero	new analysis
Lowest Rayleigh values at fixed cutoff	**T1**	—	CCM Prop. 3.4
Even-sector ordering at three certified finite matrices	**T1** (finite)	extend certificates and obtain a uniform continuum sector gap	interval/spectral analysis
Eigenstate and compact-uniform transform convergence	**T1** | (H^*_λ)	simple, isolated, even continuum ground state	CvS Thm. 6.1
Selected zero branch and fixed-cutoff digit limit (Prop. 8)	**T1** | (H^*_λ)	simple isolated transform zero; noncoincidence with γ_1	Hurwitz/Rouché
Growing-cutoff quantitative rate	**T3**	convergence of the continuum zero and the CCM/prolate transfer	new analysis
$n = 4$ prolate singular-value asymptotic (K, 9/2)	**T1**	—	—
D_{Primes} leading K: CCM floor transfer	**T3**	constrained h_0–h_4 quotient comparison and prime sign (Lem. 5)	new analysis
D_{Primes} leading K: reverse cutoff asymptotic	open	archimedean–prime cancellation	new analysis
D_{Primes} CCM floor slope 9/2	**T3**	transfer of the fourth-index Fuchs law; arithmetic inertness	new analysis
D_{Primes} matching offset (P_m, c_m)	**T4**	test $C_{\mathrm{conv}} \approx 3.03$ in the saturated limit; construct Z_λ and prove the uniform comparison	new analysis
Shannon-like coordinate $\lambda^2 \ln\lambda^2$	**T3**	prove the rate inputs and identify the exact time–band limiting operator	new analysis
Smooth Riemann–von Mangoldt density and $K^{(0)}_{\max}$	**T1**	—	phase counting/Stirling
Scaled CCM equilibrium realization	open	normalized determinant/logarithmic-potential limit and external field	new analysis
Archimedean matrix spectral distribution	**T2**	uniform spectral distribution and arithmetic stability beyond the fixed-cutoff bounded remainder	new analysis

Continued on next page

Table 8 (continued)

Component	Status	Open item	Type
Common-edge interpretation (two scales)	**T3**	determinant-to-zero and spectral comparison; universality class; deep-edge constant	new analysis
Archimedean rate $\delta = \pi/4$	**T3**	finite-state Gamma-envelope realization, Problem 1(ii), uniform tail control, and doubling	new analysis
Closed exponent $q_\gamma = 2/(2\gamma + \ln\ln\lambda^2)$ on $7 \le \lambda^2 \le 100$	**T3**	calibrated-domain formula validated; unchanged large-cutoff extrapolation falsified	empirical model
Large-cutoff transition map	open	whole-cutoff comparison of exponent, onset, and logarithmically corrected coordinates	empirical/analytic
Onset constant $c = 4/\pi$ on the calibrated domain	**T3**	derive or replace the fitted argument coefficient; joint large-cutoff fit gives drift	new analysis
Conditional Gram representation (Prop. 9)	**T1** | (H_{RH})	unconditional substitute or controlled off-line contribution	new analysis
Divided-difference transfer (Prop. 10)	**T1** | (H_{RH})	uniform limits of $S_N, B_{k,N}, R_{k,N}$ and cutoff-uniform tail control	new analysis
Corrected displacement identity (Prop. 11)	**T1**	bound the signed-complex remainder uniformly in k	new analysis
Multi-root cancellation and exact remainder (Prop. 12)	**T1** | stated finite hypotheses	uniform root matching and stencil conditioning	new analysis
Verified-height complex-tail bound (eq. (46))	**T1**	certify coefficient/energy inputs and outward rounding	verified computation
Three-root predictor, 121 distinct retained pairs	**T3**	larger frozen windows and independent assembly/ordinal certification	numerical analysis
Nonlinear common-field defect (Prop. 15)	**T1** | finite hypotheses	certified nonlinear stability and indexed matching	new analysis
Indexed dual mode identity and resolution disc	**T1** | spectral hypotheses	uniform primal/dual tail and root-derivative estimates	new analysis
Nonlinear new-index test, two of four targets qualified	**T3**	unresolved windows retained; no uniform 1000-root claim	numerical analysis
Stored-state ordinal counts and absent movable branch	**T1** for stored data	transfer source-error bounds to the exact CCM state	exact polynomial counts
Complete operator-predictor tests: 1362/1362 pairs	**T3**	uniform parameter-only rate and source certification	two complete prefixes
Finite analytic ground certificate, $C = 250, N = 500$	**T1** (finite computation)	full indexed root transfer and uniform rates	verified computation
Two-vector indexed dual reduction	**T1** | invertibility	uniform dual and structured residual bounds	new analysis
Root-specific dual and finite precision bound	**T1** | finite hypotheses	uniform slope, scalar and arithmetic residual bounds	new analysis
Mode-ladder ground eigenvalues, $C = 250, N = 750$–1500	**T1** (finite computation)	indexed transfer of all point-state root errors	verified residual and interlacing
Symmetric movable-root pencil	**T1** | block hypotheses	uniform min–max comparison and resolution threshold	new analysis
Mode-ladder predictor edge failures	**T3**	uniform remainder and neighbor-resolution condition	complete-prefix tests
Analytic zero-shift pencil, first 1000 roots	**T1** (finite computation)	uniform cutoff/mode remainder	source intervals and root count
Arbitrary-shift error decomposition	**T1** | branch hypotheses	uniform channel and resolution estimates	exact finite identities
Two-source predictor edge failures	**T3**	closure falsified as a uniform replacement	all 1362 targets retained
Monic zero-shift polynomial	**T1** | finite positivity	varying-weight asymptotics and indexed leakage	exact finite minimization
Cardinal-subspace resolution bound	**T1** | positive measure	explicit C, N, k leakage estimates	min–max inequality
Finite-mode continuum action	**T3**	prefactor, discrete remainder and source transfer	5/4373 displacement passes
Monic source resolvent	**T1** | finite hypotheses	uniform source-amplitude estimates	3332 exact rational checks
Trial amplitude and remote-energy model	**T3**	discrete prefactor and edge remainder	174/4373 full-trial passes
Uniform Gamma-carrier factor	**T1**	full polynomial amplitude is separate	693 Arb component checks
Joint monic mode transport	**T1**	uniform indexed update and ground-source bounds	2275 exact and 585 interval checks

Continued on next page

Table 8 (continued)

Component	Status	Open item	Type
Prefix counts and interlacing	**T1**	zero-shift finite source; uniform rate open	4902 analytic brackets; minimum $N = 1091$ at $C = 125$
Specified accuracy budgets	**T1**	two finite zero-shift sources; parameter law open	18 000 targets; 12 495 exact minima
Analytic ground-root lifting	**T1**	finite sources; uniform predictor bound open	2000 analytic root/error certificates
Saturation control, $C = 125, N = 2000$	**T3**	uniform indexed convergence rate	1000/1000 analytic error comparisons
Cutoff discriminator, $C = 125, N = 1500$	**T3**	analytic ground-source transfer and uniform rates	1000/1000 point predictions
Finite deflated Stieltjes bounds	**T1** for stated finite hypotheses	replace measured deflation by input-only control	new analysis
Global germ and Padé shape tests	**T3**	uniform moment estimates and independent validation	new analysis
Central Schur bracket and weighted moments	**T1** | block hypotheses	uniform CCM bounds and assembly certification	new analysis
Energy/transfer increment split	**T1** finite identity	fixed-cutoff limit and uniform bounds on transfer variation	indexed budget refinement
Gamma-polynomial and finite kernel identities (eq. (47))	**T1**	uniform relative arithmetic weighted-kernel estimate	new analysis
Scalar Schur jets and positive innovations	**T1** | positive finite blocks	uniform relative innovation profile; floating-point stability	new analysis
Unfitted corrected estimates, 100 retained roots	**T3**	a priori cutoff/resolution law and assembly certification	new analysis
Finite conversion-gap formula (eq. (33))	**T1** | $(\mathrm{H_{RH}})$ + **T3**	controlled remote-zero tail and saturated-limit asymptotics	new analysis

The minimum lower bound is unconditional; the equality direction holds under the quantitative noncancellation condition of Prop. 1, supported across the experimental program but not proved uniformly. T1 | (H^*_λ): the stated implication is rigorous conditional on the explicit spectral and zero-branch hypotheses in eq. (29). Yoshida–Bombieri establish the positivity component in the small-support range [16, 17]; the remaining hypotheses are stated separately rather than attributed to those results.

11 Conclusion

The indexed extension remains unfinished. Sections 8.20 and 8.21 add exact finite accuracy budgets and analytic ground-root certificates for two complete thousand-root sources. The uniform parameter law and edge remainder remain open. Section 8.19 certifies a finite zero-shift resolution threshold and interlacing, while retaining the failures of the continuum integer budget. Section 8.18 now gives an exact joint polynomial/carrier update; controlling its indexed size and derivative denominator remains necessary. The new saturation control in Section 8.17 rejects an isolated Gamma amplitude correction despite its improved earlier curves. Section 8.16 adds an unfitted curvature prediction and an exact finite bound on the zero-shift/ground-source transfer. The full amplitude trial still fails most displacement tests, even though one remote-factor model succeeds for all first 1000 targets at a sufficiently resolved configuration. Neither that success nor the proposed turning point establishes a uniform C, N, P, k law. The remaining mathematical requirements are an accurate discrete amplitude, control near the indexed resolution edge, and the precision-dependent remainder.

We have given a single law (2) for the convergence of the CCM zeta spectral triple. In it the minimum structure is forced; the measured prime ceiling has leading coefficient $4\pi/\ln 10$, matching the rigorous fourth-index prolate singular-value asymptotic. Its transfer through the constrained CCM trial state and the reverse inequality are open; and the mode budget, the convergence rate named open in [1], is measured as the logarithm of a stretched-exponential. The fixed operational map $q_\gamma = 2/(2\gamma + \ln\ln\lambda^2)$ with $c = 4/\pi$ predicts independent ramps through

$\lambda^2 = 20$ to approximately one digit RMS and remains the validated caller-facing formula on $7 \le \lambda^2 \le 100$. A new $\lambda^2 = 1000$ sweep rejects its unchanged extrapolation and places the effective exponent in the convention-dependent range 0.74–0.80; the data do not yet decide whether the correction belongs to the exponent, onset coefficient, or coordinate. Identifying the calibrated root-error exponent with a sharp eigenvector coefficient class remains open, and direct fits to finite archived eigenvectors show substantial N- and window-dependence that any continuum theorem must explain. The rigorous Riemann–von Mangoldt backbone and the measured convergence surface motivate the conjecture that the mode ramp and prime ceiling are two scales of one CCM spectral edge (Section 7). Under (H^*_λ), the selected finite-section zero converges to the fixed-cutoff continuum zero. Along a growing-cutoff path, vanishing numerical, Galerkin, and cutoff displacements is sufficient for convergence to γ_1; the explicit rate $D \sim (4\pi/\ln 10)\lambda^2$ remains the quantitative model tested here, not a consequence of the fixed-cutoff theorem. Each claim is stated with an explicit rigor level. The rigorous core (the minimum bound, the precision slope, the stretched-exponential reparametrization and local-slope identities of Prop. 2, and the general simple-zero displacement Lemma 2) stands on its own. Conditional on global RH, Proposition 10 adds an exact finite divided-difference formula for the eigenvalue-to-root conversion; it explains the measured conversion gap but is not part of an unconditional route to RH. The tail-to-transform mechanism is quantitative at fixed cutoff in Proposition 4 and Lemma 3. Proposition 3 supplies the exact bounded logarithmic remainder, and Proposition 5 identifies the late eigenvalue error with logarithmically weighted coefficient energy. Section 8.22 now sharpens the indexed root displacement to a linear eigenvalue-error transfer with an explicit remainder and positive even-sector denominator. Section 8.23 adds explicit local rational root bounds and interval qualification of all nine previously eligible retained mode-pair displacements; four other cases remain unresolved. These certify the serialized states, with analytic source uncertainty remaining separate. The coefficient asymptotic and cutoff-uniform constants remain open. The variational bound of Prop. 7 supplies the unconditional trial inequality but does not identify its quotient with the prolate deficiency; and the remaining open items are typed in Table 8. Of these, the *named lemmas* (including the constrained-trial comparison and prime-sign hypothesis behind Lem. 5) are precisely posed inside classical theories and are natural next steps; the *new-analysis* items (the determinant/logarithmic-potential limit and spectral comparisons behind the common-edge conjecture, the endpoint-compatible coefficient law and crossover (Problem 1), the coefficient-class subordination behind q, the archimedean–prime cancellation needed for the reverse cutoff asymptotic, and positivity beyond the Yoshida range) are where the construction's genuinely open mathematics lives. The paired eigenstate/root archive also isolates a more economical candidate for the finite-cutoff offset: the depth-to-relative-root conversion approaches about 3.03 digits over the largest precision-clear cutoffs sampled. If that constant persists after saturation, it favors the asymptotic logarithmic coefficient $P_m = 9/2$ and explains most of $c_m \approx 9.5$ without a growing conversion penalty. This is a new empirical constraint, not a proof of the target-transform comparison. Separately, the exact finite-band edge (3) and its Riemann–von Mangoldt capacity (5) quantify how many zeta ordinates are geometrically available before questions of convergence quality arise.

12 Explicit non-claims

The per-component rigor of the law is the subject of Table 8 and is not repeated here. Beyond it, we are explicit about three boundaries.

- **No proof of the Riemann Hypothesis.** We characterize how the CCM/CvS approximation converges; the link from the construction to RH, continuum positivity of QW_λ for all λ, is the open conjecture of [1, 2, 4], and is neither assumed nor advanced here.
- **We do not establish that the construction reproduces the zeta zeros.** That the

finite operator's zeros converge to the true ordinates as $\lambda^2 \to \infty$ is the construction's own open premise; we quantify the approach to γ_1 *under* that premise through the growing-cutoff criterion following Prop. 8, and do not prove the premise itself.

- **Scope.** The law is stated and tested for the first zero γ_1 of the Riemann zeta function. We make no claim about its constants for higher zeros, nor about extension to Dirichlet L-functions or other settings. The reach quantities $T_{\max}$ and $K^{(0)}_{\max}$ count the frequency band and its expected zero occupancy; they are not asserted to be a higher-zero convergence or correct-ordering theorem. The fixed operational constants (q_γ, c) are validated on $7 \le \lambda^2 \le 100$. Historical reference values at $\lambda^2 = 150, 200$ remain excluded until their exact ramp provenance is recovered. The closed map has now been tested at $\lambda^2 = 1000$ and fails there: all fitting conventions give an effective exponent near 0.74–0.80, rather than $q_\gamma(1000) = 0.648$, and the unchanged formula overpredicts the measured ramp by more than 100 digits. We therefore make no claim that the calibrated q_γ or $c = 4/\pi$ persists at large cutoff. The minimum-budget structure, cutoff ceiling, and stretched-ramp family are separate claims and are not retracted by this failure.

- **No unconditional general finite-cutoff hypothesis.** Statements requiring the full spectral and zero-branch hypothesis (H^*_λ) are conditional. Yoshida and Bombieri establish its positivity component in the small-support range [16, 17], and Suzuki establishes a positive, simple, even continuum ground state for sufficiently small aperture [5, Thm. 1.4]; simplicity and noncoincidence of the selected transform zero remain explicit assumptions. We do not claim positivity of QW_λ for general λ.

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Statement on the Use of AI Tools

In accordance with current publication practices concerning authors' use of generative AI tools, the author discloses that Anthropic's Claude and OpenAI's Codex served as aids in preparing the exposition and the Rust implementation code, to specifications set by the author. AI assistance was used to improve prose formatting and readability, gather research material, propose candidate arguments, and validate calculations; all such output was reviewed by the author. The author originated, directed, and conducted every computational experiment presented here. The mathematical development was led by the author with assistance from AI. The author bears full responsibility for the contents of this manuscript.

Empirical testing and code availability

The results above rest on an extensive empirical program: over 5,000 high-precision CCM experiments across the cutoff, basis-size, and working-precision axes. Candidate rate laws were tested against the data and discarded when they failed [51]. Each budget was isolated by varying one parameter while keeping the other two nonbinding. The fixed published formula was then tested without per-cutoff refitting: an independent implementation gives RMS errors 0.47, 0.66, and 1.18 digits at $\lambda^2 = 7, 13, 20$, while a new $\lambda^2 = 1000$ sweep supplies the explicit out-of-domain failure in Table 3. The breadth of the resulting dataset ($\lambda^2 = 200$ at $N \ge 2400$

here and $\lambda^2 = 1000$ in the reproduction [50]) made those parameter-isolation and extrapolation tests possible.

A calibration state is admitted only with an artifact identity that records its numerical route and dependencies, a precision check, an archimedean cutoff or independently validated closed-form check, eigenpair residuals, structural symmetry tests, and reference-zero provenance. These gates are part of the reported empirical law: precision stability by itself is not treated as assembly correctness.

All tabulated results can be regenerated with the claim scripts supplied in the companion repository; each script writes its complete console log and structured measurement records. The paper source and reproduction scripts are available at `https://github.com/TeamXcelerator/ccm-convergence-rate`; the shared high-precision library (Xcelerator Toolkit) is at `https://github.com/TeamXcelerator/xcelerator-toolkit`. The integrity-bound high-precision reference-zero dataset is distributed with the toolkit. No proprietary input data are required. Both repositories are source-available in accordance with the licensing provided with each repository.

A Self-contained structural facts

This appendix records the two structural facts about $\widehat{\xi}$ used in the body, the closed form of the transform (Lemma 1) and entirety/reality of its zeros, in a form that depends only on elementary computation and on the self-contained proofs of [1]. Its purpose is to make explicit that no statement in this paper routes through the Carathéodory–Fejér machinery of [2], which we cite as the historical origin of the reality mechanism only.

A.1 The closed form of the transform, from scratch

Write $t = \ln u \in [-L/2, L/2]$, so that $V_n(u) = L^{-1/2} e^{2\pi i n(t+L/2)/L} = L^{-1/2}(-1)^n e^{2\pi i n t/L}$ and the Fourier–Mellin transform is the Fourier integral over the interval. For a single mode,

$$\begin{aligned}\widehat{V}_n(z) &= \int_{-L/2}^{L/2} e^{izt}\, V_n(t)\, dt = L^{-1/2}(-1)^n \int_{-L/2}^{L/2} e^{i(z+2\pi n/L)t}\, dt \\ &= L^{-1/2}(-1)^n\, \frac{2\sin\big((z+2\pi n/L)\, L/2\big)}{z+2\pi n/L}.\end{aligned}$$

Since $\sin(zL/2 + \pi n) = (-1)^n \sin(zL/2)$, the sign cancels:

$$\widehat{V}_n(z) = \frac{2L^{-1/2}\sin(zL/2)}{z+2\pi n/L}.$$

Summing over the eigenvector $\xi = \sum_{|n|\le N} \xi_n V_n$ and reindexing $j = -n$ (using $\xi_{-j} = \xi_j$ for the even eigenvector, or simply relabeling) gives

$$\widehat{\xi}_N(z) = 2L^{-1/2}\sin(zL/2) \sum_{|j|\le N} \frac{\xi_j}{z - 2\pi j/L},$$

the closed form of Lemma 1, in agreement with [1, Prop. 5.9]. The derivation is a one-line Fourier integral; no external input is used. Entirety is immediate: each apparent pole $z = 2\pi j/L$ is cancelled by the zero of $\sin(zL/2)$ there (the removable-pole computation in the proof of Lemma 1), so $\widehat{\xi}_N$ is entire of exponential type $L/2$.

A.2 Reality of the zeros: provenance

The statement that, in the critical case, all zeros of $\widehat{\xi}_N$ are real and coincide with the spectrum of the perturbed operator is [1, Thm. 5.10]. We record why its proof is self-contained there. The proof chain consists of: (a) [1, Lem. 5.2], the observation that the relevant matrices have the structure $\tau_{ij} = (b_i - b_j)/(i-j)$; (b) [1, Lem. 5.4], the elementary commutator identity $DT - TD = |\beta\rangle\langle\eta| - |\eta\rangle\langle\beta|$ for such matrices, with D the diagonal mode operator; and (c) the conclusion of [1, Thm. 5.10]: when ε_N is the smallest eigenvalue with even simple eigenvector, $T = QW_\lambda^N - \varepsilon_N \geq 0$, the perturbed operator is self-adjoint in the T-inner product, and the transform is proportional to a regularized characteristic determinant, $\det_{\mathrm{reg}}(D'' - z) = -i\lambda^{-iz}\widehat{\xi}(z)$, whose zeros are therefore real, being the characteristic polynomial of a self-adjoint matrix. Each of (a)–(c) is proven within [1] by direct computation and does not require the general Carathéodory–Fejér extension of [2]. The distinct qualitative passage from finite sections to the continuum eigenstate and its transform zeros in Prop. 8 does invoke [2, Thm. 6.1] under (H_λ^*).

A.3 The independent classical foundation

For completeness we record the pre-CvS lineage on which the truncated Weil form itself rests: Weil's positivity criterion; Yoshida's unconditional positivity for test functions of sufficiently small support [16]; Bombieri's study of the quadratic functional restricted to a finite interval, its minimizer in the L^2 unit ball, and the eigenvalues of its finite truncations [17]; the operator-theoretic strengthening of Yoshida's theorem at the archimedean place by Connes–Consani [18]; the prolate identification of the small eigenvectors in the ζ-cycles paper [19]; and Suzuki's screw-function framework [20, 21, 5], which unifies Yoshida, Bombieri, and the Connes–Consani(–Moscovici) results by means of continuous functions and de Branges-space methods. The construction studied in this paper can be founded entirely on this lineage together with [1].

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